

CTL 密钥封装算法 x86 架构实现

自评估测试报告

算法名称：CTL

算法类别：密钥封装

细分功能：密钥生成/封装/解封装

实现版本：参考实现版

安全级别：128bit/256bit/512bit

目录

- 1. 概述..... 3
- 2. 测试环境..... 3
 - 2.1. 软硬件配置 3
 - 2.2. 测试工具 3
- 3. 参考实现测试..... 4
 - 3.1. 功能测试 4
 - 3.2. 性能与资源消耗测试 65
 - 3.3. 传输与存储开销测试 67

1. 概述

本报告针对 CTL 密钥封装算法在 x86 架构下的参考实现版和优化实现版开展自评估测试。测试内容包括功能测试、性能与资源消耗测试、传输与存储开销测试。

功能测试用于验证 CTL 算法密钥生成、封装、解封装等核心部件是否完整，验证封装端生成的共享秘密值与解封装端恢复的共享秘密值是否一致。测试过程中采用固定随机种子或固定测试向量，保证测试结果可复现。

性能与资源消耗测试用于统计密钥生成、封装、解封装三个核心操作的平均耗时、平均 CPU 周期数、吞吐量，以及算法加载后静态内存占用和运行过程峰值内存占用。测试次数应不少于 100 次，实际测试次数根据测试命令填写。

传输与存储开销测试用于统计 CTL 密钥封装算法涉及的数据尺寸，包括公钥、私钥、密文和共享秘密值等。

2. 测试环境

测试时间为：2026/6/24-2026/6/25

2.1. 软硬件配置

序号	名称	规格和标识	用途	数量
1	CTL 参考实现版程序代码	版本：__1.0__；参数集：__257-512/769-1024/3329-2048__	被测试程序代码	1
2	CTL 优化实现版程序代码	版本：__1.0__；参数集：__257-512/769-1024/3329-2048__	被测试程序代码	1
3	测试终端 PC	CPU：__Intel® Core™ i9-14900HX__； 主频：__2.2GHz__； 内存：__32 GB__； 系统：__Windows11__	编译与运行测试程序	1
4	MinGW 编译环境	GCC 版本：__15.2.0__；目标架构：x86-64	编译 C 语言测试程序	1

2.2. 测试工具

序号	工具名称	工具类型
1	MinGW / MinGW-w64	C 语言编译工具
4	benchmark_selfeval_ctl.c	性能、资源消耗、传输与存储开销测试程序
5	test_kem_correctness_ctl.c	功能正确性测试程序

3. 参考实现测试

本章针对 CTL 密钥封装算法参考实现版开展测试。

参考实现版使用如下编译命令：

```
gcc -std=c99 -O2 -Wall -Wextra -Wpedantic test_kem_correctness_ctl.c <CTL_SOURCE_FILES> -o test_kem_ref.exe
gcc -std=c99 -O2 -Wall -Wextra -Wpedantic benchmark_selfeval_ctl.c <CTL_SOURCE_FILES> -o benchmark_ref.exe
-lpsapi
```

3.1. 功能测试

测试编号	REF-1
测试名称	CTL-257-512 参考实现版密钥封装功能正确性测试
测试目的	测试 CTL-257-512 参考实现版是否支持密钥生成、封装和解封装功能，并验证封装共享密钥与解封装共享密钥是否一致。
预置条件	1.已在测试终端 PC 上部署 CTL 参考实现版程序代码； 2.test_kem_correctness_ctl.c 能够正常调用 CTL 的密钥生成、封装、解封装接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-257-512 参考实现版程序代码目录； 2.输入 make all 生成 test_kem_correctness_ctl.exe 文件； 3.运行 test_kem_correctness_ctl.exe； 4.比较封装共享密钥与解封装共享密钥是否完全一致； 5.重复执行 3 组测试，并生成 ctl_kem_kat_vectors.txt 结果文件。
预期结果	1.test_kem_correctness_ctl.exe 程序能够正常运行； 2.终端显示 “All CTL KEM correctness tests passed.” 3.正确生成测试向量文件 ctl_kem_kat_vectors.txt。 4.测试向量文件中的 ss_enc 和 ss_dec 相等；
实测结果	程序正常运行且显示 “All CTL KEM correctness tests passed.”， ctl_kem_kat_vectors.txt 文件中的三组 ss_enc 和 ss_dec 都相等。

测试向量：

```
algorithm = CTL-257-512

seed_len = 55

initial_seed =
ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 521

sk_len = 2953
```

ct_len = 473

ss_len = 16

pk =

99B4BD408EE8A95011ACEA41FEE592AF5879CD3D3EAD420122FB50AC7B94D5F15091E56DA6D
57C7A10E2766DF8144E94A2D67DE507A6139B0DA5DC0B72A106A1FA49413C8612D88938AC9316
DC8A0EFB45464AD04FDA893F76B94333FDA9976DFBFD39CDC7CDB2BC071FD0F98D5551D5D35
A4CF0EFF3C04BF4EAA49013E3A3586A562C012D6CEBB7DCB48F1B310F471365D884FD22C9217F
36615FA02CD2082B0C33D7CA23014D918331530BE5E0D00598F7AF0B93E55A9151146B3D80889682
1903579022FF90557DBA44F95E401DD029E325F672D50CC90F0FEE91799F3F49E44FCEC1330B8EFC
FA0BB0A10775A7410758235182FA57D2E9455E40C6C393502FEDA2F92745D1FD0EB1A39786C9FB
DA9B9202F4E3E791EA7DC7D953AF696588FE47907C8A1A908F5D2EAE969A9421029A9F7A664DB
8B81D059289D53F98E164C65AB0C10641641211590A98B32B56EBD084F17B58F3B823B62EA25FC1
FAAF00D6BABF8F807BFAF6062E8CFA2A79872931B5BE46164F97E0DAD625C100521488E6331E93
F9024946CB7E7D2ED3C6B7CFBF164A0D36E1521DD2EE461B9DE28AF9019A85D33DE8C72BC75E
97E47E97DC02A7F1D94B3C3D54B994B59D2F48C30B6DDCFF0253E781E573CD09BFC1D20076C45
87A4F85009894AD762C19D7EE5FD0356B546F654705DEAB6426B0916A05057E176F2A00C5B2D349
00920AD9FE9510F147A389E5BCD6BA907

sk =

19AE5D933F84538C0270359F7DED4695A38B68A9E050084EFBA8BDCDD685B942EE3FE05922EE90
1FE0E9E956242B81B7F61C15C1111D711EAEB4076508CEC087DD0000F011FF00F0011010000F00001
F1000000000100010F0F10F11F000101F0000F00F011FF100000000101FFF01F110FF000F00F000F100
0F101F00000001010F00011F00000F010101110F00F0000F000000110000000000000F0F0100FF0000001
F00F000000001001000100110F1000F00FF0F1F0F00F0010000010FF00FF000F0000F00F001F0000FF001
0011010000000FF01100F0F0F1F00FF00010100000101011F000100FF010F0F00001F00F00FF0F0000F00
00101100FF0001000F10001F010000F001F0001F0F0000F00F001F1F000F001000F00F110F00F000000F0
1000F000F1F1010FF00F01F0000000F0F001F000000000FF01001000001001F00F0FF100000110000F00
000F0F0000001F000100F10100000000000100F10100000000000F01000F0000000FF0FF010000000001
00F000FFF100000F00FF10001000010000F10001F01001010F0000FF110000F000000100100101100010
00110F010000000111000011F0010F0001F1100F00F0000100001000F00000010F1000F000001000001100
000F10FF011F0F0101100000101100F0F000000F010011100F00010000100010010101FF1010F001000000
000F000000000F0F00101111101010F000100010000F0F0010000000F10000F00000FF000F00100F01F00
0000000000010F00F00000F00F100100000001F000100F100F0F0F000F9E00D8C142FC31C9038139E882
420BFE40FBA13D1010FCFB81830C0041F3FE840FDF83FC207FFC5EBC0BF10307E00023E0B913B08
5042042040F010FE0010FB08C03B0CCEBEF3CFE0BD17DEC00BB0BC1F904204113C1C2DB70BE0
C3180F00D7C088080F7F040040F8203CF3CF47F0B2BCEB8FBF2BC101FC40840BE0860FF0BEFC007
E03B005EFEFC7F430FD0C517BFFF0BBFBE27AF3B106FF70BC1000F7FFE008DC3040EC70050BE08
51471B7F8A0760440BAFC223FFEE7CF3F177002FC403EF0003C0BC300F880830C2002E43F8203E07
D0F227E0BDFBE139FFBEFD13D13C33703E041E8118800403F03D1B5284202FBFFFEF3903F17E07C
0BA1BD0BFFFE10113EFC423507FF810FFF010C1F83E4304307BF3C0811FC07CF3F081F7E0BA27A0
C3EC117D0C107B0BF041FC900103E080D8017FEC5003F030C0F3AFC21B903EEF90BC0C420A03F0
C10FBFFC0400BFF41F3EF09EFE07F0C1140240F7DFFA1810C2FFC1BE0C207E140EBE002F47EBFE4
5FC1EFC079FBB2011880BFF7AFFCEFB042F44140F84EBFF03001E44083100FC6EFFEFEBFAF00EC3
F80FC1106FFD07D0BA08010013D14107BF7EFBEFF90C2FC3FC3F7C03C04417F1BCDFFFFC146FBD
E4813EE000460FF083F3D18203CF39083EC4FFD003F8003B07DF38000DBE1FE17EFC01BFE45F7FF3

F104004FFCF3FDC0FBD138FC4FBEE81F47083EC017B0FDFBFFBFFFAFFE04213E13FE43043FBC20
400313D1FF101FFB0C207EF7D043F40F43088E840C2100F80F020BE104246EFF0BEFC2080E7DFFF27
E1050C013DFC01C3082FC4FFEFFDEC103E07FF7AEC203AF40008EBD044F030C6048085FF9005FC3
E41F3BF41041F45044EC0004FBCFC20C1E7C08110407B0C403B0C500710307BFC1009101140045000
F40E7CF3F07F13D041FC3F80F3FFC1FC6F7F18AFBF03D03CF80FC1F3D07BFFE07E17A04300307BF
C30BDF7E18003B07A084E7C101E3BF86083FBBF84F8300117C07E0000FA044180F0000003FFFC0BF
F00202F3C08613C03D045F810BD03C08310604304017B03B03F1C10C3F3D03EFC2FC51BBF86FBCF
F7181E3DC774C455E41707090BFABEBB8B39D913143E5D144CF023713A528B8ECACFE781A0BDD
6A790AA99CAED38E79B60049E62B56C63A6EA9072FDF8E73CA10FBC3A9E1E62100E4E09A8627C
76AE0F2F774E8C46F8B893821798CFB4DFA4532F7302ECC1A801014E1991BA9FF9BBC2AC331F130
FB7F782BBCC6D4E5D28E0422935ECEA1941B347259A2D8B98B263F515ADB3F4E601340333E26240
7157EE88DA7B97EE127E64E2F4C35D894F2107D14C1E93BA9AE7C283F500D59D4B72D1D6516453
FDF03224377464F95C3D905374F1B9F32AC7F4F7DC5C8895381C272DBC90A9896BAEE611A5EEC5
7AA3C601A4E5BEA150E263DEEFC120DEC41A2071EE7DB881442EE1A0A90D9637D5D5163D00B6
9B545C6939C0677AD6453527DA1C0BB8E51C13DD7BE850CB1FE3C55D39300EC10EC8E4E23EA2B
90B8BAC11A16F2B9E1E4E4DDE8477FF3D384C535ECFC4B7DDAA077CF43AFD9B2D2D50EB40F6
BBCC4C3CB4E2A17FFD3992A0E4BF3FC4C7C1B51B29001BA980C37B77831B28399F83AA21FDC4F
BE4F587C3447E4152F732F363D82E9B11F30DEEF9B585BC50B26F7B931D358E673C3F0AD6F15304
985D589F0E03D7347953C1DEE42A7698FA40A259F102BB46EBC14C3AF080E78212D9C5E236C6AE
74D6EB961AC1F2F50F3B8CE74EE0DC098877C095DB98D7C2EEF936BB641A13CD375B975E8650E
E10C6178BBB594528CD07FE2414F50C28D3FA846D85F727A3422F8A593648C711902929D0F9B4355
6C61F2FE5D19D98FB2C035DE1207B868C167B79A3A2C8DBB442A5DD0F2E1D881EC644823F4ECC
261EFF4C447B5901FF7D99327B256010BD9E3A08355C4D1A16609B577A086669A300695960CF0645
C32BDC7C1542E7BAF91FCAA7D5F3D1FA0E8DE9710B2C621DAE60146090204F1558BCF0EF3EF
D330715EB74E23C0505038201FE5FE9C8E8ACD19218E72A08924BF70F87691E3B0039DFCA0C52B9
2AD5937CBE61C4760C93002AEFB87DCB3CD75B9A37021A865DE1DEF7B2FF8A418AE3FD019E6D
7EE6D8FF2E22672FCF0C34605902868A3047E941339372F7931FC3C9A0C6CDEBF3AAF909E6B12FB
5081F13C5005575EDE294963171792A2C58B13458EB804BC7D0FF6912460C42ADE17F082B8CC851E
2F5F7639C1E4F2EFD38FD3580D4DB3A4FDCC8130AF05FF9A506CE6F5CCEEF0406C47B0B0174E4
C33B6D854DCE9000D30329181EE9F8300D1272587C4713E25760A737E07B5A2C149CA0FAB9253F6
275F1B4970ABA14B6EC75E9697FB461CAF12EE92ADB6AD023C31560F5AC457A792F5C77598F7B4
D9023080C689C484C8887AD8E1B9E191B6AB7E55FE441E88DFC64BA36C16D8EF91F27902A8F1EE
CE20E7259230D0D06F16D3023E7CBC8D897192AC1F2FDB8B0C53025F082324E99283C1901BF025B
0AB1DD3957A787A3A41426F763FC86598105816CC8F2FCE1DC6F4CD7547020F4E83FC6AC2906D2
1B38C81D33F4C31B628229BDF3C334312A541C7DBFD9EA72886E45C3FC3CE28CD84253641FBB02
B4BD408EE8A95011ACEA41FEE592AF5879CD3D3EADE420122FB50AC7B94D5F15091E56DA6D57
C7A10E2766DF8144E94A2D67DE507A6139B0DA5DC0B72A106A1FA49413C8612D88938AC9316DC
8A0EFB45464AD04FDA893F76B94333FDA9976DFBFD39CDC7CDB2BC071FD0F98D5551D5D35A4
CF0EFF3C04BF4EAA49013E3A3586A562C012D6CEBB7DCB48F1B310F471365D884FD22C9217F366
15FA02CD2082B0C33D7CA23014D918331530BE5E0D00598F7AF0B93E55A9151146B3D80889682190
3579022FF90557DBA44F95E401DD029E325F672D50CC90F0FEE91799F3F49E44FCEC1330B8EFCFA
0BB0A10775A7410758235182FA57D2E9455E40C6C393502FEDA2F92745D1FD0EB1A39786C9FBDA
9B9202F4E3E791EA7DC7D953AF696588FE47907C8A1A908F5D2EAE969A9421029A9F7A664DB8B8
1D059289D53F98E164C65AB0C10641641211590A98B32B56EBD084F17B58F3B823B62EA25FC1FAA
F00D6BABF8F807BFAF6062E8CFA2A79872931B5BE46164F97E0DAD625C100521488E6331E93F902
4946CB7E7D2ED3C6B7CFBF164A0D36E1521DD2EE461B9DE28AF9019A85D33DE8C72BC75E97E4

7E97DC02A7F1D94B3C3D54B994B59D2F48C30B6DDCFF0253E781E573CD09BFC1D20076C4587A4
F85009894AD762C19D7EE5FD0356B546F654705DEAB6426B0916A05057E176F2A00C5B2D34900920
AD9FE9510F147A389E5BCD6BA907

ct =

D98D3DAAFB6D25886B47F191AA1E96323CB0B48B185E1B7F0DAB551ADE7C6A0B7DF36741428C
BC7DE0FC30FEF88B3F62F4C11C82E28A7BA66E94BC4B099E72015F2CE8A95EC2A54371FA6535F1
696A7D385D9373B2705133E05CE35E3D8D350E5B06CC20CC451368F31D1C6E8D29B7F63C7F6CCA
2A19397E594B6FC51388B3A575C29A25BEEE271ADCE66477ABC252B7DDB9ADC3151663D8499EB
3FE32FBBE8EBB722E47DC16F542D139B939B635C1A77B9245E84EB4FA7DE57494A9EF6674707763
4CBFE531C6D231C23A292E39B4A3BCF5DDEAF27463F2AA1BAC9B643A840E93FCC54F4323F33E
B780A8AB5309112A27C953241AA31A78325D41204F0AEA2F7B8FB2C2EA3111831DF8052347E3C9
C32C8FB6A06B1F663D2781299AD9166F023900EAD92DEF85D3F05F201E9C4688C934E00C22F4062
9D59445A4CFE6F739D72B8AE7851473BB20B7DE12E167D0F9EC51FEF9F9467E21C5B0FFA09EF88
8F319CC8311745E685099A7CEC230B73F937E496B2D60B39F7FA2B419678B70A02114A91843DE800
F7AA58F6FFFA992AB3F847CC0A1E4B54DBCA405E65BBF0D577D01AD73F84D3E801A99157AC2A
3A3AE1E78A39284A9A0736242014B56AA4147D930F8C2826D080B9CE114A873906B90B1AE9E9876
E054B8D0C284

ss_enc = 460FE91C8D752E7BA74838C7B2B891DD

ss_dec = 460FE91C8D752E7BA74838C7B2B891DD

result = PASS

case = 2

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16

pk =

991581121AADB24C53C4E3C3ED26E30087DDA5E64C6C24CE94963F7C4BE3EBEA5C89F63B141F3
B9B1D37FB499539AD64C53B5502F1680A15631F6C98F730431373C9A54EFBA63DAB090E9725C196
1EE1CD771F4FB93A9E5786D8F1FA74BF773C4C4E48384E6CAE5392417570B9B85E49E1800BC4469
7A73EEF59C550B34D8509FD3E3B878A74A3BEA2E681C900DB223F2BDE9CF48A56A945729486E7E
A53567320E28719AB60AB6DC218C85822BD0E5814AB3828656A180B93F22B6BEE3851234EE91440
7E4B59A6E7D59A604C8CACD5233B5EE7D5DEBCAF0FB8949C917DA888C0A366767058E64022C0F
6DBE36AA3DE5E0EC9C9891C11E86319FB71090EF54E588B3A30C14626FAB7888FD04440C94A178
B7FA5590D86C40CF4F9A8FC5D97E7BEAF4898385C05FE44C0560D7137DADEAE8BEA2A003F406
AE1B7E66F847080A86F56ECE1142826DBCC33D392F46495CB9B6DA18C5A16E38DE256C3CA1F20
AD186A4207A91B5EC6BBC546C963B9D645CA14C5A8255935277F87FC4993F6AEACCEE1550915C
4328A34F2D50E6A758035CC20718A915A679B31460BA273B3F64B19DBC8372A0999949AD412C290

E3205752F8AC15AAECF02D06F776FDD508418B8DA406229E53301F1170D1D026F8562B13E3B6C10
CDDBC2266F66AF635742D3E3568ABF55F0B437B732B81FE89A9D4A126DEC5EA5CEEB266EF9DA
60851CCBD3E522E3429600E8D552AD2

sk =

19C10DA35D778BACBD25A9A47315A4F312EFD4EC044C315DC4CA00F59E297A9F8D1A79ACA146
51B4FC7482B26D2F7125FCBEED20025EE7947A5AE5FE04F9B6681EFFFF000F0F0000000001F1FF1
00F00F0000F100000000F00000010010000000000F02F0F0F10010000FF0000F00F1000F1000F0100F
0010F00000010000000000F001010001F0F0100F000F01E111000010F00100000F000F010FF1000000F
001000FF0110100011000F01F0001F0F00000F000100100000F01F010F0F00000000FF10000F00000000
01000010F000F10000F0000001F000001010F10F1000F00FFF00100000F0F000F000F0F00F0000F00F0
1000F00000000100F10F000001101F0010FF0F00000F1101F011F0000111F0000101000F00001011000F0
000000000010F000F1F001000F000F1000000000F00F11F0000000001010F0FF0100100100FFF0001010
100010FF000F0F010100000F000FF0000010010100011000000F0F100F0000F0000F000100F000F010000
000100010000001100010F01FF0000100100010FF0000000001000F00100010F0F0010F00000FF00001000
00011000000F000F00FF0001110F00100F01F0000000000F00000010F001000101001FF000F00F0F0100
000011000F01FF000000F0000100F0F1100F1000FF000000000F0101FF0100010FF001010011010000F11
100100F0FF001F0001000001F10FF1011000000101200100000000F00000000000FF000F001000000F000
F100F0F00000100000F100000100F00F0000000FF0F01100FF00F000001F007E13E045F43FFD102FC0F
C1044182F01F09100041F81005FBE1BDFC60831BB180203E7AE830FF0FCF00F8803D1000C6F7DFBA
180F3FDBDFC3039000EBBF3EFC1F420BFE84FC3FBA0C4E42080FC0043FC100517B0042BEE00F86
E00E42100086FBC142F031C518207FF0B08603F1BEE7D003F84FBDDBCFFC081E3DE46EC4E860F50
FCF3DF421BDEC1180DC30B907A04707FF3907A03EEBEEC417DEBDE8A1C3F87F7D082F4603F0C2
EC317F07D1400010C0280F06FBD10833DF83E80FC0084005187001043040EFEF03FC4F00FFA07FEFF
FC3FC5EBD07AF8503F1BF047F80EC007A041F000FFF85FBFFC2139F41FC4FC30BA10113E0C5E8C
EC208407C03C043101F010010020F807F0FF0BE0030C600310314703A03FF041BA03BF05080FF7E7F0
46FC2E7C201F3CEBCFBCEFD3084FC1002088F02FFFF7FFC207BF7BFBE044F3ADC213C03DFC5
004085042084E80F830FFF02F840BDF7E0BFE85080F410BB03DFF9F400BE088F3A1C0F82FC21BBFB
DF7E00117AEFCFFC1BD083D83045F7F07F00118403FFC1001F3AE7CFFE180F3F042FBFE7B0BA100
13DE8004503D1BFFBF038FBEEFA08213FF81081EC1FC10C113EF86041FC1080E3D1401C3F78FC4E
FF002EC813A042003F7D2060BFF00F7FF400C4181F3EFC313B080F7AFC2F7EF4413B084000F82FC3
0C103FFC1083FBD07EF8113DFBE1FDEBF0001C204203AF41EBEEBD0380BE1BE0C1FC303E1C3F4
0183FC3F42F0310213BEBB039E3A2840860C517E1C404207F03C038043FC70441C1F04F79EFA03FFB
DEC3183101DC3E03FFB0C3FC007FFFFF03003F87FBFEC1108080039EC303DEFE0C40C30011780BC
0BEE85039F3DFC0F8220117E03BF8107AEBEEC3FFDEFF147042104040082F3EFF9FC1203FFF13F08
3FC0FC1EBEE03FBBFF90BC041F3FEFFFD080103040184FBDFFBFF7DF00005084E470C20FCBDF3
EF7E0BF08A1810C4FC2EFF080E82FFF041044FC0F810BDE7B00B0030000000FFE87003140FFE17BF
3C08403FF8307DF7E0010BDDFFFC0FFD0BD04103BE8207D0800C113D1BF046FFD17FF7DFBC0C0F
8013E002F7ADAF5EFFCF43C9857EE67782577FA9C56DDF887EB8E81E2A1A782E8A877F37334013
DC195758408CA5EEDD3CAA1234F96D60351C7B58EC1472519B6125588EE448E89D1B3ECD369B3C
20B59BA1CBF7929804ABA0821CA3474168C0641118DEA40981458704F3AF5E69237C4E21AAE7F92
F18EBF8517D48F15896878F985DEA161081E88770BD644A72588732A7984021EA06721819C95DA2F
672FF9F495C497E0C19EF59648C3BCA269F5FC19FB7B077CBF484E6F39163892D44D9AFF60991B05
FA582102CB86B198A345E5D7340579D770CE5DFFCD8126B013E13A29C5D41E814703121B6240408
09F0707FA297AC2DF76134319E5DACC5F789C23A56225441A0D1843F7A9A0589FCB064BC8A117F
CAC3223F30882EB1F101FCF50B96B49A86D48D1926FA9BE7DDCFB405D6F63A088304436D16F65C
0B9AE1FAA0345F9F2B85B3E30550D2C96203D0505D7D31347CD1B12C46F40D11A11F1CC511DD2

C1E4F6FD30980E160B09810FDD71F14764205CA1E08004339473EFECE2D5C7CDFF731088BF327FF
0C9A5D6E9D0111208A687DF31DDB88830B44C0A50D2EBE4F905FE7C14621C8925DB8C4C07FE373
E1433838CC27DC222503AB81C87BF0D0D05E3DECE5161F1FCA357F39F0477AE3F3A177F93009C4
4E17D254F5AD313F3BD1F3186FB37CE8AE035CDE817F0775BC1988CD47B6D507EFD2522FF66AF7
EEF40C65DA030E71316F5397A346EB9115C581776494CC3A58A1F72CF22925258D741B383B3AD77
E715E92E876E3B45DD2F47A4009BA651A2264C7A92FCC8A39183E9867D5FF957ECD3290A1B6F34
1E852500A701A59835E820F0FAC093E909C39A7CD10BE9337C8E7BED451E3D0A16B15E39BD4251
978378A6DC0AAAE5039F333162E551817631BE34571785E025B28CD877BC054BD0D71797088D0FB
092F51C7BFB7735189DA07BE82F47BB5A5B1408EAEF0AFB2F99542D29FF284CB9DFBFD3035F43
B7CBA19BBDD9D971540366BCA762B2BF9FBFB6EFE2E0051BE8A27C2FFB0FFD4D78B2DC20A45
E83DCFF72F94AB8816C2312999CE85FA20269C7E764133591FB576084A779E3EC7DDBB4F442190C
C54FC4BDFE4D776800087FD309321BE7C2C9BF9842AD23C8BEB8630D9FAF232881CF49D50736B1
81C7FBC4D8741AAAC4417181F57E79F8F196F17DF4A3745E2204DD917D0E46A02D52D1E123AFD5
8DB6A62CA21534FEFD4D244CE5D94F240AFF90C9C283829C46C05FAE8E3BF8E5A79E6846282143
1EE68024E5EDAEE2376D6CCA96283F024B15B47CE59FE9526F763EF35B3A49CBEF45EBB7F5AFE
21C51ED26759F832BAC4E1A7BFD7B79ED9E5743DFCE472F3BB3E5F0392279D1FDA83E7A3256CF
9A1BFB22063DF5F09DD54FA622E3F3B9468488F09479372F5690894C59D96ED03EDED120185C2337
297E8B5142FB9D1762AEB67783906E5866C6B9831B82A1DDBB73DE4FB3405F4DE739D8CB697688
2285BAE0A3CE09842FD0332605E90AC8020620671FCD6EBA4B8BB42635F015E88D81994486615811
21AADB24C53C4E3C3ED26E30087DDA5E64C6C24CE94963F7C4BE3EBEA5C89F63B141F3B9B1D3
7FB499539AD64C53B5502F1680A15631F6C98F730431373C9A54EFBA63DAB090E9725C1961EE1CD
771F4FB93A9E5786D8F1FA74BF773C4C4E48384E6CAE5392417570B9B85E49E1800BC44697A73EE
F59C550B34D8509FD3E3B878A74A3BEA2E681C900DB223F2BDE9CF48A56A945729486E7EA53567
320E28719AB60AB6DC218C85822BD0E5814AB3828656A180B93F22B6BEE3851234EE914407E4B59
A6E7D59A604C8CACD5233B5EE7D5DEBCAF0FB8949C917DA888C0A366767058E64022C0F6DBE3
6AA3DE5E0EC9C9891C11E86319FB71090EF54E588B3A30C14626FAB7888FD04440C94A178B7FA5
590D86C40CF4F9A8FC5D97E7BEAF4898385C05FE44C0560D7137DADEAE8BEA2A003F406AE1B7
E66F847080A86F56ECE1142826DBCC33D392F46495CB9B6DA18C5A16E38DE256C3CA1F20AD186
A4207A91B5EC6BBC546C963B9D645CA14C5A8255935277F87FC4993F6AEACCEE1550915C4328A3
4F2D50E6A758035CC20718A915A679B31460BA273B3F64B19DBC8372A0999949AD412C290E32057
52F8AC15AAECF02D06F776FDD508418B8DA406229E53301F1170D1D026F8562B13E3B6C10CDDDB
C2266F66AF635742D3E3568ABF55F0B437B732B81FE89A9D4A126DEC5EA5CEEB266EF9DA60851
CCBD3E522E3429600E8D552AD2

ct =

D95C4D11C8C700528D3B2F5F2F30BA1ED3D4476B23F7B92E711B412B5C6BB72EC5FE7578B9B891
236EAE25F198AD9793241FF28DEA7C2A0DE09537CC4C23B82E5E6E98818970CD76606F8E673347
1F26A476AD257A8C7F5D54A733F1EA399A25AAEDA311889C11E23C200418DD74EFAD09624A98B
15C678CC993378009150D5981D6E1F484BF3A415B35B39E5EB6A3291DD176A4B95494D50E71656A
3E03C3E60381A16A345C5E5A2C0658011ED55C698C4B2884B37B2388160D2CFD94D3345547A61D
A6CB362094DCDF00B4A1A6A8BF3ACECF8D9579AE56BF913E41570A9A585C7737B42E383D225C
B7D4203E724C980677626D35960AD4E77BC6410FE26125D54B86AFC262CAEE30C05E318978AE2A
44481B36874A523FE49982F5AC86DEBB091F88DB27D348B2E24980E6C399991D4DB683AAACDEE
B44FF53110514E0C411203821BCA3AAD4203BC7E34FA9921F9FB5BB579F3291084ECC17EE96B5EF
5D0001294DC01EE082E985488602A77E6CE77D6398A0610954D766206FF25E75BDB502CAA6C027C
84AD5A06317A13EC69AF9D92534A2CB42D06C65B70726876EB50FC41322FFBDBBCA706004B2D8

DA89DB7E9D1D7431B97918445CFB3194DE3B9AB6E22657CEFA61B3702446613FC2E0600FA319C9
6CB3DBAC268

ss_enc = C5FD9F8003010BDA9A156C81FFD8B31D

ss_dec = C5FD9F8003010BDA9A156C81FFD8B31D

result = PASS

case = 3

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16

pk =

998B97C194C0DD837E7803595B40DDC4A6E11507EEF4270CBE8C74BD12509A80E06A0D49D50B6
B83A05093BB09D63F2D1B1FE7E45FC0AF742D03BA43D094C18B1A60E65C5C3698E5C735BD74905
F6BC51E078A16D9C3A45D42818B587777FCCCF083A0236BD9E6627D33BDE2AA0AADE4B16C42B
AF89FEF0A7ADBFF99C8269DF716D77328CF0C5BF54CE6A69F9762D763C40DBE52E0C015EBDA70F
597F50D71AFDFC7BF0BA48237EAB8995CBE98FA201220A7CD7A7E1E652C3257C1C015BB52FDC
B1907304337AC0B78F087DBCEAB3972F0E8FAAB5D2678410C9E7F91FCC38D60F245C3C55BA9C0
4A0B6DC42C6D251AEE4839E22788C68173BB3AC39058DE6133368F3CB30C08AD70C97D14DED9E
845F591EDC9E5342EB19CCCB628E46FD7C58D259CAB627B2C3A021731D2C9C2572BBDA40C55F
A4712303EB94524AE92984ABB93E3D813F2B3049ECCBAF59B1473427A2C9953A0C52657A411E672
B97FE45C7EF11DF968A77295EBEF47075BDD336836DE4739F194B782091B227EEA110821ED87C73
1E0294AE18E7DCC620662A0515BF41F5D9B7EFA31BC4527364DF0BEA2B6FCDA41E66913C585819
861A16AA9DFB0FD5BFE9828E98FD6DBFA4CA5BA7CA521715B9DFC0BC37CF0C43DFC44CFC1C
04F333B75398C17893147017A80066BB96FF3BF454A40C888588FE873AE8570A11FF7071A70992C00
7036414C094B0DF61E6044E1447668A8074A2

sk =

19DB89239991D32C3EB3108A23D1421031A8D9487C3780776C30D933302760C3740655D84B01E4B3
1CE8F5FDEE479BF63F572106F3D0DC0062E114C2569F6225BA001001F000F10100F0011010F0F1000
F0F001000F0010FF0F0F001F0F0F01F0000F00F0000101F0F0E000000000000F11F01000F01000010001
1FF0010000F000000100000F00010011110000101000000000F00000111000F10FF000000F00000000F0
00F00000001000100F0F11000FFFF10FF00000F00F0001001000F0000000000F0100F0F00F11010000F1
0000000F000000FF000100001011101F1F10F0F00F1001000000000F0F100F10FF00100010001F1010F00
1010010101FF010F1000000011F00F000000000000F0000100F0F00F0F0001F000101000F0110000000100
0001000101000001010F1010010F1F011101001F0000110F000011F000F0100F0000100F000000101F010F
00F000F0F0000000000F010F001F000000000F0F00FF0100010100F00F10F000F00000100000000100000
00000011001FF00100F0100000010F10F00F010FF01F1000F0100F000F0F00000F00FF02010010FF11110
01F1000F0F0000000000000000F0011001F001F000010000110F00000101F10000110000101F01100000F

[illegible]

E0B1742D45D0B1F0A96E0782AA4780AEFF0ECCEB70F188910AE595E9B8EBC6E3B0B5108D6B9D2
6070ED24D077B8B324681017D6EA6575E0E29B2C14F0C6246E543D8C1DB41346F51BE003E2A9D78
CA4158F98510E761034F1E7FDEF831976AF308CF7FE15D1749A3DF571829F413B29569F58F7B27EE
0A81CEA74A47D67430B211D1FA139E36EF55669CA056C864DFD110876533474643F8B97C194C0DD
837E7803595B40DDC4A6E11507EEF4270CBE8C74BD12509A80E06A0D49D50B6B83A05093BB09D6
3F2D1B1FE7E45FC0AF742D03BA43D094C18B1A60E65C5C3698E5C735BD74905F6BC51E078A16D9
C3A45D42818B58777FCCCF083A0236BD9E6627D33BDE2AA0AADE4B16C42BAF89EF0A7ADBFF
99C8269DF716D77328CF0C5BF54CE6A69F9762D763C40DBE52E0C015EBDA70F597F50D71AFDFC
7BF0BA48237EAB8995CBE98FA201220A7CD7A7E1E652C3257C1C015BB52FDCB1907304337AC0B
78F087DBCEAB3972F0E8FAAB5D2678410C9E7F91FCC38D60F245C3C55BA9C04A0B6DC42C6D25
1AEE4839E22788C68173BB3AC39058DE6133368F3CB30C08AD70C97D14DED9E845F591EDC9E534
2EB19CCCB628E46FD7C58D259CAB627B2C3A021731D2C9C2572BBDA40C55FA4712303EB94524A
E92984ABB93E3D813F2B3049ECCBAF59B1473427A2C9953A0C52657A411E672B97FE45C7EF11DF
968A77295EBEF47075BDD336836DE4739F194B782091B227EEA110821ED87C731E0294AE18E7DCC
620662A0515BF41F5D9B7EFA31BC4527364DF0BEA2B6FCDA41E66913C585819861A16AA9DFB0F
D5BFE9828E98FD6DBFA4CA5BA7CA521715B9DFC0BC37CF0C43DFC44CFC1C04F333B75398C178
93147017A80066BB96FF3BF454A40C888588FE873AE8570A11FF7071A70992C007036414C094B0DF6
1E6044E1447668A8074A2

ct =

D93290FD94A2C273DB3BD306FEB89A4D160733DF7E87361E96248F182A67CDC938FE415CBE4340
31D288090BDFD18F12F341862F0FD3C4F19AD8880555BD1A750D46D6FEE6FA81846BE3095768FE4
11A42F53A19FBEC2E5E38466678E43A20C2A5A3540007D34FB273AF66CA99F6AB26528C624DC85
E590AB276ECE6774CA65D9FCF8B196ECCD8454BF3D771999176FDB7659ECC5180A2D8CD9569B7
8A626E331B1046817D639D17C7D6CF45D02271ED0999DCB775BCEBE53D0EB4943271876A8904EA
67CD03FBFFEFFCCC9C7C076BD42B150274D131D2722DD668B33934153C6248AAFC12D50A3F2F0
6FE104B2BFB2E3B171C7109FE7245A34E73D301BC1F6A07E129A109FD182681A5780E7C8A3AB6D
039ECFE8753AF3FC83EE0C6D52E79BD47B6F1522384B42E49ADA156F2EBC6762CB3B32201C79E4
9470D978268353A553162CA343B3FD01BF33E1EBC7141DCA9F19E2DD103220921CC8EBD905E58C
804C9F632DD3560BF78A21513DE403BDEDF4AA95EFE492C96AB37B739263ACE66BF0383915A036
D9F97E91D1F285C87688B0B1E793B8EC478517DFE074B3D456896C53392A846259B88C45BF71AF8
1200DA5853B796F9117AD03450DF117073203B1EBD19F7B67207808F6F71CCA8ABC211921C98F8F
96C6D1B29

ss_enc = 28A57455611066AE6E52E5E7A2604445

ss_dec = 28A57455611066AE6E52E5E7A2604445

result = PASS

测试编号	REF-2
测试名称	CTL-769-1024 参考实现版密钥封装功能正确性测试
测试目的	测试 CTL-769-1024 参考实现版是否支持密钥生成、封装和解封装功能，并验证封装共享密钥与解封装共享密钥是否一致。
预置条件	1.已在测试终端 PC 上部署 CTL 参考实现版程序代码； 2.test_kem_correctness_ctl.c 能够正常调用 CTL 的密钥生成、封装、解封装

	接口; 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-769-1024 参考实现版程序代码目录; 2.输入 make all 生成 test_kem_correctness_ctl.exe 文件; 3.运行 test_kem_correctness_ctl.exe; 4.比较封装共享密钥与解封装共享密钥是否完全一致; 5.重复执行 3 组测试, 并生成 ctl_kem_kat_vectors.txt 结果文件。
预期结果	1.test_kem_correctness_ctl.exe 程序能够正常运行; 2.终端显示 “All CTL KEM correctness tests passed.” 3.正确生成测试向量文件 ctl_kem_kat_vectors.txt。 4.测试向量文件中的 ss_enc 和 ss_dec 相等;
实测结果	程序正常运行且显示 “All CTL KEM correctness tests passed.”, ctl_kem_kat_vectors.txt 文件中的三组 ss_enc 和 ss_dec 都相等。

测试向量:

algorithm = CTL-769-1024

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

pk =

AAEDB9E409B79C235DEBA6776DAB251B5E7B60A973DDFAEC4C60ACC7529CA05E43356A1FB9
B066A2C9773E8E084FE2FCC05326C2C93DAA35B0A4145BC72AC251CB8BD997ED518CAEF173811
FED4E5BF320AF90670B5F8FDD7949E4C62EFF971F575B76D80582E3E29352DBE23AE1AA85205C1
6C849DBAF150FB8E831E018A503C24781F00F3EBF9D34F32C38DD71572794D152102F5AB2DFD3E
F74EBCB4511151C61A6ABE00263AE7EC0B717CF11D5F56F1A6F291E176ED9DB1E4653ECFF719F8
B8EEBE16419EA4DF190604FF5EBAFE4D056CEA3B82D1F82B915581559175E0658682906A377A30D
B0CD67F1708CC7C62C00BD9E8754E7A2433A94EDC2F94546916ECEE5916E822FB478E941046C2F0
5B8D09FA1AFF6343B0FA9D472B28FF30383A70ADEBDC9DB655AABE051A81B7679AA875509571
88A29AAC3C81F677295BDBA9CB3CCAF16F8D5A6BF2EA3046B18F149A4FF7F778A0D3BD458C0E
ADAE4E5F89699BA0807727FDD6685A428FD128F5C9A9134FE155B0C198F31C9071410C95BDD1A2
2495894E1AFA8C7294A0FFA20A0385343812EFA856FF5BF1E0687435A0BD621FCF6857CB8E44388
05486C2D07CC650DC876A91C31159D72E3152D3D25CA5D141A2E6D3B48435BA64738C0E606E883

9E89B36DB62B98D8A5964CEA454C9AADBA2E31B32EF359AD24C3562516B6DBF869E0D12AC947
E7130C268B49B9529BF23FF1F6E3D0D7CD17E843E2C9F30A0183007A2ED8CB1B881A246C8A8AC5
7F143797E8C95503CEA0EAF1DEA02BA8524149206BC4812D0D55BFFD8033E934FA764B6CFBB428
885133F9C7911DE1A7A8F6C5479689BD348116115FC7611DF8ABC70B0FF5409D41435A5783E0F292
177B55BAB65576997165C50636929CC24000EA99C99A619B4104F1D384CC839A834FD3CCABDAF3
46A79C467D41019F100FD766B6AAD4E25953FBB766B69378EF6C5C57B511C7B67C3BF61D67318
79DFAC50C108109CBBD0D4364833DC6AA422F100CFC195FDF9DFDBEBEDD297F3D0F9783B7AE
1F23BE3D09BC53AD468B52EB86517E864AB50CF61E58D3CB069D6E74FC397BEAAFFD818A7D3A
CDB34EC54CB051C40A18217190E298D5A3A5C2F6EE9FC666106C7392B21E96310D9B674590A4110
955727AD6EC77E2A4EEDFD050F2ABB4C3CD23E1AB84663096D85883E5E9AFCD979DDDE255E2
5883C76B78C6453AA7979280F3813E4416118F6717CB33D5C27FB6E265FF1461A1C18EC30F3357D1
E54417A0CB5044F1DD7F25E0A9B472E6FCE08E3824DE1BF9794655A61F7374303DE6597DE207311
019328452D0D0F07635FA51B73D5FFC6ADF3CBE785B5CB6939E88896CD2C331C7FB3240789B6289
3DECAA564F72F85CF26AC69AB582F43470CC0A3C0BEED8B563A779D742465924D8D807C83928A
AD35CB06BB88D3401A74E87C0441B47FA3C9560E17D6C52BD3A37CF48C9FEB80AFE263B2E6849
BB4B820DF1A3AB8934BE17BF52F110CA57C7A6A9AD6DF8594F51608164BFD0601B07870B864FE8
B1B088D11031E51910D1E705027AEAF1FBEE109773709A3EE7298A0319DCB7720870C3C03755FFE
9A332F0D73943078EFD56367A98702B582BBDFD7A4BC44649C9C8CFC584E3A9A29E95E24A73445
75DD2421CA8834BA68027BAD42D2C26AB847E68CDC1B7E6D5EAB2D5501044685CFBC9D90948E
A3199FC10902617964B454F0BFA290CE439721374A59556BB01B1

sk =

2AE87E15F7E6CAB66514C99994C2F5BD92BC52A8554D2C081C0FDE814A57C2964B88F2D26D1A7
6C7C4E199431B7C311A7BEDD9E26E7BACEE31120166F61BB2756B1110FEF01101001011FF10001F0
1F1F0F00FF0F001000000F10FF1F01001F0001F1100F000010011F0100F0010F0100100000110FFF00000
020F00FF0010201F000101F100F00100010000FF1F00011FF10F1FF01F0011100F0100F0001F010001001
001012F0F00F0F00010000F000000F0102001100012F10100100F010FF01FF001F00F001F0001010F0110
101100F0010001110F0000101F1FF000001F01100F11F01100001F01F10F011F10000F0E01000FFF10000
10F000F001011100000011F0FF011000FF000001000F00101F001000F000FF0001100010F010F10F1F000
1010F1100000000010100F11110F0011F011101210010F0FF00F0100000F00000F100000F1F11F0000000
00000000F0100000F00110FF0000F0F11F1110F0000FF00F0010F0FF000F010FE0010100000100F110000
1000001000F0FF10F00100F0000001F11F0100010F0000000F00F00F1F0F01F01FFF0100FF11000000000
001F0E010F1000000F0000000F100101F000011F01F0F0F0111F20001000F00F0100101F0011000001010
0100000010100001110F1F010000F0001000000FF1010011010100010F01F000000010F00F001101001000
000010001F00F20F0010FF000F00F0FF0100F10101F200F001F0100F00010000101001F00100001000000
F1000F101F0FF00F00F1000010000EF10000010000100000F00000F0100F01000F00000F00F100010100F
0011F01F110000F0001010000010F000010F000F0111000F010F0E101000000001F0FF0100FF000F002F0
00F001F10011100F01011F00E00F10011F0F00F1000FF0F0000000100010001E10010101100F001000010
0021201FF0F0010F0F00F0F000101F1100F001100FFFFF100F01001FF0FFF01000F000000010111000000
F0011010000100010101001010000F00001001F10010F0100001F000F1100000100001F00F000010FF0011
EF00F0000010111F0FF0F01011F00000001110F00010001F1FF01FF000F00001E01FF0100110000FF0FF
0E00000F00F0111F1F00000000F0000F000000F1000FFF00000F00000010F010F0F111F0FF000000F1100
000000001010001F000000FF10F00FFF10010F1010E001011100F000010E001001F0F0000010F101FF0EF
F000F100001F1FF000FF000F00000000000101110FE0001001FF01011100000100F0F0F00F00011010F11
00E001F0001100010000F00000010001F001100001001F0100F000F0000F000000110000101F1FE0101FF
F0F0000010000F00011000100F0FF000F0110FF00F0000F0000110F001F001F1FF000FF0F01010F000000
0021100F010F000F10F1000011001F0000001100FFFF01F0F10F001000100F000000FFFF011000F1F0FF1

10000F001000210000F0FF00FF000000010001F11000000F000000F010010101FF84C6D343FFCFC0080
AFCF84041FC8FC80BD080043DBB1C60813370011CAF012BF0FD0C2FC6142F7A3BDEC708BFF8DB
EF3718827FF0007924104017E07D0471020BF047F3B303180F85042EB80372B71F8EC70301BD00103B
0BE0FD081F7D04B2090811053FE13E2FFF89F7C283E7D100F45EF2F7D33BEBD1BFDB3E3F0430FA
1C2140CFDE7B088DC4049F42EB9008F7FFBEFCA203F0224117EF36F081C9E00EFB3C5F071C217EE
391CAFC504B135F7C0450852BCE393BFE870C0F3CDCD0BAE04FC7FC9FF80F5EB2F89E0B108FFB
08223FF02E7FF41F42EC120400B2BD341F830861862B61C32070FC04303B1320C1E7CEFF088DF6E39
F4303814407CF7E17E07EF880C2FC4EC300013EFC8011F84D7FE832FD178E44D88F0827F0C1E781C
23B9D803C2F0713E03DF3A181004F8903DDFB147C8E03FF3F23D1C2EBA148274DFBEC414310207E
20BFFBD7F17C0C3E87FC6079B0823ADC2D87FC1104F80FBAD80284FF728EDC1D79F4CEF9E7BFF
FD031400AF0BC078DBC2AFE3B306FFC007F7CDBAFC8F87FFFF8618B1BB33DF350CA207FFBFB5
F4D2FE13D0060C8046182087EB608E0B7281D7FEFAFBFD81FBCF862F217E138D45E43F40F0003A0
7213BF3D0431C4DB923A0850CEF7EFBE1FAF8BEEFF051F8F040C2E4103F14B03BCC70F8FC007BE
4AC48F4A0C4E7BF4617C040081D42E040FBE46EFAE41FFF0C4EFDFFE0BF000FFF0B91062440C2D
870B01780C410C140E48E7FFC7F03D871420BAFC707A0BFF78E77102EB62C1178FBCF3AF46DC504
0D8B0FAF41F3723F189F811BFF08F3CE3D180FFC27EFFAFF2E72FFFE40FC1FC103F001173D8F2F1
1792C31BDF080020BC0BF242FC20FF174084F7B042F35F3CE3BF41E7F2C4E372BEF47EFF081101FC
31B6F82FFE38B1C937E006D03FBA1061FE1C2E360C3D81035F3BEC6EF8F86F43186F0510CF430C1F
F9E0ADFBF7AF84086F8C2FEE76F3FF03EFDCEB9D00FC20B937CD40FCFF7B1BE047F3B28D0F7F7B
E05EBE43E042EC1E7BDC917E038E5017F0B5CBB207FC81F7DFCFBEF7937C03DE7CF7B079003174
179FB9F4023C144F43E432831FC201B8400C08403CF79E81244F440FA27BDC04C5FC40B1FCAEC623
DCFD07C1050F8EBA03DD7903C0FA0C10B7F87F031F3D83F420CC03F045DC403C1390CB0FD23A0
C8185EBF18700B03BE84B46E791C20FEFC0DFF33FDFBCFD286F07F81E41046EBCEC4086005FBCE
CE23E1C5EC70BC144F38DC028807BE3BFBF07CE84089FC3F39F0608613CD7F08000427B1B9187EF
CE4604FFF91C307C07DFF6D3DEC107B14220A2BAF84D07F41FF6EBA0C9107F79C0538520B27FF4
207F139002EFF07C17833E13514C2C5079EFA238FFEE411F507DCC5179185F7A0F9E7AE0A27F1420
FC04207DDFFD0113B0C403AE76DF4004E46184186FC0002C7F141246D4510920207F0F7007F3FD81
DC2F461FDF07046342FC917CE421881C6EF7DF8134CFE186F4B17B13AF09E46EBDEFA00FFF8EC0
E0220400007FE832F7F30F39EFBDF08207E0840FFF84FBDE45083005DFA138F770C417BF02E79FC
34450C40C4EF907CDFD10AFFC17DEC00FA0751C01890FC07FFC20FF0C3202041139DF71FBFB9E4
CF38379205DFFDC2085CF9F3DF80F861820C1186F02145F36139DC12BE03D0BB0BA180104E38F022
40D34FC2F44186039DB4387F8307AD830B803D307201DBBF7303F087105FFF3082FEF0FFFEF81072
DB604100910203D23D041E46DBDF032420DAF48D3607F042103087048FC0DBDF3BF8C0B8DB9EFB
F7DDF9FFFEBCF010BB042CC20CA23CF41CFBE801060C0ECB040078FC20800BF0402081FC1BBEE
AF04F7A1C1F830C73FEFB9F85E7CF4428E0C0E06FFE2851070BFEB AFC30F40840C5F002FE10317B
EB818E177B80DBC8F80F8217BEF80C40BD182FBE0371432772FF000F79188F7E03AE4400013E13DFC
6DC603EF790022BAF820C4F86F71EC708A139002F8DE3E045088F7D100004084F3EF45EBED84F85E
84E86E3CEB407F07B0C713AFFAE7407CFFDFB0030F9004EC53B80C9188043F45F7CF8727B24000
CE82EBD23D0BCF06205044102E4CDFEDC913CF431BEF3FE42FBF1412FFEC3003F43FFADC8EFFE
7EE7C000F88287E4700003F17C1FD1F5FEDEC2004F7E2C1C3D30C008205179F0739515B21A3ACDE
DAE3E71473FEC1A26331DEC5B8AB35A581BC3B0873FD704668AE2B2F8BE09C94246BD63B9AAD
FA80C0963EF1813C601904712B3BF0E51D81C6831D3CF57F94F3845E24BF0CE26DEFB87103E44281
5FCD5B3BA63763D0B0E61FFB819EC1AF02088F2CBD4DCCE4E8BD92B4845C4482336669E44429B
61D0020237A8D01C366E63D35E925E4E89079A9E7E5FDC0C92FC32963C7E85598249A0463D37D96
EF03E0BCBB80D180E5778A5626ABE6B881E2F0B17E0E6DC7895907C53E1886C85E980633200473B
FAC37B8EBB8D1B188158595693F2DA6255D30A13E9BA14BD5FE20B5E8E638ECA0DE7E87AE24B

5008676B8F1761A64CB0F5B683D9EF6591878819172CE0C8DC87DD561FF6C5D2645EC74D80F14D2
B86A2C442845C962F2D0272BFF09E17DD6178C2ECDDF063EE0F349560FB74FA82AECAAEE263D00
940468DF68F0E8B440BF3620D44EACE096687CA608057F747C663C676FE84730DBD3D3F3943A690
465FBA2FE218C3520DCF7A4641B752BD06C7D049C851EAEBA767E84940BE8CC3C3B6DFFB0FED
8DFD11A3829D36A31C3BC00689CEB74F61B4AF2A1F1903FD58E0EFB172E8E09F33044A1ACC2D7
C33FA397A125F4F8FCEE43E79988135450EAACD41FFD2F0AFF6B108825FE1FED582EBF00B783B4
E53AE1E2250C208523B2012899D3C312107D1FE5138AC8D29DB38C163F1233E5899B5B12E4FC1BC
C34E1CB2E6E70CB20BE93C6E5CA7E03BC4D8225A911AC6FB6371567A6EE28076CFC8D7CA0FCB
D68B952371B1BABC93CF171FB785D7ECD604DC6C96300F9BDE5EFDEF057039020BA1E168201648
3F04639E4C9EAF6F1C5FE126D3ED202602F205B5CD634F8A5C6A95BBDDCA3E8D91427CD8D89F6
7078E0B5030692C33CDEC6E3442A102930669B0E7D8200D2629550047B96AFA4ADD71F93030B397
D2F8C3DC729C4396FCCFA3E188F89445B85FCCD93989D87D063CC06DA3BFF6CCC059F50D1BB1
8633829563AE2F55C01C97DFE9FA33096580C5863902C202D148D824012180D302A7FCF001540033B
BE7805B846CCFE76DE635038936EDDF7B0CD88FA803743E4F96EAF67CE10FCD7E888D9E6F30C48
2B7BCB0356E87CC78AAFBEOAFF617E45B1FEFEC0D3C71FAE0F39E43DD3FF056C6A7CBEFD232
CB069267F97261EB4CAB719FE7A03DD8C3092AE82A3CD77CE51FFCF6EBAFDEDED5C74537AF8BD
38C6FA71C72F3E7DE73F12661E31769129C4EF2BB14CC546011DCD53DBE78071A16AF38DF83E93
E92A6B344C3425E3235B91A1FB8E74247CFBF67663347718D31C427E4B5FEBD018D6F9A1881AFE
CC25238C7068599F9549FD2A19908604FD4C7B7F7E87803B07A6825F07C1FCFC9EFD8261B6A07788
26977B0C5721107A816A8D2699FC033AE63D14FB9D8DCCF61A1BBE874B9D666EDBE950836809A
E383B7F196FC3559B7FD2F8EE612B3CEBC584DA5E09BD80D15A6332384E69B73BDFEDB5C962E7
A731DED96921ECB7B08F8F9DE7BC1E2297BDB1927882C138DF992BEA019F3C0EC67C18B42FA44
22EF2491A6C95B5AA52321C9A907938BAD86C776F32BB78980538355D81D00D7E679B23B33F7AC
76E37FA8AF59B07EB3FC1BF264CAF0F9E7FF61C0BE11FF9055575A99FC4EE90A0384BD620557FE
3AB68C84276814680B7A9F19074FA5C01C4238D3D395CC3A3C3D478D5CF757C0D3936578C80CC6
9FB9B413C1FBD148D2DB65B20948A05BC2AE72EBF3F77283BE16018F6DEA7E3D0B46383332F4F
A9810F1F23828DDC478A10CED743691F3A3E28DB770A0E4C0132671DC0DC9DED20C056C22491B
7C5D8CDD5E458BF51784E8263AEF29BD23BDF15D0EB13CCF0D467F4C2C511A2CB383445B8EE5
E63DBB88AA52EB31C8805EFA7CA2DF30E8C6082BB1FE4E8202C8E3B0C98FE68D2F0D782C5BA8
BBCAFE2D92E71C8EDE858A307F204C6BBB361DC1E8E9E4327E2D60D3210B205AE7A3D66793F5
DD165275875C46BF45F98F9CCF5DBF9DEC1A0CCEBC3D7E34339190DC1C8567BB09C17AE257387
524B543155FFD139CABA33ADAE39A9EC80E190A21AA0E75347C2E3362F609EA5E4C421FCCEDD
B2ECD07E59E3B0B144F99988827584A4C39E74B7450C5BA4A1099F4AD735C1B968EE5A8B0372115
B9952B1E15BDC35166B46DF9419EEBCE8E4ADDDDE511F103C4279430B39911686C9DAEA5160B9
A4206863EFC2FD71F3C7B18C23DE3F79B9C5A074106551D89D136E4C24A3D5D39659867F45C42E3
7C2E8FEF337A83B3A3040005E5B9EE788A7F0B3945F7D5D3CCDDF3020F0D3152BA49AF3219CF93
897802E82F2EE37DB5A8624C6FEC287857ACCBC5A92443EAB1B6A75E98AD2FB77E2BBDBDED50
F7CEF01F06F002C4E389D1E48B19384A2F3B8E9DABF9D9E3CE0CED2E9B9C470061EF1C3E814581
90D9BF3E6459607375BFCFCE19FC8E757DAFC6CD40E4A64547CD3F9C0234E921DCFBFB1A1E82E7
646E807375FA6FAE3E0ECFE8C62FD74DF25F8F0C3B4DB94CBD16777EA85E960494C37F6BFD3F98
3D1350BFD97F63BD0C02A0E99B1DD4CE432209321E673A8FC43FBB6EF82980AACEE9880E0EB96
F33B36BAFD9E51CE577EBF6E27C0D528DA790F86BEFC2B030D57A85CEFF61208D36AE7313BEF7
F3CA3653050F34995B4783CED76253B6DDB241F82160BEC053475C62B6193CD46C380FA71362937
AFA6E995BE22E19ADB4EC3A64D20484F1E9CEC362B31FC8C8508B5BDE4089C8C26C7EBB91B83
36280E0575B46EFEE02DDA5F10E0493F8CF8C9D0EB72C9AD9EF21EE3B62C084CC82DCBCFA302F
CBCD7209471E9D6FBB79CAC3C86E6206BF37AD245C99FF7768B07F1883D34FD7898D5F6413C3D

B268E8CB03CB5647DDD4C6812F7593230C729AD97D7C23A8723EED19126473E0869270659E5B8B2
4962FA2109799144532BF3B2E33BA03877B7BA04FF5F2E6B3895E25FAFD0307BFD4E10FCC1BF311
E79543D0E3F6FEA87F57B7F5400983E09503B4CD58BA528F91EE7DC018447C59DCCB874074A0DD
D601243A1288849E0174F1B192A5892986A1109A0F57982A661411CFC2E984B4A17AA1CEDB9E409
B79C235DEBA6776DAB251B5E7B60A973DDFAEC4C60ACC7529CA05E43356A1FB9B066A2C9773E
8E084FE2FCC05326C2C93DAA35B0A4145BC72AC251CB8BD997ED518CAEF173811FED4E5BF320
AF90670B5F8FDD7949E4C62EFF971F575B76D80582E3E29352DBE23AE1AA85205C16C849DBAF15
0FB8E831E018A503C24781F00F3EBF9D34F32C38DD71572794D152102F5AB2DFD3EF74EBCB45111
51C61A6ABE00263AE7EC0B717CF11D5F56F1A6F291E176ED9DB1E4653ECFF719F8B8EEBE16419E
A4DF190604FF5EBAFE4D056CEA3B82D1F82B915581559175E0658682906A377A30DB0CD67F1708C
C7C62C00BD9E8754E7A2433A94EDC2F94546916ECE5916E822FB478E941046C2F05B8D09FA1AF
F6343B0FA9D472B28FF30383A70ADEBDC9DB655AABE051A81B7679AA87550957188A29AAC3C8
1F677295BDBA9CB3CCAF16F8D5A6BF2EA3046B18F149A4FF7F778A0D3BD458C0EADAE4E5F896
99BA0807727FDD6685A428FD128F5C9A9134FE155B0C198F31C9071410C95BDD1A22495894E1AFA
8C7294A0FFA20A0385343812EFA856FF5BF1E0687435A0BD621FCF6857CB8E4438805486C2D07CC
650DC876A91C31159D72E3152D3D25CA5D141A2E6D3B48435BA64738C0E606E8839E89B36DB62B
98D8A5964CEA454C9AADBA2E31B32EF359AD24C3562516B6DBF869E0D12AC947E7130C268B49B
9529BF23FF1F6E3D0D7CD17E843E2C9F30A0183007A2ED8CB1B881A246C8A8AC57F143797E8C95
503CEA0EAF1DEA02BA8524149206BC4812D0D55BFFD8033E934FA764B6CFBB428885133F9C7911
DE1A7A8F6C5479689BD348116115FC7611DF8ABC70B0FF5409D41435A5783E0F292177B55BAB655
76997165C50636929CC24000EA99C99A619B4104F1D384CC839A834FD3CCABDAF346A79C467D41
019F100FD766B6AAD4E25953FBB766B69378EF6C5C57B511C7B67C3BF61D6731879DFAC50C108
109CBBD0D4364833DC6AA422F100CFC195FDF9DFDBEBEDD297F3D0F9783B7AE1F23BE3D09BC
53AD468B52EB86517E864AB50CF61E58D3CB069D6E74FC397BEAAFFD818A7D3ACDB34EC54CB
051C40A18217190E298D5A3A5C2F6EE9FC666106C7392B21E96310D9B674590A4110955727AD6EC7
7E2A4EEDFD050F2ABB4C3CD23E1AB84663096D85883E5E9AFCDA797DDDE255E25883C76B78C6
453AA7979280F3813E4416118F6717CB33D5C27FB6E265FF1461A1C18EC30F3357D1E54417A0CB50
44F1DD7F25E0A9B472E6FCE08E3824DE1BF9794655A61F7374303DE6597DE207311019328452D0D0
F07635FA51B73D5FFC6ADF3CBE785B5CB6939E88896CD2C331C7FB3240789B62893DECAA564F72
F85CF26AC69AB582F43470CC0A3C0BEED8B563A779D742465924D8D807C83928AAD35CB06BB88
D3401A74E87C0441B47FA3C9560E17D6C52BD3A37CF48C9FEB80AFE263B2E6849BB4B820DF1A3
AB8934BE17BF52F110CA57C7A6A9AD6DF8594F51608164BFD0601B07870B864FE8B1B088D11031
E51910D1E705027AEAF1FBEE109773709A3EE7298A0319DCB7720870C3C03755FFE9A332F0D7394
3078EFD56367A98702B582BBDFD7A4BC44649C9C8CFC584E3A9A29E95E24A7344575DD2421CA8
834BA68027BAD42D2C26AB847E68CDC1B7E6D5EAB2D5501044685CFBC9D90948EA3199FC10902
617964B454F0BFA290CE439721374A59556BB01B1

ct =

EAB233302F386B9B5F6A2E8B8638D9DE6D2AC9DED819F9D1D8F1ADEE93FE15CDE54BB82C98D
C40CC05477F9982BB42A17DAC659AB8A33B4ED64B08C2867074C8870FB19AEE95AFAC4C488DF9
92C6C67D4B566CA0BE862C68B0427FD1C7B3471978610F86EB702AD256C0D845E0481F96350118D
A47D131B01E0CDD6447B85773CB87964037B657AC377B90FFDDDD094DD547506A4C94D32A1CB62
C6F6CD6BCCFC286BF587CDA481FAD312795F8F8A3C407E38151E2A361AA6B8CAF986CF006DF7
E43DD9130A6E19B83A985FD700A972B5C6096E9DCF174EEE1C0E375E57C73B10530996413A5DB1
EF3EEE53B3247BF5881D100920E885FF46F8304266470AFE8A82ADBAB28C508E6EC95544F982DF0
4E8B41334E79592316017621929370C047A0B5BDB506D7A9C1ECD6B94EE169822902E5CBA359417
E668CCD255642928BB42631D412ECF5DD10A42E03DB67316E0A85C845A179C58E46C45BC2390A7

02AF4B8646FF87A2CC392A4EB0368A5D909C5DE01561EC866CCB0A4DC93AA70A9EA16BEA2D7
D0138ED523B298F2EAECBFEDF5DEFD61F37CD11F9FFBDCC4C871EEE9EE2C54715F21BBB24E5F
1E002FA42A384D8400EA751E045E7000A398825CC36D28B00564D3150B05FEF37EA09CA0B2A653E
645D708C26092A30032E1B4D770D54ECF0718E931892F4697720210F29E33BDF9F1F7BA82BA7F6CF
BA7A236D04164CD2A7D1CF51E901E3BB673E0E2A4A1A5A292DDA652B459B1AA31380C3563906
AEAF1F93051D7F41F763335A76AC0C6469E9A1E9BF97EDD666AAED9039ACD8551070881A1402F1
FED40293F927411AAEDBD1A3957CEB6245B013FDB5E219D0ACFE8517B958784ABB85E6EAE8697
B87793FED2A3DE158FD32D415567C51484FB2A10512B30544AE0D612042A697CC9BEFDAB8A9A0
F02BD5B2C629CDAF4CA08F4E3A659DF05F38B956BE566641ABC8473670D80CA294EF9E76504B62
8FA14E81C2807B67D259756B7A7CEBBBE8D7C40806E6D1349D874E34B3067DD97B08DD6BD50B0
BB05ABA90D4191B6A466519A6BE35B11DE155CCE0E318405FC927B1210E165C79A9D5C829EAE6
E1C5327F2A678DF9635E057FCA2B298FCAB991BAD35BF322C893DCEE5E88CE9E9C2A6696F5A65
8F750102752AAFC9D9CA7CD14FDC48089A5B1F710D75A648D10ED30E97E5812BD831ABFD93454F
361D656EF3DE5B2319DDD71E7118120D9428809BACEF62D19CDED7DD01811947FF744F3EBD2338
1F80535B2B01ECADE25D439EA99450FBF0690340EF71E41B05CD4AEE7495062533C5888CEE031E
C84C0627BD8BCB48C885B9B7C86BB17D19AD9CEDF5455E4A443E24D72269281C90B4EAF4EE9A
6C4CE0D1A0EABDA73F4E35F3A701AF4C122FF3D9833CB42C4A0BC5BB0EC7715ED5C53205DF86
B

ss_enc = 1AEFFE6D4470D34D66A5A7D3AB59573E5B6B9176ACD1A34115A8194B3A8998BA

ss_dec = 1AEFFE6D4470D34D66A5A7D3AB59573E5B6B9176ACD1A34115A8194B3A8998BA

result = PASS

case = 2

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

pk =

AA87E98B0AB7936BB4D21CFA9E189536E3CCE8FBCBFB115DF9B1E1D162523C05D05EEEE0D4FB
2B5F79C425C9C00B3B06D242C2140A4B8B73B6D973E12460DA4553B01B7B1F58282924199EF1DA9
51B122696291CF1542FB3C32419B72C4353275725DE7E47DAB08BB3CC1B619F8B2CD93849C84FEE
1FC340D19B1A26814C9462786B388C855B57066FC3AF4B7B7BE4598D2BDDFD05F4714189D0469B8
C653B04FC5F5B643FD5BC5AC94CCC6BAFBCE9BDA9DF157FDD54123379B90C163D824C41804D6
AB75FE68FFDF68C25F82667172379A01940A39F2F586807B5D909F3F564D8F4D0FA77F98F433E82D
1360A7679E6772674C1FEBB39C21F268B2174B708421C94A20A699CC015CE8BBDA3DF7A96AE8C
A762836704A1E5A6F662A2989BD0DDBE43456F6573F94CD7B4DE3C2EB31998D4227958F6496ABD
49B89375E9CA356E84CBA1C17D2269C283CCE351EC523BFC95F94A4A4A45D9F0991F6D65BE8AA
43960DBC02A2DE85BFC876E4972BD478893DAF2A8EEC6A9EDCB4BB1C129D99E8A63F1CBBF
B69351A55BC172E44B225DAF18B045AC486FABBBBD9DB5C8DFDBF897B1250E722B5A2EF38F815

8847BA11AFAD525F45F349BAD71BEB6A2C938C6FE110C9C340D513B221AB55C38292C646CA42
53F85AC9249CD87B03E9EBE7243B200C78717AA37E42DDC0068DB9E35463E309492E857E0AAF7E
42D1F8C30B4AA73A5EC097FB5D6D933C16E4B5160D8D1C8C733C9B0D152634CED4D9726552796
8AFC58304880C4B0148637D0048F4018095D7A05AD55CBFD6EFC1EF122D90B5E757E6F420B1C4C
9E4638B8DA61A8199C38B01301A01BE16AEA0B7A8AFF6AC88E538AC369586C2DF39F6AD5B15D
3AF8AEC7A86E5A392DD40D853E0F01915A225830D4ACDFD34BCF34A083E79FB26F31FC611AA2
F788A1286B13841CC3BEE866D271FC9B3C46E850FF1077FB3BA10814BEFB82FD6C33F6C071DE53
194E647213E9D75D2742501E14B13DCF30F18F6B96F3DA530BAB465AEB8D179D2CBFEFC665C0466
45CFB4CED3E625754817881193977A559CD09DD4BD91F48FA71F76A495987EB99CE27AD73186C8
945A43A8E667B8D9AEE6568FD05D16DAED29AF438FBF03525315AD176D699D9D33C2D433FC1C0
ED192F9CA4DE28AD9A89CD854A4FAECE58E5278E63F5EB67BCB2A8F50F68F4A040664437C5A54
0AED49C65759A4001F4CB419FAEFA2509372BEF5A2403E41C3980B9B9A3F20007CF3D5EFF168928
A9D1B9019AF8E1B75C966D5FAB7F600BD72EA31CB9F9F3168365954CA48A3DA8FDB06F2809AC3
498545AF7A65B68E22C2A123FA2BFB56DE15EECE8B1CAA8C80D4459D346968F667F53DE12AED8
CAADF9171F032A6F58300E42B4BFF56D7EA044FF9BA0CEFDDB2AF65D8111681A0017373ECF7DA9
826FDCD2980B58B8D9C3B513D8B651BB05A069CA439F7AB2A17D56BD5A6871249D1D5EC822C3
4AC28468665602A9A70B37811020510C4F53770FEB34DEACCB3F8053AB4CDAB015427278A146B
4127CAD5EC62603C3817F2D22E2DEF57340E2E95173C6DC3C1112CD3181A3821ADC0C7351468765
CA6859A5FE9B24F5E3EC4F2EE94610C2AA19981DB8E3B34FC29E037562B7326DF3EAC6A70A8438
B8139E20125708F1013BACFB9761ABD8B748288F6A3D67A9468EBE1DB8A7E0DC6BA4A57B25089
F0F9B472C489CAB9CA2F1194CB922374F1F2648D097616367D8B56838C0F

sk =

2A72D9791672F23564E6557E9AD061B58BF0D53125D1811588BF8A125D4E6EEE7F17AAD3BB3C6A
781523024D0E3C5A0E300CA4C57E3B4007BE15584DB9ED89B6EDFFF00F0010100FE0100111F11101
00010100001101F01110F10F0000000000F010100010000000F01F001F00F0F00000F00F000000FF100
1010F1000010F001000010F0010101000001F0F10F1F0F101FF0000000100010F0FF002E00F0F01020000
010F0011000F00F00001F10F10F0FF00E0FF0111001100F101FF1000F0FF110E0000010010F0F0122200
E1010F0F0FF01000F001020100110F10EF0001FF011F01000F00F111F0F0010010F0100020000000F0101
0001F000001FF1F1F00000000F10000FF0F0F0010111FFF1F101101001E00111F100F000100FF000FF00
000F00F000F210F1100001101F0110F000000110100FF000F000100010F010FF100F00F00F10F200F002F
010F001F0000F0001011001F1F0EF0F0F1F00F00F1010F00100F101000100F000000F0001001000000F0
0000010101110F10010F01F11000F00001F1F1000F101101110E1101F0110F000FF10F0F0000F00001001
00000010010F00100F1F00101F1F0100F110F01F10000F00F00011F00000F0010FF10F100EFF001F00010
00FF11000F0F00110100000110100111000000FF11000001FF1000011F00000F110F0001F0000FFF00000
00010F100010F0FF0101F00FF01FF0F0000011000F100101FFF010000000F000F000100101F0010F00001
1000100000F0110F01100F12010000110F0F0F1F0110000F0000011F1F0F0011011100000010FF000FF00
F00F00F00000F011F00F00FF10000F0011F000000000010000001FF0F1F0FF00F0FF00000001FF0011F1
00F0111FF000FF0F1E00000F00F0010F10F0001F0011000F000FF1F01000110F10000FF00001F00F1000
00F100000F10100F0F000F10000000000101101021F10001F001002011000010000E1F100F010000001001
01000100F000000F01101000000000001010011100F0F10000F10F1F011F10001F100F000F01F000000F0
0000010F0001000FF10001001000000F10FFFFF1F0F00110F0F1001F000010000F00F010000000010FF00
0101100000101F110000001FF0F0011F0F000011000011110F00FF0FF00011000F0F1000001001001F01F
F0100110F0F0F000F00000000F0F0000F100F1110F01010F1100FFF000F00F1F11F0F000FFF0001FF1F0
0000F0FEF00FF100010F1110F0101000000011FF1FF011000000100F0F1010F0E00F100000010000101F0
F0010F000011F000000F1101010F100001F01F1001F1000F20F11FFF0011001F0000100000F01FF000000
0011100100F00000000100100FF0010FF010000010000F000E001F1F0000000000F010001FF0FF0001000

10100FF010002000F0000100F111E001F010110000F00F0000F000000FF0EFF00F0100010000000F1F010
FF1000100FF00000F001010010101F000000E10001F001000010000F0100000110111D75CB2081FBD07F
0F60C50C4FBA184FBF101F05F4117FEF9E04D7A1022F41F50FFF88046EC6104140D81FC5205002005
FB8EC5DFACC9F0D14410507AF04EFF0C413FF7733814038CE3CDCC37A0424391FAE44F821CC183
10AF40FBFE881B8F79136D3E2C0FFF07CEAFE3A185DF533F079F83E3F23BE3CDBD13F23C27F13D
F47F35E85E81FC8208FFFEC0080F000330030F3DF208413D245E0ADB6F0607C100F031BA1C11F813
8E8F17B17A1FF00A149EF6FC01033000C108008AF790BF07A2871B708CF861093C81321C3F3BFF81
4427823D086F0503B141F8933ED400BBE7C3C4DFE0B924203E0B2F371B837F17A2000C0E7A2AF33
D2420B3000E750F933D103486EC1059F96FFF036FFFFC0F3EE78D460370461C71CC137D41FF5FFED
BCDCAF41EFE27EF8A03FE050432B520F001C46F80379E7D3BC005EC8F0D038DFFDB943CF3E145E
3DEBCD08DFAF860C8DFC0B71CB07FB8A17CEC3E7FF3B1381C5240194F8D1C6DFF086E090C0E7
A185044E44FF4EBB03EFBFE06DB2240E85F87FC5108EFD080F0AFBD343F82E06D0ED14EBBE4AF
82F851BEDFC0FE03E1C304214A1480062BF04923BF82F8420CE84242D85F86FF6F41E3BD8DE84140
03408AF06EC4205285FFE0BC1030B4E7CF7EFF9002234FB4E43CFC309FCADC637728123728AF361
C5E78FBAE3F0FAFFB2010C40440FF0FA23DF86240D40400DF7146EB013FFC7F81FB7E3A3C8DFFF
4D084EBF183F02009E410FEFFBF3A27AD87FFDF4C08FDC147917EFBA082001248F401F5F0303AC3
E13AF7F1BFD87FBF1FEFB428C04117FE78246EF7F3F27323D079FFAF01FC10F90BEF41234F7BFCA
003EC114FEF8F05F7CFFE104E05DC1E3C0402FD140281DF4EC917D2FE0BDE7D2BC03EEC9F7507B
E861443FA1860C91F91C433C041EC108407F339140CC010033E2FDE000BDF8FD32840F81071FA03
C04E281F7723C04103EFFF142E021030BE0761C4FF6F0807F2F6F40FC4085F72FC120513B07CEFCF3
6084EC1F86179FC3FB6179F36E72F8ADBCBF81B70BE1F6E4CD86F7F43D1C70FA0FE043DFAE85D
0707C2C3CC8F8AF44FC41BCFFB102E041C4DF923EF352FBFFFFC0107101338F7E141F3FF7F0FE0C
4E7A38304007DF42F34E43EBF0BA1BFFBB077002EFA0BC03B283FFB0FC24303EEB9F0E101406E81
EC9144201E38189080F04F8423FFBB07C47D1FCE0003D20737FFC4EF948327F0F9380037FBA041E3E
0C0F780C2FC51450FC23D2FBD45242F83F4337C1BB040F7F13D07A000F80E370BFEDD242F83B4917
E0BF045F87E8AFD03B8FBDE791C0E3B0FEE02FBAEBE0BF0450C1F4AEC937D0C2EB90FCFB8086
EC0E8ADFE083EB5075F46E8C07CE45FC7000086280FC5CB3003E0A0F8F3FDB9F4C042E7CEC52BB
176EC50C22B9EC2E4404333E080E3F0C01FDE3A07AFC60040C90C9FC4E840C50FD0C0084FC2E400
7EDB907403C1FAF7F201F38144EC71FBFC033E003EB9F3AF46E3E03DEBF041EB8182E44E840FF17
E145EB80481012F7104040EC40F81CAF49EC8E0610AD81EFC07BD48F363013BC101140E05285083E
BB0C0139E411F537DFC3D47F7CF050BA23EEFE07AF3DF7DF811400474050C2F8EF8203EF0200719
2F7BD7D1C5E3CFC4C380F92F9280FB7188F04E41EC1101F3D077285F490BCF411F62061050BEE490
BBF3DD7EEC31800BBFCF0C01451F3E43F0A07EE3E2F7E0613FDBBFBCC72E48D7EFFB0BB07BF4
C3C5039DC60410840BE3BE136185EFAF05D81141043185E3F0012810C12884C62F518510223D13B23
603B174F75F85F3B183F842051C81C7E4117EF3D246F04140036F78040F3CD4A23C23920324630BF87
1C0D8517FFC4FC213EF3D043086EBFD80F0917D2C21481041421F91BF0C10FC179D3D0B9E76DB80
4A27E181EC31C20C4044E03F89EB9342DBED80E7C0F2F3C1C3EFAEFCEB70C1F0B03ED82DB2182
03AD44243DC017D27FEFDEB90FBE01CC0DC80C3D0507AF01F09144DC21B83BEEF8037F03FC3D
B627B2BB203F0217CEC213A03B0450BF08204AF03DF4EB9F45037F410C430000903CE4503937FF44
0C4178F4DF3E182E09179D4BEB4FBBF5203A0841C538FF87049FFD0FEEC82F3FC2FFC34633DF3C3
3FEFFFEFA24615023E1090C423BDC31B2109EF8BFFE46E3B0351F71C3240ED5DFB3CC43D7E12C16
659431384AFB82ACD69FB77CE0CB881B156676DE488997ADBDB4D03880C2667552185F9B22F0C2
F940DEF658AE481BD1C1E41A5E60063196FF818C7A60E54318B3FA9B0E519DAFB76EA6DC6201D
8F65BEFEDBE9013C964818A2B045822EAC9231C501C12E3D0A4F9EAFDDC9F163183204BD38CE4
1C82D88A92808852A7A83EFEC011A81151CA973F6B71F137A565EE5E49B7E1EF36F65346136BE61
CF18344937301B7B35EFEEC6DABF522F8F4B240E681826B688D4C8CEE3B5EDA929794332AC40E0

1EAF0D7074D7BDAE606466C117B0843C611DF2840E2FE5BD14A08C7EBC4BA8BA5D661B3B67863
99899F021450615B07941B6A80836F437380AE28600EA61043FB6B9FAD1D71E15BCBEE9D88442224
13C632671AC5220D16983C8BFB98D4D21D9882FB10C503BC14F8301C11A83FE0DE55684FBFF0B88
B035B66DF27D6F38BE8CBE0EC5C72D846D11F93D7DC231A91A8EBE6B90B368DE541716E7C003A
FB4EF2765B29E0F04087415ECA11520831B4FBFC473412911497B4C283BF5EA927BADBD5C9B2FB
2B4366B14F9D8223532F008E08037E65B5402E726981FBDE7341BC31C274CAFA49BD363672F20A25
A29E230D89DE3C2D9BB416DECF0866042ABD5F41942F159FDF99EAAA230E85A27F8EE089B190F
82F14D3C28350FF7475018A8BA7CA1177EFD9D8A07C10743DBF928AF7CCF9006EEDE6445B2EF05
BC53896F558FDA03BD5A31EA7F6583D9FD4980AD66641F5959C424E6940CF08116FB53634B477AE
43A8BCD42DDAB910184000469F393080094FEC65260E1C70F77D07904E58CA51ECFA32D4A26261
C87629ECF00A3047E7517E99F154E078FE088140E76D99E2F1CD646C2789D81D9CFFB7A17ED8BA
DD950E186D8D5A4D19A0E6BD7300B77C10522BF4F130EBDCF36B44E4F28E1E1673D9DDA7CA21
D394773B887A87B0AC4FB56F937DA876312FEE5E07C7BD3DB2E0D05944DA0615B8EFC01F0A6F6
A22D92E5F26732BE883B2BBDDBC1E93DFE011BDA131DDED968CBA865653DE67198F1B3C6004402
EF288F91499AE132047C37BBD3A327D06BDF3D68954E5504E48DFB3EFC659BF2F2A6DE7D381D81
B6E2DB2B79E0FF445C39606649063F43AA0E7CCD9E98B0374281423005E17E7FA65BECAFD122079
5755947AA12916680E3CD726B3430806FAC412EC7A37D8946FD3858E1A65D25908B24315BA24DB2
37DE5A989C924823172167BE4BF3A44840525F50AC22481AB5CF79F2D74CD3BFA4E3466197CC19E
3FCDCFDDB2E77848ED4584FDFFB9B50F146FE7DB13169C2ECEBDDFCF2D9BE833CC3EDDF78D38
7D9920430206740DF87F119B1EB1996ED2F28EFE44AFBC5C32A770277B4C118FA8157B1095B98B5
9292DAE201930B665CE11E0E2F31045BF3CE7ED58E01E6BEF525B9CFA659648F6255C22911156D7
C7450DE3753BA6912A0728FFC07032405EB0A3FD9011BF6A0658390AE274E1FA576368AAC9E8863
5586426BFD7C18C011E287D18905BBF05FEFC66DE4985E0680E90712648FB461813F4CF8464FE1B4
71C1D140063D4B4A4467CFA700EBB48BD3D21EF74A71FD3633E0432F2819C61FCC7DB08EE0DC8
3FF0EB897B9CF55102CBF771BE9161CF3EA79BAE484A450A2361FF49A383B9C1321964ABD71865
CFD9D10464B759DE7CF27A0F02BFFD462EF0CAD4F5D382D03F68EE7D185F0B9439F4ADF55EF22
5B69F064EBF2F76E36F6029A0B1FF6F8F4473413A9ABD92888AA4CAD20D57E2FF9D927B4425C79
103A1DCB35DB1E1D0F3CE871F33217D02BF37F0C07715F0569CD5FB19D103B8A5973C01F025441
D21B073D2CC7E6CCBC7E1FE8F2EE13E77F455DFFDF29B9974CE1278F07C4360093B3DE6304521B
D1386B7BDE0F12E4CDE22671BEA1651CE2AEF6800F1BFD233FBB81BF481363671EDF47FBFBCCF
1DA7D02FD78350B35A192CC9B690734AADF9E71C83886CB46B2807100D0B67BA71C1DB17D0161
81E6BDC1F9027E3F1EDC24C1D033B2FB446E9B49509E06A2963E4DFF3D8C0C31D29C143542B3A4
1FE3A3E9DFA7EFAB8B6236528A0F888AE6E18CA311250C22A189B8FC5A634671BA44891C2D21D
D4F2B7638D0D492679D33DFA29FBD47E833C3C0B243C522B2780EC5A2D7B5818CA82DD4D8F628
5DDF27ECFE4E3F6605D0D10149305F8BF796CDF972C48B3D9D8DF3833E84B37A6A25190DB265E7
37EAA3873199A9C63BB13202E62B17ED54BEB96551F15108EC5CE39A1CFCE6131A18CFDCE435A
DAFC1CF1850B98A0E2A39F7EFF93FF8528C7BA629A8E39BC46AD05FF87F3C3AF89D6189762D97
E48DE3B3FA4F1AE18A69BDF311E3110F07278BBC24F0B1E03C46ADDF9EB14A518068F6E66FAA
DB3110FF32B0C2062104DE270FA264329FF953F1D2F96BA779AD1A40502C9D35FED1FAE98C4846
FA9CBFEF672946F3085BED6C053F73ABCCE2BEF6AD7CE7AA9FF12E222B0658FA79CE21679EF1E
5D005107E93D78FE5FAEE06766ADE91E164C07C4B459FB9DEC6F2F8034127D9DD23EAB7054CBC
0998792905172A7C43625D4221378E747EDE2C2F3372017D889363668CFCF7648F6E73F886E5927CA
C38C35939E29C7885E34679D87BF2A50A88C7C9DF645194A6739341BA7355CDF65F400C817A6D1F
1088DA94F20444E5DDBB18204628020A6C18F8AD98120E331D280114DC39B218F2EDE821C163A88
5D8A7C12E68857F201E19EE4C8E446C592A7582EAE609583E527B0EFD4699822C2BD01866D4E266
1E846B45810C5B1178244E1C76CFD80686DE00B843AE3092FEA81804EA10F3C2240CE4B6E77E8B9

347B8D803EA33C1FCCB63E21CB279007DB220CE77334301F229C86AE73E77DE9AE32D84385228C
3A996B20AB7874480A31A6FFD9AF75053C1E668AA2C4480BECF0FE2A6504CF1E9C82A57D95A31
8F0830C7A1A4DE8DEC391A3BD913FD02CA376DA07A85C02EFD92193C9C4BDE7AFB2E92F9D60
A84D9EE82F83FA690386794C73092815AF374BD3FDCE3DE28EBB1E41484D6718A68790BBAC6119
CCFB44B2BC11EFAC4F1C54D89B90E904118DAFC4CE402EE87E98B0AB7936BB4D21CFA9E189
536E3CCE8FBCBFB115DF9B1E1D162523C05D05EEEE0D4FB2B5F79C425C9C00B3B06D242C2140A
4B8B73B6D973E12460DA4553B01B7B1F58282924199EF1DA951B122696291CF1542FB3C32419B72C
4353275725DE7E47DAB08BB3CC1B619F8B2CD93849C84FEE1FC340D19B1A26814C9462786B388C
855B57066FC3AF4B7B7BE4598D2BDDFD05F4714189D0469B8C653B04FC5F5B643FD5BC5AC94CC
C6BAFBCE9BDA9DF157FDD54123379B90C163D824C41804D6AB75FE68FFDF68C25F82667172379
A01940A39F2F586807B5D909F3F564D8F4D0FA77F98F433E82D1360A7679E6772674C1FEBB39C21F
268B2174B708421C94A20A699CC015CE8BBDA3DF7A96AE8CA762836704A1E5A6F662A2989BD0D
DBE43456F6573F94CD7B4DE3C2EB31998D4227958F6496ABD49B89375E9CA356E84CBA1C17D226
9C283CCE351EC523BFC95F94A4A4A45D9F0991F6D65BE8AA43960DBC02A2DE85BFC876E4972B
D478893DAF2A8EEC6A9EDCBBB4BB1C129D99E8A63F1CBBFB69351A55BC172E44B225DAF18B0
45AC486FABBBDB5C8DFDBF897B1250E722B5A2EF38F8158847BA11AFAD525F45F349BAD71B
EBC6A2C938C6FE110C9C340D513B221AB55C38292C646CA4253F85AC9249CD87B03E9EBE7243B2
00C78717AA37E42DDC0068DB9E35463E309492E857E0AAF7E42D1F8C30B4AA73A5EC097FB5D6D
933C16E4B5160D8D1C8C733C9B0D152634CED4D97265527968AFC58304880C4B0148637D0048F401
8095D7A05AD55CBFD6EFC1EF122D90B5E757E6F420B1C4C9E4638B8DA61A8199C38B01301A01B
E16AEA0B7A8AFF6AC88E538AC369586C2DF39F6AD5B15D3AF8AEC7A86E5A392DD40D853E0F0
1915A225830D4ACDFD34BCF34A083E79FB26F31FC611AA2F788A1286B13841CC3BEE866D271FC9
B3C46E850FF1077FB3BA10814BEFB82FD6C33F6C071DE53194E647213E9D75D2742501E14B13DCF
30F18F6B96F3DA530BAB465AEB8D179D2CBEFC665C046645CFB4CED3E625754817881193977A55
9CD09DD4BD91F48FA71F76A495987EB99CE27AD73186C8945A43A8E667B8D9AEE6568FD05D16D
AED29AF438FBF03525315AD176D699D9D33C2D433FC1C0ED192F9CA4DE28AD9A89CD854A4FA
ECE58E5278E63F5EB67BCB2A8F50F68F4A040664437C5A540AED49C65759A4001F4CB419FAEFA2
509372BEF5A2403E41C3980B9B9A3F20007CF3D5EFF168928A9D1B9019AF8E1B75C966D5FAB7F60
0BD72EA31CB9F9F3168365954CA48A3DA8FDB06F2809AC3498545AF7A65B68E22C2A123FA2BFB
56DE15EECE8B1CAA8C80D4459D346968F667F53DE12AED8CAADF9171F032A6F58300E42B4BFF5
6D7EA044FF9BA0CEFDB2AF65D8111681A0017373ECF7DA9826FD0D2980B58B8D9C3B513D8B65
1BB05A069CA439F7AB2A17D56BD5A6871249D1D5EC822C34AC28468665602A9A70B37811020510
C4F53770FEBC34DEACCB3F8053AB4CDAB015427278A146B4127CAD5EC62603C3817F2D22E2DE
F57340E2E95173C6DC3C1112CD3181A3821ADC0C7351468765CA6859A5FE9B24F5E3EC4F2EE9461
0C2AA19981DB8E3B34FC29E037562B7326DF3EAC6A70A8438B8139E20125708F1013BACFB9761A
BD8B748288F6A3D67A9468EBE1DB8A7E0DC6BA4A57B25089F0F9B472C489CAB9CA2F1194CB92
2374F1F2648D097616367D8B56838C0F

ct =

EA2A642C685994E095C3E936DBB95AA26D436C279B166ACB66C2172021E58F6F2761F137F3ADCA
5AA534C6ADBA6949AC9CEA6216F889B4811AE95E704F6645F4A2A499657F2F9AC48E19434F3E20
B27B798AFFCB26642C47A67C3136DEEBEC20208DBABF74F6DB11C759051DC804257907CECE511
53EA208FE448400B4EAFF1E15386B56EBCD84C0BBF7EF8AC885488699FF922CE5CA03C667116030
78F9A95B72A8C00A53054AA41C110AF354629031A14CBD1F52BD3D4FF5F01DA86BF901B99CE196
A71A08D31316AA7BDAA726F292C7EF7182E7765FD9FCA13CF4E971793194358354A3C82C71BE7C
EA00F281137A915D51BE40402059ACDA922AAAF633B1F0AAD39E5376B925B91309010B73D4D5C
37D91F5D6A4886B99BA0818416B1E34863A83B4EE415CF3832E8FC856EF1EAD8B0BBE1A7F22301

20B85B90C8DE3AF39AA52E7954C5B7A8C5B13487FE8AD74D8B917D1B115A60E7DA9084EA895F
C033F6422B0A1DF4E176044306C73B6A024C1A8A9A0A65C8469D6FC944FB992858E9FEA2F9AC90
9E54C206B5D110536F094838BD2A8145A46D28A90E776DE46428FBD58438A9EFB46E9EB98130E2C
52196F511B6AA406BAD349C6203DAA0D65AF3655FB762574BF0AAF724B81F18FC78D1D3E8635D
B8E0890B532210C922CEE15A5E587268EFD288E0342EB2982792F7823CECD03C982FEEEE35ED6DE
C109F88214F4B65D08738DE2FB41B51D1B216E3C60F804718EC65CC9C4AD3226B8256A2023A5BF
C137EE04FF2528F04A6228F42A55369EF4545DE7FFA8D4BDD308685EFFDA41917F5E25B276E26EA
4ED7BEE7170180FB5B0D59B772BB348445A0D29ACAD7165EF910F43C2367310294B6BC3602B7020
16FFD8ED1C40F64C510B90F17B1D68D7DA18912C2D81B763CBE82A9C8CACF91B78B53344733A6
D4A2D39E2AD13A436243DF1CAAC05BFCE6E77FEC7444F341A37A3216942C225AFB66B3016DFC
D9B48CB0F6D7D58649FD0297E183A532DB20910362986A406E762BA9474D50BD9A50D9923EA27B
4B0D21449C3641DE3F23C3567411450830A1FB3068A6C76652FDCC30934477FEA0020DEA3421E628
79FE35B50509F4F5AA1507276B11068DEE5E0BE1D2474C6A7FE60B6273578B5D38ACCD12B40A85
B6FC10094AC99BB85D329201E0CA0444A8688FF40894E3430A35D7E6B6C391F2986EE640CCA016
DE62F61BECB833A7B1646B7938396CE36C6208C9D189A4C48386E60DB859C5ABD6F78DB3543041
C2DCD843718869D6426B5CE6DE757EC2800B16D5C5C235B56B3521B6B6E03C15E2CF81467847674
7A38D647C269E4F67B8A8A98A9898985E1349D9ABC035CA55312E283CB4F8A9E40CCAA1696946
A3167DB1E1EC4010BC40E2C05903EA3D9260CB6C2BDBA889162EDF98D62CE99

ss_enc = BB7E10C333862EFFB7C4C78F68CF8A2CB878D9A5017B00ECBF8A4F4FD5B72ED7

ss_dec = BB7E10C333862EFFB7C4C78F68CF8A2CB878D9A5017B00ECBF8A4F4FD5B72ED7

result = PASS

case = 3

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

pk =

AA473471EED82A7C2D0BF4E2C41F70F9CCEF3F98260F6AD127B1F999BAC39A516E37B48DE0D9F
3765ADCC89923738E584C6D3B5483FD493A57E4134DE6E58B0ECB8B5C623AA91C21787460B71B1
8F471BB2B4E3B630A17EFF7652D1346DB6054AEA18D7BFFCBFD22F132289C7084213B59FE7E3F4
A8AF1A381D50F93ED9C207FCF371E9C6456B4B10F22505A7DEF44CB03223B37F2989073D24FA226
0398378905048995E9D8707F5BCF470FF588C9863CC82944EF369E8ED03E39C9E1CFFB0FD6F5D83C
77B5C83078C19A7E22BF0D9D24C3AA434D1122F3CA363D5F2E4C7E72DA29924E351A108D8A79D
51C25A151B867093F8576DAE325DBAF667DBB02CB5967C4D007793F2E29E7589CA1A768EDA881F
503B2CDA05F0D73AFF0AA48F0F58ED125AFA2AA43535D56CF1AB0AF61B17DDD5C3279C157D1
3DBA93657CC33BA616A368D04C969DD23012C2BF383E2EA4DAE3D3B3B08E56D6BB76FF659492
E2EC03E1366DF4BA20118DDB06AE069F8F1EE5E3779DC9F37C37BF1F0155844FE3E300D80BF2A3
2A5A598D0ACD6D3964A9898DF80E73B81A98FC2C09FC81BE91C13357026559E66AC2FFBC20241

D97CFB871754296E4695771AA6D4237621C93AAC5DED221F892AE7C4842C3267E51C018A2A513D
001D4F417857AA2C22C32256149535305FE2DD1814B032172AF1E4F653623364E102F36BA8783DFE
84BCDD7E21DFDCF9708AFC045A5BCE16F7C56808AD09303DA83499E8FE7DE10C5A8972DA016A
29BCB5E4C0F881320058C5BD636ABE0C4F672A93D7F29C7B13329AAF109031D12AE9EF99629C57
74052E44707C08873E051F1A1DD26EE6EEC658B6990CBEB6FD979839A50DF2CBC77DAD9BB27E6
6FB223817B396B9D490290168831A5689F92E5349819034844A0714C66E16B327CD3715AEB1A99762
B44ABB23867A3DBD80F148D139B7F388E09918F39D3914A19176B75573C541C14FFE45AE4E28366
85A9019CD3D3B331F5EB0418FF092D0C41F1C7DB3A565E36B5BD01AC075B494E8C2EA6AA0D93
E3F3DE4000F1DF0F012EFFCAFB4F0C375624C1AE48918C0EBEDB2A66478A85FEF39EFEE32AF29
B69F4D5E072BA2F289C556B8DF59EAED7333A2B9C410BBB703CF595D6DFDD4D8340FAF221E52
F8E53630E57587DE4E1D90561DB108F73649E2C7F3C11200B0731100E3A3412F1D9DFDF4F04D0602
1DBCC44A42989D89973D8DA0DC3E9C659F71292E815A2F9FE9C93C65FD68D32320A5ADDD6D72
18A2EC53D12341F7F60CB9CA778602EE60A47550460D0D569B0239E44019E7C0C7C98E28F711DB5
03AD457C94FC7FFA7FB62F2614A9286B521676ECF10195D139AF6A50AD6C0284D8EFC750C0331E
BBB3CE7CE0B89499BB2C35C7BE3722E405041AFF20CAF170CD9C88695ADCFD7525494085F8806E
DFC7A6198E4D17A03C43C228DFEA4F585ED6BB7309BBC093FDC45CDB69A956CE94AD5C27AA
2976612DBCCB9D27352EB34BF003B29DE0EE007C889106D4DA518F4169C7A3983FABAFABAA5E
53A7F0795A52A738E77C90CFDC05381142903B9E44AEA9A069177975FC6D983CFBD4E1EED628E
C72F0BF7CC872568B9B6EAE1912EEBBEC269C982C133F84BFEC7E6093CC38CD1B968D91382E5C
EDE1D0CE8C5544966807202C575DE6B252769A9D480F37CF3F43C041D34511E98588CE0E65C7AE8
C856F22BFCF511477ACC3C7D3DE359E6075B496A1497BD1CA5894

sk =

2A95DDE6A32C32F48CAEE5F84DDDD36DC06C4B4C64056C70B422E4FBA1430273B3672FAC947C
C23A81C1744E5146A1BE60FAD95A0E40F8B0121E50D5585BD5D540000001000F100F1000F10F200F
10EF01F000F00000F10100000F1FFF0F00FF01F0100000011100000011F110FF100F01F0FEF000F00F0
010FF0000F1000F00001F0101F1F10F0010001F010110F01F000F0101F01020000F0FF001F1F01F0F0100
101001F001F00FFF01101001F0100101FFF00000F1101011F10011010011000110011F00EFF1100000001
F00E100FF0101100100100F0000202F02F000000011000F00001F0F1F0F00000010000000F0FF10000011
0F1F100F0100010F00F0010E01F10010100F00020000F0001FF001000FF0FF0FF000F000F1EF011000F0
FF00120001F0110000001F100001FF01F0FF00110100001F00002001F0F00F0010000000F00001F111001
011F00FF010000000F01000FF010F0000100F00F000F1FF01100100F0F00111F10F000000F0FF0F0F0F
000002F010F0100010101110F0F00F00001F100000F0101120100F1F00F000F0F00000F10001FFF0F0F10
000FF0F000F1F0000F0000010000000F0000001001000FF0F110100000F0101F0F10000010000011010F1
1100F0FF0F001F10001F010F01010F1100100F1000F0F0F1000000F00110F0F0000000E0F0F10F010FFF
000010FF1000FF0000010F0010F00F000F0F000FFF01010000F0FF020010F011010F1F0100FF00000000
F1100F0F0101001F0010F00F00FF0101F01000F100000F0E100F01000F00E1F0100FF0F100F0010000F0
1000000100F011000001F00F200F110100010100FF010F01FF01000FFFF010002FF01FF0EFF01001F1F0
00F01011F00110F1F010F101F001F010F0F01F000000000F01F0F1000001000111011FF1000000F00F100
11F00100FF0100000F00010001010100000100F0000000000F0100000001010FF0F01F0000011FFFE100
01FF00101FF01F01100F0F0000FF002001FF0F00FF0001FFF10010101001FF000F0001F0F0100101F010
F00100F000F10FF0101000E001F0000000100010100F00011F01000000F000FF0000F0111101001F01FFF
00F0F00FF01FF00F0001001FF000000000F0000FF02FF0F0011000100F00010F111F0FF1EF0000F1000F
00FFF00000100F0001001010010F000FF00000110000100111000F0F101FFF00F00010F110F11FF10EF00
F00011F0001010F101100000101000001F0110F00000F1F01000000000011200F00000F0FF0E0001F0000
F1FF00F0000F0FF00010F00000120000001FFF0010E020000010100E00F0F0010F0010000100001001001
010101001F1FF00F1111000000F0001F00F000000F00100001F010010001010010FF1F1F0F01120F10110

1F0FF0000000FF0100FF10001001F0F101F0100000F0FF0F00F100F00020100110F0010011FF0011000F0
000F001F0F00F00F01101001010F0200F0100101F0F10F0011000110000F00000001000F000148E4213CF
C018113FFF9F00FFB1FD03BF3D145FC8F36076D0614147D139F3718CE46DC8241F40144079F840F80
FEF44089FC11FDDF4F3E23C53F003F3BD4A241078045006FCFE3F1331F4136EFD081FBFEFBD891B
EF81EC5E021461762780811BE00DBC300AF8BFB8E43D4207E4FEFBFE8A182EC200DFC608CE401B
50BE07C278080283E33DC9F03FC81012A6000FBCF3FE7D006037EFD00010107CFB2E7E18810907BD
7C2020C2009F0237C140F47000D42FC7007DC5F6EC3F0FD205244FC108023FF74370000FF8FFCDC2
F46F7EDBEE802002B6DF2F81ED5108E3DEFDE080841C5E44FCACFDEC60831BFE7D241EBA2491F
3087083103E41DBF33303CFBB1BF17DF80EBFF3B07DFFA1FCE71108E83FFEDB7D390C7F0F086E0
8D051BB0C5E041CCDC814317AFC0044E81145085105F02CC907A084138E3DEBCF06238F77DC713E
E80F351BBDC4EC3F3FBFF44F010FE20513E08410BE04309F90F830330FF03A2C207FEC20BCDB5F
8B0C10C60411350CC13E204EC51BE040D4AEBE202087F7F04203FCFD07BF800840C92F6E3FE3E28
5FFF38517307F0030C403D18304C1C9276143EBFD48FD0E0207BDF4EC603E0441C0E3CD47040C852
341781F9003E86E06F7DEB9DFAF831F717BDC404ED41FFD085EC5EC62C3201FF5FC6FF61870BBF
BE1C1183DCEDFC1C3243F3A0BCDBAF8107FD8F0BFEFDE84FF91F71370FF048038EC0C7FF7FFB
AFB9042FC4DFBE76DCB38043CFF4F880463CF1FBD7E03F07B2BE230FBFF7DECD1F5F841B0F820
7D285045040FBF03FE09E0BDC4EFC0791431F913DB8CE462CFD08F7BEC71F9E09006FBB0B827B3
3E0C6082E430FE280187D830BF1FCF78186005FBDF44FC9EBAE86009DCA03C172EB9C8213F2C303
DF7EEF9F0506CF822091FD1AFD7FFFEC000407A04617320303A0C014008603FEBAF4427B1072830
C4103E3FF87045FC3FE9CC7180FC61FAE380FAF46F7B13DD3C14D040247E00183F7A18333B0020F
6280FFAF44045F8410A0C698504908CEC60FD006E820B6FFE0032C5EFBF7E17916E145F77FC3F3C0
7F1FFF7A13F0B7FB2FBF147105FB4EF8EFFFFCF03AFF1FC7E7728EE7AF80FBE146042E4308403E28
71F8E7BDC90C2E4ACC1148DC0D82F80EC40C816FEFF0D2F02F35A3CFC5288F3CF710860CBD3CE
393811C1EFA044240342FBBEC2FF3101E05F4403ADFCF76EB60890C8F0037D133F7A10008123F183
039FBDF04090E780C9F46EF7F06FC4E8FF442FFCC607D10F0FEF701C2F437FC0810031851C433BDF
704138FD7404643AFBA030C4A187242D81EC2CF73F7F38DBCFC218C035DBD1C4136FBD0FFFEFF34
0073CC2FCBFC203CECA001EC0F85137F45000037D420BB07617BDCEFFEEC50FB1C6003002D851B
C08A1F0F80E8310CFC8F43F7AEC90FFDFF171EFEFC5E0A145007002DBDE4613927D242037FC6000
F082BD0CE2861B710107D07D04300D139442E47D80FF50C6F77EFC0410832840FE17D1FFEC9FC1D
F620B180B7CF3F08233BFBA1C6E850CD0FE0BDF052FEF3EF8314523BE3BE7BF40E3FF8AFC4178E
7613EF37F7CFFAE400FE3F6E46E39ECCF04FB1286081D85FF8F481C7F09EC1005FFB144F7912F183
201F78ECB2C1D3D1BBEFF0410020BFE81FC4146105384DC7FBC0801FC0B813AEC18613BEFAE4
F105084FBCDBEEC70FE06F04ADFC043F83EC7406EF7FC0EBD0CB2C0F7D040BFF100D3D0FD1BD
1B4101E7B2C3DFD1751431B8080F8904923DFC5F8513FF8AEFADBD084FBD0BF245000041D040000
3B102E37E83F05F81137CBF03E18CDDF142DB80BFE42D7BDFD184235F41003F892FFE8E306EC0E7
BEFE040EF71030C5FF3F82385D06F49F3A0791BC4FFF3FFFE20513FE8513FF76F3FFF904513834833
BEFDFFECFA1461070FC006F3A1032412C0282F0010308AFBF17F079181E4014B183FFDBBE004DB9
386FFBE8017508BE76EC0078E371430B90C007F2BE08317E13F2FEFB6FC20FAE8622C0C4EB9E8738
128FFC1FB1280FC1FCFFFADBF83182EFB0BFE81F4617B0C70C5E04F77040E40DC1E3CE3E2CA1
C3EAC13EF9028407E1071B8CB6134DC43842BBF3DFFA1BE07EC76105189E84F3D245104E3B000D
7C1C8303F360C063913C13BE8AE8DE34F30F85F82D411BBF400C6343F3DD8107F141037311014EE4
D845FDAC0D3D0B9B36E9582263B059CB809D9A4EA8EF787ECE78F37DE8DD3AC06F2C8676650B
F9F27CD7B47F5E674C6C9FC3742B1FA1AA0FE2872A33A71E670D63213103294FC25FEAE3EE7347
D5FA24C220D9B75835687613C4760B5E3F56BE6C9CCACE9C98DAA7DAA0EE721B177BE35C0A7C
4F3FD893DDDDEA90C9046524C7FA03F672AC161B53CDCBF96732757C1E38571BF87ED5367B7471
B7BD07380016CFBE3B3F78DE7EA7674051266473ED0FE1623AF0241553876077A493AC6F65D36465

53F7401FD4719AF7225BA599B4E63B79EBA33602786A1CE2D18A77AA604E4E1B16866BC3979C89
7C71AF0FA81097826024ABFB2E8A13231280443FE6A38281940FFD5D89DFC69A0144BF35C851B8B
E7DFCBEDE7B91AA063AFEFED033D9129BA928600332A99359F170F9915F5DE77368C6481F91145
A864EC518448D40A0092DA7B0DC1855F07217193CC667914C92685D844BEC64AA256A137EFB8340
C2FFF877B2E74BD4441DCE11020646C0E45A8C1C716200934C4FE4C3C508BB3FA559608360DEB0F
048ADC96A23BD17FD43D65D1FEBFDDFD11C04A7B52BE902EECF46C6394A5D6057FFFCE768218
12A1D897DC8EB615932EAB77C17443FF816279D91C385CB39656EB7015993DEC075DAD14F6BA80
76EFD6F0D0AA0419B7EF81D8B1ED592EA47D47B31CC6A719934B962080E23439368462417C23C77
673478D6780CE11AD19642C69C058B13537E56ECB92C737502EAEA22B18F250A416CC89E94B4341
52E28567F4A76FABCC80E3A9EC848B57448401772ED51B323FA193183BF7638B0D0477EFFF007573
CAA7A17BFA9476FB05257DEB3DAB661324144FDBBAF3256A11C72A23F13DB1F1105B1D602A58
09189305FC2E034FB06770AFFC37A017D3399B11538496BB2D9ED7B91DBBEF225EF03ACFE6ECC
AA504B77EF99ED347789E4AE9376093D5740CA3C5CCDEC01013669DEBA42ED4F82D6F867EEA80
CB96A42CCC0C1535D566FA0B09617CCBBFACB16DE8BCE79C9A0CF566E733A938E72D43E07558
4D5E29E5FB377AC88B0558D302C18F0DB3FA23E8FDECC264A57C7A38DA6DFB696818D6124A732
54E8C37772BC6083ECB4FB088B4CFE064D02FE5C5B04E474B2386D079A46985EB82DEF6FFEBFF3
0BBC0BFB8E973985723DD40C67D1A58B7FCBF545C179FA88CAE15EF30F5C7F577C79A81D933808
E9DE484C1B4E0FA5667840140DB1BAC5D81B714DC584FE71C2DE7218D5A0063F1A4800153A009
DF7B2203EE7A72BFC7227DF52E727F41F00A10060404DEF91BDAA78DA8CFA5882A7002EAF3E68
322CA75798DFFCC9952E785C3F532E8D9E5E5877B00B9D1BF47EF1072939323E9C7738C7075A2155
0CF41E89B955C449F2E7F9D99FD007B824D451F0CB0F18A3D44336CEA853E94BE73ADE1E6551B0
76E6E0814B89E2B018385109D9B7F2EDBC168693A36E5EEF80383389D534CEF9F27148A1576290D0
832C6A19D3C70EC21A7C0E688120ED8C769458C4C1C70DFEC9165D4EBC62631CFC9FBDE3EF158
B70F65D83C4031ABB9AD352469A4061F396ABDA0479B2BF89881A3F492811381399C717078CE8BC
F11780746E79495D5F79E0518FEA51F245712146D761A8A7681F0141D0ADC82197287F73E76D217D3
F638352E05C62D1EA66964D9005E3D04600A8B3676414DCCCF000DE8C22A6D8C3A62E67DFA6A6
033BF9398C713AA3A32D41CED02D07F2724B53C2479B86FD769BFBD37B8781F407E37AA702876A
827619FAEED361A74253DBAA3070D322F69E0AC097B763C5B993BC8E8623A0042F73C988C79D9C
6075AECEF8BC99AC22DAA566D4B0716191AE8578779B4C9A6146D6BD8BEF868B21302202071778
D48FC2960F6C73AFC5093C4CCEE5F20338992E6ED005DC10B6FAE21B40D15116DA4F993E29F3FB
399FDB7A0407C8B17E566AFB6AA3B5433B600EB535A30577ED5297F23CA995B27810C8F7B742CD
A6042FA8E13D087CEFE7FB10727C37FF40DFEC6D05F8AB2374F9EADFE7128B4744409D2F71CEA
6DC6BD0823A6EE54671DDC80AE3F890C490D9A740C7D3C239DAE29DF48A23886E9ACFCDB8920
4648613D169C00BEB58A67C3F21918B6D4C173BDDBA9E07CBC28114E12B50027B58499F8D359BE
C03375C3F4972D4B111DC7FB199553D8BF9338F123A5B3EF946C16B866E39791DC3776B020802E18
2489EB5BC2FAFF24B2D2E7F32F0DF3CA2C2955116D2FEA0677E31BF13C331E91B8BE1A9A6AFA
BB913F7BF19836AA6424A1C53CCE8E9CA9F28420D3635C1999478ABFDC6DFFA570708A3FE700D
D6B889D43C0A3B0C40CAE66DF737E014E91EE7FE2A0E07D82BACC5FF64C3446DF9410E24084C4
91BD8FF4DBF984B744211CE76D5A469D8F1ECC72FC2A24E54BB589A146C34F595B83F791D6EBE
6D403BA784D993C1392F77010AEE2485CFA19856279DADB42BF2B0042B42D27D7152A9C8FA34F7
31C29BD3500BD1FF31F9B84E90CF9A8FD5C5D7C5DD351C6AC56983975FF1D17C8FE477AF0DEA3
4E615B61305431CD6D77AF4C3A6650FFF12F8D0CFABA7E72F67810E3450EDA528D00C0C5C8687D
C13922D7002C094D254112835F1DF05CC994E0170B6B1F4D3D3C47EFEF0084B3A4A9CBF0FA2B9C
99368FC1008FA65EF727C6B2301F16A2F11305FEF9389E01A8C318ED6C0021DFE06A04791046440F
3B7133FCDF903C59D1E719334C5E381FDF6C48BC981847FB610262F1A40B010477822ACF9524C029
E3BD2C616CDE3299DC832291890B78D0987A7ECE5E4E68035DA8B8F283F666CC85FD73E1928E4

0CC8CC83005E61B0BF48E0A249E3D959019BB22E3D739EC2FFAA33904193C8328B82114B353D0F7
D2DB191DB448E4CEB3D4292F3B456BB9AC0C82FC1FC6A8629963F129CD4CDA8B884D366DE53A
C95B9A04C3C9188C2C014DFDEB15E2F17AB6D8BB101DBF694C8F60787196C8AB791275E5BDD73
0595ED42F72027F483CED83F2498F081782954E0403BEB398BFF2A9F79EF8013A49429828CAF49E3
BD1EFDEA0F2813C1B9D1652626EF61366362F0EF0ED6508896E4706E473471EED82A7C2D0BF4E2
C41F70F9CCEF3F98260F6AD127B1F999BAC39A516E37B48DE0D9F3765ADCC89923738E584C6D3B
5483FD493A57E4134DE6E58B0ECB8B5C623AA91C21787460B71B18F471BB2B4E3B630A17EFF765
2D1346DB6054AEA18D7BFFCBFD22F132289C7084213B59FE7E3F4A8AF1A381D50F93ED9C207FC
F371E9C6456B4B10F22505A7DEF44CB03223B37F2989073D24FA2260398378905048995E9D8707F5B
CF470FF588C9863CC82944EF369E8ED03E39C9E1CFFB0FD6F5D83C77B5C83078C19A7E22BF0D9D
24C3AA434D1122F3CA363D5F2E4C7E72DA29924E351A108D8A79D51C25A151B867093F8576DAE3
25DBAF667DBB02CB5967C4D007793F2E29E7589CA1A768EDA881F503B2CDA05F0D73AFF0AA48
F0F58ED125AFA2AA43535D56CF1AB0AF61B17DDD5C3279C157D13DBA93657CC33BA616A368D0
4C969DD23012C2BF383E2EA4DAE3D3B3B08E56D6BB76FF659492E2EC03E1366DF4BA20118DDB0
6AE069F8F1EE5E3779DC9F37C37BF1F0155844FE3E300D80BF2A32A5A598D0ACD6D3964A9898DF
80E73B81A98FC2C09FC81BE91C13357026559E66AC2FFBC20241D97CFB871754296E4695771AA6D
4237621C93AAC5DED221F892AE7C4842C3267E51C018A2A513D001D4F417857AA2C22C322561495
35305FE2DD1814B032172AF1E4F653623364E102F36BA8783DFE84BCDD7E21DFDCF9708AFC045A
5BCE16F7C56808AD09303DA83499E8FE7DE10C5A8972DA016A29BCB5E4C0F881320058C5BD636
ABE0C4F672A93D7F29C7B13329AAF109031D12AE9EF99629C5774052E44707C08873E051F1A1DD2
6EE6EEC658B6990CBEB6FD979839A50DF2CBC77DAD9BB27E66FB223817B396B9D490290168831
A5689F92E5349819034844A0714C66E16B327CD3715AEB1A99762B44ABB23867A3DBD80F148D139
B7F388E09918F39D3914A19176B75573C541C14FFE45AE4E2836685A9019CD3D3B331F5EB0418FF0
92D0C41F1C7DB3A565E36B5BD01AC075B494E8C2EA6AA0D93E3F3DE4000F1DF0F012EFFCAFBA4
F0C375624C1AE48918C0EBEDB2A66478A85FEF39EFEE32AF29B69F4D5E072BA2F289C556B8DF59
EAED7333A2B9C410BBB703CF595D6DFDD4D8340FAF221E52F8E53630E57587DE4E1D90561DB10
8F73649E2C7F3C11200B0731100E3A3412F1D9DFDF4F04D06021DBCC44A42989D89973D8DA0DC3
E9C659F71292E815A2F9FE9C93C65FD68D32320A5ADDD6D7218A2EC53D12341F7F60CB9CA77860
2EE60A47550460D0D569B0239E44019E7C0C7C98E28F711DB503AD457C94FC7FFA7FB62F2614A92
86B521676ECF10195D139AF6A50AD6C0284D8EFC750C0331EBBB3CE7CE0B89499BB2C35C7BE37
22E405041AFF20CAF170CD9C88695ADCFD7525494085F8806EDFC7A6198E4D17A03C43C228DFEA
4F585ED6BB7309BBC093FDC45CDB69A956CE94AD5C27AA2976612DBCCB9D27352EB34BF003B
29DE0EE007C889106D4DA518F4169C7A3983FABAFABAA5E53A7F0795A52A738E77C90CFDC0538
1142903B9E44AEA94A069177975FC6D983CFBD4E1EED628EC72F0BF7CC872568B9B6EAE1912EE
BBEC269C982C133F84BFEC7E6093CC38CD1B968D91382E5CEDE1D0CE8C5544966807202C575DE6
B252769A9D480F37CF3F43C041D34511E98588CE0E65C7AE8C856F22BFCF511477ACC3C7D3DE35
9E6075B496A1497BD1CA5894

ct =

EA795729021761F21A5CC6AB23DDA390AD476909415955B773B52BD7F370878837C56CE27EB25FB
3DA421FBE278747F2AC4D684BAD47EC65EE7701CCD553DB8546E207FCE4E8A14F4943B0FDE118
D5CABC51A929AEE90CE2FC402E7D1144944B88D1798C493A4B93D12D58717748C2F1F6E6F53671
0F3FF7D3CA85B36AE97A4C745079C0C03998862CD11E7C59472BA3896B39670377121A462FECB36
AD09CF4E6AE82571110C8D96509F556EA87E9C5AD45D12048C65F5D6411D194F9FEF3B3F0832F4
DAD105D59057BF03676775A1047662A1EAC654B4947986CBE3D0A988048473BEA28D7C45029DA9
2458C9CA8B567548BEECE45D3EC6BB0A0030CA7051AC5D83077DBE3A8AA6A01A441C62CE73C8
3689D1C50BA619A086DE729AF97811882616E85AC55B6C8FE56E82D92CA5A920359882E44AE405E

E30BF248DD43CD4168737225F56D9D80AB9AB8AE0F04191CD23A68400F477EBFDDC01FC6FAA72
970E4B2A97E5C3E40349893AD436EFBEFA3FBA8A80050C7CD1A5B8B776630A7B7659A07359A921
5E3F5F6A1D2972256EA14FA4367BDE1B4C8046D329ABEDFBE2EA984F1205BA7B6888401BFCC5A
A40E45FCDF50F5689B1C9115B2A96C0184F8B459698975A0156D3B5FBF442B876913CA5A2A1F4D6
31B3D720F38892D15C38B2320D76758429922666FC13CED9B517A76492080102B2623F288412280474
2F73215E204BBD9385C08A59E164DDDBEE34206151959045487FE30A9F5102F1273F920A58F270F103
25D6DAEC354914B5CFFAB4FF3751593D9997179574635FA5F3DC32D410FDD64D2B44F7971EE826
1E7CF2FE0E9011FB2F682AD218B363A6EE087741E5A2CEE4DFE7D5EE67D8F21CF1DB9A77B785C
12EBAE3D0ED053CE53ED0F58DE52FC0F5D88826E1DD46C9E0E50C2052C2D7409E49E441E76D7F
2B7229E23AF5E6E52C825C06EF43D2E937D0478602412AE21A702941B2D6B002DE68E162DBD6A5
CAF2B5E4D259BB4A31A4D759AECD2B63D7C316BA8CA05994BC28380C1FB24050432F42488E2D
A0701954453D8AE2C229AB33B4939D8A14287BF5918691BDAE58ECECF0835F0B079359F391102369
C471B110E24D2A46BC0AEBC0AC9647F276BBE22CCE42E061312A9929F154611ACCC19835124999
D03AC8626E9D051994F703C4183005D653EF1A7B7975A52276D6DA42E0A0858E87D48D7E8C58FA
B2993DC359F866F964D5A64861DD4722ABDC6A0E3105ACC9CBAAFFEE5DC9A2B6EF03AB31A41
109511A6FC5E1B2B89973FFE44CC62CCA21456EF41A792FFA03AB7432F60B9500A70D26B3D37218
8363EA10EAFAB4221BB89F286B916427EE03B526F54791D1715489985E8EAC0B246984C311F48152
68905DBB91389A0D3BB2B333D14D9CFC3A2041B45B4CE500BED723F20E2

ss_enc = B11EF9E97DDC18BE412371D4279CB7626625E9B8459D5E25B0D625C289DDDAAC

ss_dec = B11EF9E97DDC18BE412371D4279CB7626625E9B8459D5E25B0D625C289DDDAAC

result = PASS

测试编号	REF-3
测试名称	CTL-3329-2048 参考实现版密钥封装功能正确性测试
测试目的	测试 CTL-3329-2048 参考实现版是否支持密钥生成、封装和解封装功能，并验证封装共享密钥与解封装共享密钥是否一致。
预置条件	1.已在测试终端 PC 上部署 CTL 参考实现版程序代码； 2.test_kem_correctness_ctl.c 能够正常调用 CTL 的密钥生成、封装、解封装接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-3329-2048 参考实现版程序代码目录； 2.输入 make all 生成 test_kem_correctness_ctl.exe 文件； 3.运行 test_kem_correctness_ctl.exe； 4.比较封装共享密钥与解封装共享密钥是否完全一致； 5.重复执行 3 组测试，并生成 ctl_kem_kat_vectors.txt 结果文件。
预期结果	1.test_kem_correctness_ctl.exe 程序能够正常运行； 2.终端显示 “All CTL KEM correctness tests passed.” 3.正确生成测试向量文件 ctl_kem_kat_vectors.txt。 4.测试向量文件中的 ss_enc 和 ss_dec 相等；
实测结果	程序正常运行且显示 “All CTL KEM correctness tests passed.”， ctl_kem_kat_vectors.txt 文件中的三组 ss_enc 和 ss_dec 都相等。

测试向量: algorithm = CTL-3329-2048

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 3009

sk_len = 15617

ct_len = 2353

ss_len = 48

pk =

BB2A133D8802A109CE74A90B6C0848479563E4BD364A0A5D6C720C77388B1144BF3F4A6AED8686
8F864FBE3F2455578176AC6B39ABCB271BA38C0B08192022817BDC8A29728A8BBC4E168ACBC11
B4CC62A90652C53810485AC9A0A8CAFE5AB68B25D0B52A6E15C2DEAA9755E0A8D86DEF601FA
1E9B05D372D0151BA806E5A680DB5D238E96F839352B6A7930E7CBCBE1B7766A5704C975B58779
8930512C13B61388451DA7705AF69F5E81D96C294338269617029A1925479ECB631BA846D377356A
B04F57A528B5B111604E173B062B5573BE62E96E53916DBC0724259F9AAB0DFBB6F88A29B5A22A
02E63A6B53CEEDB7AE8718DF037D852261865358B6ECFE602273BEBCE28D2280E776F64AC04573
0C1D60E2F853F2131E03A7DD17BE55418519367F8A8F942D26EDC9C7445DDCDD93517B7940751F
C118B7C0CB5489DAEDFD47C00E62DC79C6459D48554D124720E6E47B80D32846C45022C02D6703
99DE532A89FFC78701C3B18F6DE44145E99871340BB8FF515E64F05686008CEDE77A5F6628C45A34
AEE2B3A2354D71BD3DF5D44FB561ED236D23A22741E056F5ABC9B4E83D8EA52E4C2BBF54BF01
8044CFC2CDF6CC2C28EBDC2626495BE821C8FDAEB26E08ECB1B877FE5AB5B1A3E9DCDF9CE78
8D72D49B177E2C2D5F571D27F886923ECA50E45EBE6707A7362251CD579EE56E0DBD8152CCB94B
0BBE442071A2AA0B5E959DF4A0B9006E5BD9E1337B78603C268C81C7EE3132C969F2FFEB628D8C
A334DD34EF624D7DB536D79ABDADF3A4C335054CEF00DABE90DBF2B77D699908A2A5BB8A32F
12A4483C90A43D653A6ECA694743B9E5E1D4FD7AB42A4A3B00AEE1123DB9879BF5AD883AE50A
341451637C622B7B97E7880E72BEB46691535F090A9C47EA78EA696F1B7F3E423D15CCF91EC0B44
E8EF4AE1D4065282ACE10F898ECF0FC86EC3CBB8CE3FDF50A82FF01BA8B09EC5C3904D23A83F0
6B99C2C7EEFE8FE1B3C3FCC4B4AB2715E32C22E3A015CBFD15F99902F030696C6D98EF6F66F
BECDD77B75D26B44FA1C86D672C78A85ACF37CE5897124A56C1F34E74563850452C04BA941B41B7
0CDDDB9668D5E8BC4BDD10E0CC19D28F93D3FADFEEA95A77AC3668E675D0139637FAF0A23E6B
73C0DEE2B499E789216FDCF3E61AEDACA0EC00AF51B6F3E966A4460A0DF380F44AFD5FE707BD
B19CCD2E748482C5E3A0F67D0B3D6A64D95E6D94DAB4D90270C3F2EF5D2332F0A17767B402EAE
DDEACB8D575485C3CBF3B56E0E0FF732D68A595888D200377A2FC589D47F0E18308FA2CEDD436
BE0CB8B2A38D93C8A0351205A560B8E6517D396730F1A6398A4A301E886A7DD898688ACEC3636B
DDB680DBB99507582FCA4E60D18F70FDB6966E1B9DF30AEC316E78DD81FA017ACFE0432D2F556

151BC4EA9EE5D4331CAF8976D18CAE683EE9279AE222B2049278E61FC4222463272AECF9B8D53B
A1EC01FA28951FA0740E5CA1C2D0BEB78F0A1BFE12C8ECF18AA35F39EC36DEDB664EEB0D0171
624D15AF9814973C4C0273BB557BC05C527C3C6FAF6FA1F5EA0D30E776FDB89436A6EBD1A7550
F216B7CC78E6FD8119A3524D0BFB445FA09DE7D6F5FAB1202EF1345C0A012E6D7B01150BC3F767
748E84F9F3B0BAE0DD6FF25339989EBBE6B27E4A2CF743170979D244807C6262110D8BF23605FA9
EC74E7F69E206CC73001697A79F4AF5C0EC00A2E30167C8AC392663B522EFF48B2FE521433E9E
FBAF61BA96D13FCC69AFB2F0B5B64BC8AD09E8E797CC540FA02D510AB00029A86E701AE2457B
32CED213B2FDEB126DA522CB6C258FCDD2988011C42CE704F967B73A10A6E9DA0A684F28A262
ADA1EB1BC02C5D95B7235C00AB11C3AC4239065311E7A8976F0872ABC21DBD06085D219212DE2
91F7EE5DCC6A5C6E5028D1A04442A83F53B3BFBBA39A7C5C4A2777F31C874518D72A39BF4DD40
21971C585E587275E65589A86F0647A2F089D396046B6DBFF292E9E26D6D07BF354DE03F923D6280
2E3112FBFD910190226CBA1746C10CAC2FB2A5127F9597FB8AAEDA4B2F959B39236EEFC4553D47
127D63726FF04A5B2F50926BDC2FD9761F80CD79AC526C0C5D1FC948C463E1C91EF9975289D1FF7
68D0DF6E629648A550AEFD87992A70A9E93EFC8A8B423062AE389DF3F8F86DCAB5CA25BF7831A
B1D0E8484D1189A0A032EA7DF8B8A446F2A04CE63A4088077C46560B450CC14AEC244B5CC0FD8
A9546E180C4F6026491021FEBE80E376D444B8DA2211210E65414981D48417ADB21C06B4B32877E4
14B8016ECBBF6F95545AF9B6E6F806AFE9D48E9AA378CC1E5AE9ED5F01F1892C0949D68F4838
A77BA1317049395B230D44BA2FA403C0C4C63C091C4683CC138CF499751F1520B8C81FF3D98E419
317B99A4341413632DF1D86B48BC0A8DEED7186C6E225EB3ADA8002C40F32A2991B49C039E5BE2
0B626705D3C5A5A502B88D2B2E90B4117A0FBAEB6691802EFE52F4858EE544750EC4DF8122E9880
D2A2CDFD735AB0CE15B6BF968282EECCCC7636179E64852C3C48E03D95F164A243A6FD6A37B98
4018C4F9D7A6D086E46A619539025217ADCB842D68DAC0B485021FEF4A62E817AA3DA68D549384
3365AC838035E627399E29BE49F6D082A45F96F092ECC5C6AC5A4ACE980BD7D82B9DC73531DA5
59003AA5C809B43FF4C4AFD2E72D44649B160A1DA52D0B56B62121B84E25425A870959A9537D0F
FA3C9E835F70AB2496DCD9F55C9C6EEF91E7247093482E5168C94CF5997D2E4642BC4D029F58293
D29A7DB64FD267612FAB7B2286939DAB844A7EEF1B8E814915ADBE7E6871A182FB6AA94539519
8B4F31F156206A22CF67CDED5A93C75CE54F7F53EEFB5D418F142B8234BB814F09F079456CA0F9
A7132FAA1E9C4C17B7CD4D5F12046097AABE785F3C9A4BC1256E8896F57681E801286B09212EA9
866C8DCB136AF28031480610EBE4842BD220E6E5225E7E4D5272C448CEB72A482F8107DE6DE783B
16732523C64A20ED178077A275B782B8214EF52F2DD0BDDBE939473458EB80245838D04207B608A
C338C8D6EB838452118D4DD8A2D769689DFC12A80CEDFB53338CCD71297B242A9085B2C61B02C
176AA60CE5A05E733124DE0AEA4D92FCBF7BBD8EFE703C1C6E94C6A15A73362447CF56940A459
E8A5AA6FE04D2EDAAEC50938E31677C519DC095AA47A676D3E2885648E3C0C2ACA3FA2525752
09B005B28AE7F13CE074281716E21013BF9370C80D1C96ED667C751DD394AFD08ABC1C85FD81A
C9982D922635458BA72BEB88EB4768C40B21B2078E63F9A2361B8473379201FEFE56C3A49E19088C
515433D179D4E37D4CB2C25A911EF8A500BBA444BC5FC9EF7BBC2DFAF5D263A601C672798B8E6
0326BCE78A1EEF4ED8877CB9B61A31D80E8355A96F8B267AA6F796B6F9FB7F44C9419F5BBBE995
3CC1A7DFF0AF04FB3D6B54246C1301D23578D225C73952E0AB573768C15D82C25BE70CCCB16369
6596415B3A4E3E96CB4CAD58334945A7753D2793036A3A3ED45560E18D59F504F3FC49B47F5939F
D9A259EB6D0EBB767563E75025C51DAAE3BB353AA6427202166B7801AA40D86E3CE11B60794680
5A626189915AE2C112CA31FABE0D4372940AD23006EBF46E433E5154571D74BA1A0C02D195A425
0227305B948ED14798BD6A4369384E210247010A83150EE891469A16CF3278DFF67AEE9A26C3C7E
AEDB17AABF00A021B368168FA6775316C128F8262577DB7586437D2C531FD03A23F5BABBE8D036
90997FD43A288162C7B89111F2ED58BAC3CFA8773E6D0B6148D901CFCC9F79708F3CA96F079B10
1311EA6F88D205DE884DF6C0527B825F4D742294AF3747A6904E47097D48F9F79A7661010E066F66
C82CDC9B31A3F66239022253E2C805E27037B3FB2C7595FD101ABACF53E07E349BF73AD541EFA

AC79E543AB5127B96F6BF3B74E91940E810CE460B6DC20A14813AA0B1A17F810AB1DB16ECC4D8
1A93BCD1C532C359A26829F58BCFCE27C306AA3CD8706E9FFE41C25BDAF527FDB422A7BDF526
8F40B263D16358AB4F824CE8F279D051C39EB6D4FBBBF6C89A9C1AFD7FE7DFD29612A911B6DE
C04F67624D0BD91F73A4A726F37A19F40D751622076C09BBABE51B0B87CD595314DC0A52BAE88
7500BF24AD9B79C2FDC8310E8A305B22E36CAF35D85E9C3E675051921C0A96FA08FF6F759C68342
41C873609C05128929EB59B330C11B807D26EF4A

sk =

3BE87E15F7E6CAB66514C99994C2F5BD92BC52A8554D2C081C0FDE814A57C2964B7447314C8B8F
DA60158D613DD611F95C2C77868DDE0AB22B1C1014F3E3549DD20000F1F1F0F111F01FFF0F00120
F0000F10F001E11001000101F2001F10102000FF0F11F0F10100102F120000211F00D10F10F1F10020100
FEFFF00201E1FF00F10EF101000011E20FF10000000010102001EE0101020F01F1E00FFF0F0F0101F2F
00E0100101FEFF010F00010101F00F101F010000E0F01000201E0F0F1001F200F011E111001011002
F100F10E020001000000000FFF10EE13100F012011100000FF0FEF1F002E10F0100F210FE111000FFEF
1000F11010F01FFF11F20F0FE1F000F1FE00EFF20F20002F0E0010E001F000F1F1001F1FFF1FFFF0E21
001F112011FF11E11F0001F0FF01112000FF000F211F000F011100F00100F001010F1F00F01F0F10F101
EF000F1100D00FF0100012FF011F0000F1011F200E11F00F0001E10001000FF0000FE001011001F1F1F1
01010E11F0F0F00E010F00FF2F0FF11020F10F00010001F01020FFE12000011E211F020F0F0212011FFE
F200FF000F011FE0000F100F021021F00000001F0F00211F0E1F0100FF11F001FF001001F1F01010F1F1
0001F10F2111F212FF01E0011102000FF000F10F0000200F0FF00210FF0E0F10EFF10F0EF00E01FF000
F100E10FFF0F031F10E0E001001F2F00FE10F0E01111000012110F0F0F0000FF00EF121F10020F000110
0110EFFF0FF011000001E00EE0FFF0FF0E1001F011EE1FF001F0FFF101E1101020FF011111F1001FFE
0000FF01010000F22E01FF1F1000FF000001F000F00FF101011F000010F1000100F10F0F100E001E0F10
1EF1FE100FEFE1F1101020020111001F000DE01101111F010000F0F011E2100F00F0F001F00F10F0001
F00F10FFFF1F000F00F2000010001010F00F0F001F01E0F00F1000FF000211FE1F300100F110F0001000
E0FFFF10F1F10F201F1E0F1F3FFFF1F0FF000F0F0E0101F0F1112E2EFF1102100FE000F01E110FF0E1
F013000F2F00EF01FF0E200E0F001F100010EF0101FFEF00101F000F01F01100F100F1021F2000211100
10F0F0000FF0100EE11F0E00F001F11110F00101002F1E110000FFF10000F0F1101021F31F012F000131
1F10110E211000110100F00E111F00010F0F0F00F1001FFF002FFE01000000201010F0001010F1100FEF
FEF0001F0000F0110200F00102F01F12FF1F00FFF0F110F00F0E110F00F01F221F00001FFFF00F011111
000E0211F00EFFF010FE0101FF001FF10F0EF0FE10110F000EF0E2FFF011F100FE0001F00FFF01E01F
1F0110F0FFFE000F0E1102002011101F00F0FFFF00F00F1F00F032F0100EF1001FFF010E0F0F0010110
0001F0E2111000100E010FEFF000FF1F11FE101E0101212FF01100F010E01110010010FF00000F0FF0F1
11F11F0010E0F210F3F01111E00F1FFF00D00211F020301110F002001F0F1F001000110100F1F0F000F1
01EF01F00EF010F0111F01EEF0100F0F1F0110E00010F01F10FE010F0FFF10F11FE00F0FFF1F1F0FF0
001F0FE010F0E0F102210F20FF10011201F0F110001F10F01110F0010FF11F10000FFFE110200F0E02F
000FF111E0000111F0E10DF0FF211F0101E011F001101EF1FFF10001F00FFFFFFF01EE000000110FFF01
F0000FE0000F002111F0FF00E1F00FF0110000102F01E000010F120200FF0110200000F20000F00110200
12E01001E001F1FF0F100FF11D0011101112F110100E10FF0E02F1111F01F01F1F0000011FF1102F01E
F00000F0000F0101FF1F0F11E11FF01F2010012100F1010F0011FF0011000FE10F0110000FD00E1000FF
011101EF01FFFE121F0001100F2F0FFF110FE10202000000FF0F00000F0F0E1FF10FF110011100101300
1210011E210FEFF1EE100F00110100E1F020F02101F00F0F0002FF1F1110E1F01010F21F01FF11F111E
FF20011FFF0F1F0E21100EFEEF1E010000010011000100F10F01EF10FEF0EFFF100FF00F20100E12F00
0101DFEF2F1100F1011F1E00110E000F0F11F00F1000FEF21F21100F011F100FEF02FF1FF1F0E000F1F
EE00FFE00F00F10F0F222010F00FFF1FFF00F1010000E210F1F0FF0F0001F001100010E000F02F01200
FE0FF1FF021F2F0F00E0E0FF100F0F0001E003F0E0000000F101110F0F0FF010011F100111F1021FFF1
0002F010F0F00F00F11010000011F100E010FFF101EF1011000011DEFF010F0011002F00101FFD01100

F0FFFF0FF000100000F00E100E01F0FF0F001F0000000F10000010010E0000112011EFF00F1FF100E111
001F001001F0FFF00D10F000FF00101101110F0010112200FF00001F1E0F0001FFF000000FF000F010F1
010F00FFF1111022030FFF00FE01F0FF00000E01FE2F0110111FF2F100E1000FF1FF0F1F011EF1F0F0E
000F0F010E000FF1FF1FF0FF00F010F110E000000F010010F010E000F00FF11F1FE0010EF0011010000
F00101F01FE10000F1F10FE00F001F001100FEFF1F220E00F00FF100E21000F0F11C3F0F00100E10101
02F1FFF20011F0FF100101010F10FF01F0FF0000110F0000E00010F0FFE1F1002010010F111001002E21
010E200111F00E0000F000010E0F0F1F00F001001EFEFD1020F1010EFF001FF00FF1F2000F0FFF10102
0FE20F011E10F1F11F0F000F11FF0F110F001101001010FFE0001000221FF110200F0F1F1F0010F110E
E01112001012F02100E1EE2110F10101FFF00010E0010FF10F02F1F00011F0200F010102000120F00F11
FFF0D1111000E1F0000100F0000FFF1020F100021E12E01101E01FF1011E1110200F1FFFF00F1FF100F
00F0FF0101FFF10010E0F11F2210020F000011F101F101011FF2F101FF1F00E20010FEFF0201F00F2010
FF0001EFE010001F0F00FF00F20F0FF001F0020F1FFE0110F110101101F1FE01A00301600AFFB00200
D000FFE00400B00E01A007008FEDFF3FFD00D009006FF9FFB001FF2FE7FF4004FF4000001FF5FFCF
F4000FF0FF7FFFFFE007FFDFFC01501101BFFDFF4FEBFE8007008FE3000FE301801400A012FF200C
FFAFF9FF2FEC018FF7FFD00FFF001D01DFF601A002FE9FF6FEE005FF7FFEFFFF00301000BFED011F
E0012FFBFFCFF7FF401AFFCFEFFF1009FF3FF3FEEFF6FF4006003FFBFF300601500B013011FF2007F
FB00500C00A00100AFFB002FEAFF1FFCFF3FF6FF3FF7010007FF8000002FFD004000008FF801B00A
003FF5005FF3FDFFF201FFFE00D005FECFF6FED004FF9012FFCFF7FFA00D004FEA00500100900700
EFF9007FF2000002FFEFEDEFFD002FF5FFDFF7010FF9001FF9FFE00600D003FFB008009006015FFF00
EFEB001FEF013009FFA012FF000900200EFF9FF5FF0FFA00001300701800F00CFF6014FF4FFA007FF
3FF9000FF000AFF9001FFC004005013000FF9FFB00D00D00D001FF0005FF0010007FF8FF2FF8FF500E
FF7FFEFF0FF5005FF7FFBFEC006FFE00900AFF7012FEF00D00000C008009FF5010FFC007FF000DFE
B00FFFAFE3000FFDFE8014015006FFEFF7FF600000BFFDFFAFE3FF701400501C00F00CFF0FE600D
FF9004FF501E00DFF3FF5FE7FFDFFDFF8009FF4000FFD003FFB006000015FFD010019016FF6004FF
D012000FDCFF901C004005FF9FFF00300B005004014FF5000FF601700001500200200CFEE010013012F
F8FF5005000011FEF00800400A014FFFFF3FF700A015008FFF002010FF9FEA00EFFF00D007004FFAF
F601300D011FF4FF2009004FFEFEDEFF6FF0FFA00100400CFEF000FFC002002FF7FF7FFB00400700EF
EEFFFFFCFEAFE5FFC011006012001FF5FEF024FF4FFC00800000AFFB010006008005002001003005F
FB00FFFCFF800200DFF901601101DFF400AFFEFFFEEFF800DFEE00200400A013FFC005FFB010FF8002
FFE00C00000CFFFFFAFFCFFAFFD011000016003FF000200501100000DFF0004FE9007007FF300C009
FFDFFBFE6FF8009FF7001FFAFF7FF1FFA00201DFFC002FFEFF5010FF6001008007FEFFF3FF0FEDF
FA012001FF8007FF9FF9009FF1005FFB00A009FE3FFCFF4FFBFF8000FFFFFC000FDD00700D006FFC
FF2019002001019000FEE00BFF3014001013006002FFEFF4FF7002000005FF4004011FEFFCFF700300
B002FFD004FEFFFBFEFFFEFFBFE6FFCFF700700D004FF501F018006FF0FEF01AFFCFFEFF3FE9003
FF8013FEB00DFEDFF1FECFFA004FFAFF601400D00CFE2006006FE500CFFBFF5FFAFFAFF1007009
FF3010016FF2FF8FEF000003FFFFF6FF800D004007FF9005FF600E011FECFF900CFF6FF100301AFF9
FFF00200200BFE7FEAFFC00501D012FE2FECFE9007FF5FEB00B004FFBFF0006005006FFBFE000C00
DFF4FE0008FFAFFA00E0FEF00C00BFF701A005018FFFFE8FF7FFDFF7FFF00CFFE005FF600E010FF5
FEF006FFB012006011004004011FF9FFAFEE017FFE000015FEA006FFD000008FEE00DFF2011005FEA
01B00F009FFC001005FF100D00A01800500801200200C000FE8FF9FF8FF6008FFF008008FF0FFF000F
FA014FF3018FEBFFE00D000002FF100AFFA017FFCFF8005FF9001005020FFB009FEDFEAFF9FF7010
FE1FFC00BFFB01400500601400900E016FECFE9009FF901F008FF100EFFBFFCFFF00A009FF4FF900F
003FF5FFD01C01600F00D00F010018FFFFF4001012008FFB000FFD001FF6FF900900DFFB011002FFC
FF2003FFBFF1009FF900A00900AFF001E00A00D008004FFEFF4FF8FFC00B003003001FFA01CFFFFF
6004FEE00601900D002FFD005FF0005005FF401300900700AFF1FE5FE000AFEEFFB008FF700EFA00
9000FFB012FFA011FEE005002FFB00700900CFFF00E004FFA004FFBFFD007012FF0011FF900D012FF

BFF8FFFFFFD009FF6FEBFFCFFCFFE017FFBFF501800C009FFBFF3FFAFF5FFFFFFD00400BFF9FF0FF
FFF500200100AFF5FF6FF2002FEF00F000007007FF6002003FEFFDFDFF6001FF7FFEFFF00501000C00A
FDC003FF200B00800BFF200BFF9FF5FE2FFA016002FFA00200F000002000005FF9011000FFAFDFF
A000001FF7003FF7002003FE700B00B00EFB001FF2FF0FEDFFBFFB007007004011FF7FFBFF700400
8000FFAFFD00700EFF8001FF4004FF9FF0FFCFFDFFEFF3FF9FF300FFF2FFCFEC00600AFFA00FFF4
FEB009FF8007FF200BFF7FF6FF3FF2000006005008FEC001FF9FFBFF0004009FFEFF9004000013017F
FAFF9FF0008007003FFCFFB00C01BFF0FFAFFE00401EFF300501AFF7006FEB01300500100BFFAFF1
016FE7FF7FFDFFD001FF900800800AFFE011FF4007FF7FF9019001005FFBFFFFFF7FF6007FF6006009F
F9006FF9002000007003FE6FF5007FF2010FE3007002006015FE5002007FFB00D000FF90060160080060
09001FFAFF4006FFBFFD001FF1FFCFFC016FF7FF0003003FFCFF700001E016009FEB00101200400A0
0200D00200AFFA000FFCFF5FFEFFDFF7FFAFF7002FFA00C00E00C004004FEEFFC00601100900C00
B00000200900700AFFA01E009002024007001FEFFF7FFFFFF0006FF700F00B016FFA009009014FFB01B
00DFF2FF9FF5FE5004FFDFFB014FFE007FEA00500E01200600D00100B01000AFFB01B013004011FE
F002008FF8FEF010FFC00A006FFE001006FFB00A00DFFF005FEA003FE4007001017FFF019FF700900
2FF100AFE800E00400E00200E01100AFFD00EFEE000001FEA009FEEFFD007003001FE9003FF9FF200
5FFDFF0FFDFF9FFB009FF7008010010001FE9FFB00900B000FF9017FFEFFBFF5FF2001010FF5007FF
1FFD00800301CFFFFFFE004000FF4000FEE002FEDFFF009FFB000011002FF8FEDFF0FFC001005003FF
FFF2009007FE5FFB00B00E00BFFBFF1FEE00B00400B011FFE008FF9FF0FFE010005017FF3FEEFE90
0A005FF6FFDFF2002006FE4FEDFF8000FEFF9FFBFEF005FF6002FE500900EFFFFF300BFFD010006
FFBFF5008FEAFF700DFFDFF501E005000FF2FF4FFB001FFA005FF7FFBFFFFFF6FFDFF1FFCFFD010
000004005001007FFA008001001FFA004002FF600CFFC003006FF4FFD00800BFFF00EFF901F007FFBF
EFFDC017FFB008FF100C008FFA014FF801300C001FF3FF9FE600700E01100DFED00D00200E00C011
FE3002FF6003FF300FFDFFDFF3FF4FFB002FF1FF8FF000800AFEFFFB008FE1009FEC003FFFFFF9FF
B01600C00C009005FF1FF7007FFB018016008FFE004FFDFF4FEFFF6002009012FEF00CFF3008001001
FF4FFD00000C004FFFFFFE004008009FF5009FF6010002FF2001FEB013001FF3008FE6FFCFF5007FF0F
F700300B00BFF900100500A00E01600CFE3FFC009FF2FF3008005016004FE2FF5FF1FF8FECFF3FF70
0CFFFEFECFF9FF601800D006007FEFF9FFC008005FF9FFD01301DFFAFFFFFF4FF0FFBFF7FEF00300
6FF4005FFC008FEBFF400AFECFFB00DFFB00EFF5011014FFAFFFFFFE00B00DFF7000FFC027003000
FFDFF7FFE007014004FF300DFF6003001FF5002019016FFA00A001000FF8FFCFF0000011006FFF00A
FE2014FF4FEE00E000005FF901D00C009FEE01801CFF200E000FF7FF8FF0FFA00CFF502000C008FF0
00101000C00300AFF0FFB015FF3FF9FFE014FFEFEFFE002019FEFFFD005001FFB009FFC003018012
FFFFFF3FF5FFCFF400EFEE019FF5FF1FEEFE7FF500700FFF8FEFF800F008FF3FF800DFFA006FEE01
5004006FFB006001FFAFA00A00500500B006FF4003011000007FFAFF4FE5FF1FF800CFFE004FF9FF
8010008FEC00700900400EFE9FF600100AFFE0009009FF6FF8FD7FFB00DFF8000FFF00300000F009
00C019FEA00201A002FF2FFEFF300F00300B000FEE014FF3FF7FEBFE5FFB01A004005002FF400BFF
0FF4004FE601300FFFFFEAFFBFF001FFF3FFCFF3FEE00B005003FF9000003FFDFF2004FF400A00700
AFFE00EFF6FE601FFE2011FFB000002FFD012008FF3FEB002FFB000027012011006FFD001FF200F00
0FEC003FF100B013FF200CFF9FFAFE9000001002000003FFA00A018FEB00300502300D00B00300D00
BFFDFFCFE9FFDFFC005FE4011FF4FF500BFED003004FF6002FFA006FFE013007018004001FFF00DF
FE012FF5FFF004FFC006007FFD00500C006019FF3001FE200B003FF4FEF00EFFCFF600500B01B00FF
F8FE5003009FFD013FF7FF600FFD8009004006006FEBFE500000300500FFF9003FF700DFF0003009FF
4027FFE005FFDFF7FFCFFAFED00A013FF6FF1FFB00CFF7FEEFFB004FF901DFF0FFC01001500D019
00AFEE007FF0FFC008FFCFFC00000C00000A003FF400000BFEDFFD00BFF600E011FFFFFF009000FF
9FFEFF6FFF00D01200B008FF4FF7012FEF002FF900E00400300AFF8008FEB01D01600200600A01AFF
6FF6FFBFF8004FEEFFE005FFB00800300B012FF0FF300F00500B00AFF6014008FEDFFAFF100E00600
3FF6013FFE003FFD004002FF3FF6006FF300C004FFB00AFF0000FFF006FF4FFDFFB009001FFF005FE

200800CFF1FF200F00BFF1FFC009FF7011FFF006007FF5FF5004FF800E003003FFE006FEEFE8FE5FE
AFF800CFFBFF6FF7FF200C003FEF016FF0004FF201E011001000FEFFF7004FE6019FE401B009FF5FF
EFF8018013000FFDFF4009FE000D00CFFFFF3FF5FEA008FFFFFAFFE00B00201501100200B00EFE5F
F7FF4017000008FF3004FF4FFDFFA004FF700100C009FF1FF2FEE00BFFBFFD01101001BFE2FF2003F
FE00B00800600DFF200B003FFE00E004FFFFF9FF0008FFA012004006003FF6015000001008FF7FE7FF
CFF4FFA000DFF1FFB00401000400C003FE9FFF00D006FFA005FF500D00300EFFCFFC008001
FF2003FF1FF0FF8FE3FF401400900B017FFDFF4000006006FEE01201B00CFF500C003FDCFF6FF800C
FF601301801400BFF8002FEFFFE010006FFE00C00AFF900700CFF100E00C007FFEFFE002FF400C00E
FF1FF2FFD007007008000002002FF9FFCFE1FFDFF9FF1FFBFFB00E00CFE8FFEFFF7005FF4FF100B01
300400400700B002009FFAFF0FFEFEFF4003FFB000FFAFEBFFAFFCFE3004FEAFF6FFC006FEC005
FF3FF7FE8FFEFFA000017010FF301600EFF501CFF6FF8009FFD008FF900F004FF1FFC005FFC013FF9
FF9000002013FF8008005005004FD5002FE700200D003FF1011FFF01D00F012FFFFFA008FFF00C0050
13FF6FEBFF8011003FFC009001015FF9FF600EFF00A012004FFDFF200AFE2003FF8FF2000013FFCF
F700CFEBFED009FF4FF7017007001005001002010FFEFFCFF2FFB00EFFC01101A003FFA00600900FF
EF011004FFF005001001003FFF00DFF5FF800C00D002003011003FF6FF7005FFEFEFFFB009FEA01400
D010FF8FED001FFFFFAFFE021FFE005011005FE600800DFF5006012FFC01D011FFB004FFDFF6FF50
0EFF2FE300600EFF7FFFFFFF2FFBFFD000008013FFF026005FE4002FF900D00500BFFDFFDFEF006
FF600C002FF0005FFBFF00C00C018000FF5FF7FF8FF3FF5FF3FF7FFAFF2FED006000013007FFE000
01A007FFE00D010003014FF7FF4FFAFFD00D019FFF00D01D01600400000AFFBFFDFF9FF3002FF800
3004FF0FEAFEAFF1003002FFFFF2FFD012FFF00FFF600A00BFE9FF3FF2FE6FF7FEDFF5015FF8FEB
00700AFECFF1008002FEFFF3017012002012008FFF00BFE2009FF400200E006000FF200EFFCFF7FFB0
0CFEF00000F00800300DFF0004FF5FFBFFBFFF012FFFFF5FE800800B003FEFFF2FFAFF101900FFFA
FFE00FFF6FEDFEA008010FFFFFFEFA018FFA000002006015FFDFFB009FFDFF7FF8014002003004FF
700500600800E001FFB006004FFBFFC00300C01500A00400500901EFFB007FF4FFE010FF0005FF1FF9
002FEEFFA01A01200500CFE4FF400600C011FFDFFFFF5005006006FFCFFA00500D00001201A00B00
1003002002FF5FFB006001FFE00000A01BFF900600CFF801900AFF002FED00AFF00100400400FFF
1FF1008FFAFEB011FFB00600D001FDA00DFF100AFFAFF1FE7008FFD00600701400BFFEFEDFD00
7FF3FF6FF5017018001FEC00D01100B001FFEFFF01700000800100BFEB00000C01BFF601CFF7FEAF
EEFFAFF5FF8FFCFF400D009006001008013000FF2FE7FF0FFDFF8005FF800E00300DFF4FF7FEF014F
F9FF5009FFA005002007FEFFFFFFF6FF400BFF1FF7012FEF00000A01A018002FFB022002FFD005FF6F
FBFE800BFFF00CFF5FF5FF600800EFF9FF200FFF1000FF500EFEAFEBFFA01000C007002010FF7FE0
00AFFE01E005002004FEF00000CFEA006FF3FEDFE5FFC00500B009005FFA00FFDDFFF00B00900C0
06FFA010FF001400E00400EFFFFCFF0FED006FF4FFE028FED00500200C01100800F003FF7FECFF90
12FED00D00AFED008FE300700700100300200BFEAFF601300000400100F007FFFFF8FFF00BFF7FF10
07004004008000FF300100C00A012007014FF6FF7FE6FFE008008016FF301100A014FFEFF4FFB018FE
E00901600300B00A00F001001FE900500D00BFF7009012FF1FEEFEE003FF9FE7000FFBFF2FF7FFE01
1FF3FF3FECFED015007008012FFAFF4FF700000A00802000E00400000CFFBFF7FED01200101200800
900E00F00E000FF7FFF01001C00700D00B009028FF9016FFA005FF5FEEFF7FF9FFBFF3FEFFF2FF400
B01200AFF7000010FEE00700F016FF200AFF0009FF6FED008FF3FFD01B001004003008006FF3012FF3
004FF4FF500700FFFA00500EFEEFF3FEAFF7002FFB00BFFFFEDFF3FF6FEAFFEFFFFEFCFFAFF3013
01BFFB00B01600B006FF400D00200F002FF1FDD005FE5012004FE2007006006004007FF5FF1010FE10
10FF5FF600E002009FF400500CFF6013FFBFF101BFFFFED00500100401A00A016FF9000000FFFFEBF
FAFF300AFF3006FF7000FEAFE3FE5FF602EFECFEC012FF8FF300500700400DFF8FF8FF6007001FFC
FEDFF000FFF3FE900500C000FF9FF6FE8FFEFFA00101701100600B00EFFE00D019FEBFDFFFEFF500
7FF2FFEFF000DFF8FFF006001FE9FFE000FFC00300B00200000A007002FEAFF800100DFE700600000
9005FFEFFE00D001FFAFFB00201100400400E00EFAFF2FF8FFB013002FFB000FF4FF3FF9007FF6FF

F026FEFFFF3FFA01101501AFEEEEFF01AFEA00300EFFF00B01BFF9FD1FEBFF200CFF9FF8FFE011FE
F01000FFF8FFEFFFFEFFFF5015FFFFF8FFB002FFE00FFF900A01301AFFDFF4010FE7006FFBFF8013F
FD001FF500CFF5018FFC01A005FFA00000C00800F00AFF9FF1FF300B006FEA01DFFBFF1FEBFF100
7FFE00DFF800AFF5FEE00D001FFF004005FF700600A002FFBFF3008FF9FE3FF8FEF00800401201000
500AFFAFF9FED00AFF5FF9FF800E000FF000EFF3FF300DDFF7FEEFEA01AFEC00000D002FEAFF3FE
F003FF5FF5FF5FFB006FE900FFFC01D02900100DDFF3FFD001003FF6006009010FFE012016003000007
FF5FEEFF7FF301CFE3FF900500300BFF1013008FF300AFF7FED003FF6FE7001FFEFF7004FEE015007
FFA008FF3FEDFF8FF901400A014004003FFEFEFFFFFDD002008010009FFA00300DFFAFF5008FF5FF
9FE400B022007FEE001000006FEFFFFF600000100D008000009002000002FFF008018009FFEFF800AF
F000B00BFFCFEE012004FFEFFBFEE00D00901400BFFB016FE801B014006017003FF400FFFE010FEC
FFAFFDFAFFCFF9009010008FFC00BFF6FEA00800001000EFF6FFD00F00BFFA00DFF2FFAFFCFF2
00200000D007FF700CFFCFFCFEFFFDFE3000FF100700B00D007FF7FFDFF6FFA007FFD009FFD0
01004FF2FE3008004FF4FE9011FF300AFFB019FF9011FF6FF9004FFFFE4FF500CFE1FEEFFBFF9FFD
011010FDF00100200700D004001FFB008FFBFF3004FEFFEC005FF8FF7007006FF9FFC00AFF013FEA
FF9019002FF900BFFDFF0FECFF800300F00F006019FF1002FE7FF3FF1007FFF00501EFF2FF901EFFD
FFC00A013005FF0FFCFEFFF9FF3003FECFEAFEA01BFF100501F00200DFFFFFFFED00AFF8FF5009
FECFF5FFB01000500901AFFE012FFA00701801101AFFFFFE000FE8014000001003007000006FF90050
03FFE005FFE007016FF8002FEDFEE018FF9FF3008FFD002007FFFFDFFBFF7FFE019FFE003005FFC
FF8FEAFF3008FF1FEE01B00AFFDFFA008FF7FFC003FF4002002FF2FF8008FEF00C00E00DFF4007F
FB00A003FFB013FFEFFFFF400400DFE5014008011FF0FFEFF9FE9FFEFFA008FF1FFBFF3004FE9FF0
004008FEB001000FEEFFAFFE022FFEFE600800CFFCFF901802000BFE8002001010003FFEFDFFFFFD
F000FEFFFC00400001E00501000DFF2005FEA012FDEFF0FFD00900000E002FFB004007FF0FFCFE2F
F2FF6FFAFFA016006016FEE019FFCFF0FF100C00600D00001000F00CFDD009FF8FFAFEB01200BFF
BFF3FF3FF0002000FFC005FFBFF7FF4008008FF9FF1FF7FE4001008001006009010FF9002009FFBFF6
001FFAFFEFFF009FF4FE900A001FFA006006009FEFFFE003004002010FFB014FF7FFBFFA00701000
E00D003FF9014010FFEFF800AFFEFFF01200FFF8006FECFF0FE8FF6005FF5FFAFE8001017FFF00200
3FF2FFC007010001FE9007FF9FF0FE32EEC739741AD044C0EFA17F2DFA4A421B2DEB7EBB948192
654FBDF46501B808784FFF9C7653B33669AAA91AFF6028420DDAC28C19606C00E3583DBA2420122
73A8C65D2313370D4142F3FF8AE0E7C0D2E7BBFB9BB9D70710EEC3E28388638730FC4B204595350
A915E66C3CE265E02CB19A590B7DBEA38D11071AFAFE04732921BE6D32FEA819051213F7DFB35
A99E7C13811062F82DFBDEEDF8239167CA243FEBBC021DF68F73E527BC50BD282BFB73FA43D8132
80B3772DE9FEBBC3F9874B819739AD8012A08DE41837E93938FAE4AC867CDCBEBD80C13F640DC
270C30DCB0883855A04CA2EC7CE825F6528402A2315619CBB2BFC307E8A1F04D0492D9A191225E
AC5C91A21323FBCC9FF148E465094FA1361FD4C3599662FD767BF6052743C961ECAFF67C778C370
81C88EF6478B0210E3CFFE02064A49FB20A1E4C9A63899A58D4B2B5EEBB3067165068ED581F1E70
8E22648225B8051C30286FFBE987843D3F37D93CCEC508837C837F86313D1910A901109B19603F869
87014D5BB313608F1A44022BD4B22BC0961BF3E87913CD7A7DFB08CE7DEB53EA3B64553899BBE
7AF4418F244C03E57C9F3B6E8C8C9F9A10D1218D9C6849E03B534054BF3CD3B9E84F91207089784
A8EA3561FCFB145001CB044D1DA42EE748B3C5F792FC4960310B687843CD905E2F406E97A9DC14
0B6A35CBD5C4B821B42F2870B72A02B81F876EF9B1AA432941B689B7171E701C64C8A1E2658497B
87542E9014064424BDDDB2C7181D86B08B1FA62E54BE140FF6F9D75ECDBF8D1EFA46C061DDBEEF
16D45A33FB0D158B5E40460B590BAFFED95D94F1EC18DDD67069D576A53946AC195145CFA923D
A173202EB44B251C4658147AE06DC0A3680BAF6C2085904B51AE55395C9A248EFAF82EA34126F81
89C73E943F43D7FCEFF628F94DFF71C656C40325D5A42C948ACAC512A1030E6796484C0345817711
B61A7FBD6185CEB2AEEE50FCBE9DBFA9F90072FA8064F8CD1D31B89B50C0C39308332FFAE3CD
58A60F9D88690BFCD08A482AFE1B0311F2FD9E01ECEC17C378C43E80D7FD4462B3F6FB650A5025

A7CBEF740229301B1975BBBCD1F0349926C781247BE3C2647EC8F8B32F5BFFB476F58967164A6CB9
201CFCD0BC1B41FBFE6B1FF48BE1440D3F6A0CA12779E0AA42029F01E8709705394130E1DC80A2
E347360874C564002342A53DA84072D1167027F9E955D4B80FDC00F1BE0EAF9AEE96C48387DF9A3
33118B8CC4D203D0EA5C89E3F40CC22E6EB72F1A2003306239C194704105F767416C2E9059A01F2E
0914C675784FC77CA6E8A26D98CE7FF4FC6CB51A1696817B2A3CC85212219568507C331FE70A907
44879E30A8D0DE0ED33200040D8E097963AB0D58763737ED1685043941D72FC291698F1A441DBAD
A40F5F17CA33EDBAF2D917E3CCCE5AB05D42874857DD846CD83F1267FD4114806C9C3F517C24E
DBE19633DD8E4DDCBF265C563AEAED4F8B26BD00ED278181779559EF1AF49C7A8D746E3F9CFC
C7064FE7A38BFCAAFE69126C0EFBE17AB1B6DE54030BE1E514F270F08F3FB64702F22E46A8CF62
CE57EDDFB51DCCF4BA06110351A8F18D079993B9E1C2F5CA704BA3BC7096C15E4FDFE42004104
EEC50C664E18D6146B1FFD47E5275191126762A3B451C25EDA8D7CF4841DFC3B72306E94443AEB
9983EDF603CBF44865851DB397CC14EDADB62E06260D0EC1D3A2DA3D0778BA2B852029F426B09
9D0001E7A24EC5E7E4B34962A1FDA1F5A98F3C7499B2DCF76D0274813FFB5B1F6491DE384111B4
733DF199503B9A5935715706EC8BDF0424B7CCF19CDBF0C246F17353E1451391E7E6CD17FEF441F
FA63E9FA06E083AD08CDC63A8C72E05ED75FDA2320391E666243300469B186376654602E6EB5191
5666AFD327F175D9077F3809B682C6BEC54E3A9C24B84CFF11619CF0FE3BC147EF65B9040CF8C38
A9C31EA94F70C62A6AD42B8D69C7CD8CE3FF23DD4EAF07AE23D35AD20600FE7DDE5ADC2EBB
020A904C8BB8CF2A55624ABC4A5A8D91D07F1E8B9D6F9D03F5E20439C7BC5DBB0CD0897F803C
A5CD49DA9AE222F8BBC5E7D195A6DB443FC054FA410516B4047E01870C78BFD753E21BC8F6040
5F3860A1C84826E61EFF244DB004A8CD8D2FDE0C2EA9AE83E35D6A91FB249CC36FC2C6FFE335A
9B3ADF1820F778E57BF79B1D460513BBB11E771E93677A37EF582F9E364A1B8A37468317B12B19C
BE081C1C2CAEFA46740784F61E5A256715EF7CE0E2576135A98F946277C0D717CA1F28355D9BF53
0C6519A8CBFFBF1D9080B79628646FE83AC24FF131F81A5DC0813CC19C80F1979F3FD53781797F7
B063AAEA2E67C1B8CEDA61BCD389800826BC55DDD2CB88828CF1DBDCAFD86D7674D205AD7
DA8F8591B5EDC562F32ED74A380F0006B43A1B0F4EF493DA3162EB96C1340218273C077C1924843
61DA8C2B037834BCA69D6AE0968E2D70CE15FDE4CCED1AF23021FDA8E28389C9AC250F9487CA
6095EB44EEF8F30F25F76ADC2633EE3D1B607E8341E51BC9924B7FF89324CDF56CE041E69438D21
96672AC39144874D44A5B32A9E442F10CFC7DED7DCEB7718276B29CA66404664FDC77C7D2E
5BFBF409EB1FF4C261FF05E74503C259A0AEDCA743C8ABF7C105E2BF3B5C3F62F3E1AD8C1B019
9712FF76AE0B7E339F10CA8EF15FB5613DF8E0C781B95C832E3483C6CA904DF2F6197B1F87C080C
F80423F91CE461829F8999BF8DD9536BAF0831BAF74E1271B0608EAA53750D6D178442A89DCCED
EE3F53AB60A63E773A7865BCB3EA03A0225039808CC4E5BA18D15A165EB62F86CFC3A2518B74B
28FFEE9D34AF2FE72027BE4BD6DC8B64FE0226C39E9A6F086359D5D85F29217A12763219CFBD54
07217872958509136180A6FBC24582816B7E38974C89BB1B12B07A1AFC01D9839C51E13498CE31D6
96310CEEEDC42F401E8D7AF844833F0E6F10D71098564B85E0AC632A97E938314A6BEEE5C492293
57FF9E7539688CD88D0001FB6E68181DE7C3CA22523949FB614DE55EFC3ED1C0AF7741746382CA8
692778ED43A6C4D7B09A398EA09B12B40880FA84896277AE099673DE368CC4D98F62BFDC9755339
600D28498D04E8918BD023A005693A6EDC5330C5E8E018654A0FE23411E40DFEEDA0319E72B1D5
8455BC2B52F97410C44B4638CA33711E8C4B2EC18F8321F87FC888F4267BBE2FB8CB8311CB79FFE
C560B887A79D8F0D099384E83B5CDE277127CE84E7882BB8A51230F4DABC6183222FF8B73ED302
05B95D0AC379BFF6E685FFBF4E595FA8C6B607DA3875652981124413EE21076F5F4B3BAD8428C19
6BE0C66FCA45E276F08A03854457EEE2EFB86CC37FCF0A3DE0E5CCF00282FC23370835E4C7318F4
71B389A89823F66BCB19DF1E1F098AC3D530509EC40B75DE034B0FE67C3332F5E211BD09D0D250
15ABEBCE0DFB5FC82FC8FFE82F25D848901F3DEDA8F96CFA61C3ECA5A5B1B0918384B60DBDF
40DA182B97C7EA1F933765158A806BFDA8A08D480C6BD8C1903C05528D5C4FEF669EDCB438AA8
0C4D1D4F1211811674A3343A84279318B6306FE6F8A1DFBE462471C5D08C0D2D0630F468039B138

ADF5843E517C1683BC6EC944FD5FC28318628301868DF9D8B102C8B9C028998D2B23CE182A32960
2592064300E388312732F1ECF0C39438FC4CE9C93B9EF3B22EF89463DCA80323C08A5B77BE9AE2A
30AACF9CDE18C716FD4468FB924F13918E23AF704F9A0605F8FCB9620358A2E46938A3AD9DBD52
19FFFE0DC7187B3AEC96292B5FD5BE41CACAE0E4F089217EAEBE5D964298A4E3A4A722F265EC8
63181C7DD8B7954FBFAD80B063D891FD5325D1C9BF3B2B1DA671A5260CBE7919BA7F7F97857DC
D7E7C893B431D65FF97274C87862BA5F335A61A89D0DC64208F28F9DF778DC1B386C583E1980786
A506F341365A1371986068DE58F319BD6603010FDA773A9788ABCEC4C25C8DD985953D12AD974E
6E6BCCB9A5B9024E66FCCEFB018CACCE366284926287EED08A6398FADD1B23601907571C8D2BA
EA9CAA1671D0E91FD FEED4D601C1004006CEBD75168AC3DDDD8DBBF5DC658D8E6905607E098
7F60B90440E88EA62126B7E7039C3DF9FB93397A607DA0F3288E4003FB9DDA3CE2C818A5C79EE0
7A70FC98FDFB38A07866F46E7305B0267D28C31136EE8808BDBC43C37D03AB3CC4072CE3D67816
34BE85CD450D127218880D6659136927315F491C23682BB1168246677812111B2748E8D071B73DDB6
097580BA3FC6092024F6D1C3ACC68E705269A1885D1A20B0CDA185D38786E3FC4D9ED19B58C9D
B936016667736756CC09D81B937E4D43249E54727D2F4264B02C4BE0FCB69163B44C8213FDD626D
C03EF7879522C8013A2C6A7BD73DE749F8173805B87CBE9306C878971A1880E228220E8B660253AC
530F8EA7816FDD881E3F994F0496C4E69D2329961CCE5A6905139EF8E1C42763CCADB0F9E2CF9A
86812DFC731AA7719DB96042C87660C07F5B85892C0D767BC79300CAD0790FFC2CED2257A7FE73
933A82C6599270A15F4EC7665DACBAFBB9EE7A7BA4A15E33C9C04863C6C7B2970A10F5CD87108
69C0203B0E2DB476C3163E1167479B8845BC6F2CE68B9E7B272367C3BD5FC7B3C4CA558229A0015
C4128447C8F04684D68623002D9F011071E3B5CFD500F59D9E9905903C476C90915AB0A7B18460F22
5FA850466A5757E6037050C458408F9F4B19F65BB11854B7DC1B9B24736162E6C1F23E806F85E5B13
BE9D0E1F234E519AC20D398AF70E1E3E28EE19AEEE2B0F81F3E9C8648B2FCA99A2DECC29A62C
BFAAB60FA5E89D7EA7076C9DFCB3BB482330BADE056216738CD4C49FDE3022DC51AF14E0F190
02E3BB0D424A241B7EDFA968DD73128C74D8D71814D74A0D5ED5F0D1DF78BC799E6F8EF59D712
5703E7895C11238339495865E3D04D6089334A87B438FE188EC88C0DA37D42881603748A40335930F
21B88F46BC0DA315BB671CC103C160CDE561DC241F501E0DFCAC8A03BFC7180A2D623E6FC5049
5F4F4CC95002EC385E239E100F65ABFA5A83FFF1113EDC2C65C63AA51ECF732C8EF104BE2E0877
3BBBDAC78E6CCDB01C0F342698ABEE17D802AF1DF191DCD3939A964641623FC204F8E6874D464
28524860829373919B9DFBE2EF2F01891ABB11F7E981B4ECA9B4D10E02E4AD3AEE673D3373571B
802A49C3AAC3EAE81039BC592110CF515048B22FA569307E0E1EF70F66DE1D2271C46B66FDEF70
E4CB9A0DFC0DE33A6FC66710A42F73F77B3D569781B90B7DFDBE1D26887389C00C5A40108D34F
DA1FFEAF14198C84D8D424ADE0C76FE63689BE7E31D9AE7F3B2FF1F0F2EC0C9BAE8D6A9DFC
C10365FFEF8B021405F1561B86F16F6343A06C770CB8FBD7AE253B0211B8BEC41A9EAF34A6C260
450631E7F8F804A047E9198B6F4D6952BD00D1D00DE58C9A687967C055FD34993107FEFC7E09F740
0AA645DDE1E2B22D6046E38A11F065395382A9010CFA5AEC2969A4A381081696D7896107B5B1603
240677693AC5E046CB021FF14588C33085F121A3F60F8097E0CA438F3FED7BC86A4EE638D1F641F5
A87A3E90D63825EB72D3887142250CDE33B83DE0B041BFC0406AD6026EDABF8BA88C83DA3D53
5F777FCFF45A431F136D814A3EC91FDE4110C0F838B9A9EA9106F48473FCA638DA30E1C792BFAD
206F00EAAFE114097943BE5F11CFAFAF4357860405A914CEBAF800BBEEB2E780549A89D4F72C42
F3A47D49849E04095F08F4C8E99F20E2D981FDC06A58EEA402D119684C1043EB15261E2F9027C398
4E10671250432A103D236B9BA93D9EC0972F4D8E423335C45521C53210470F7AE24AEDB4CE40838
433C56EC62E181470733C9E5E568765F0BA54161320E779D88154985DE2DE675678A162566103C096
4C497AE2883C8590B1BC6453A9210D7A84AFFFE7BD6F7DF1D17A53C2103F7FFF7701088B5EC6B3
7E3781E587A4EDB9173816FA27024023214E24A87BEBF853AB2317C2798B88887E1CFE806FACFB7
40C1B2207F37EE9F2BCBAE8F125B18198F80C2A1920329FB7C374BE332CB90755C4F7EFC21A2FB2
E460C8319FF6D0BC877C688A03D4063ADD96A14080FCC221151FDD0376B5C1852A133D8802A109

CE74A90B6C0848479563E4BD364A0A5D6C720C77388B1144BF3F4A6AED86868F864FBE3F2455578
176AC6B39ABC271BA38C0B08192022817BDC8A29728A8BBC4E168ACBC11B4CC62A90652C538
10485AC9A0A8CAFE5AB68B25D0B52A6E15C2DEAA9755E0A8D86DEF601FA1E9B05D372D0151B
A806E5A680DB5D238E96F839352B6A7930E7CBCBE1B7766A5704C975B587798930512C13B6138845
1DA7705AF69F5E81D96C294338269617029A1925479ECB631BA846D377356AB04F57A528B5B11160
4E173B062B5573BE62E96E53916DBC0724259F9AAB0DFBB6F88A29B5A22A02E63A6B53CEEDB7A
E8718DF037D852261865358B6ECFE602273BEBCE28D2280E776F64AC045730C1D60E2F853F2131E0
3A7DD17BE55418519367F8A8F942D26EDC9C7445DDCDD93517B7940751FC118B7C0CB5489DAED
FD47C00E62DC79C6459D48554D124720E6E47B80D32846C45022C02D670399DE532A89FFC78701C
3B18F6DE44145E99871340BB8FF515E64F05686008CEDE77A5F6628C45A34AEE2B3A2354D71BD3
DF5D44FB561ED236D23A22741E056F5ABC9B4E83D8EA52E4C2BBF54BF018044CFC2CDF6CC2C2
8EBDC2626495BE821C8FDAEB26E08ECB1B877FE5AB5B1A3E9DCDF9CE788D72D49B177E2C2D5
F571D27F886923ECA50E45EBE6707A7362251CD579EE56E0DBD8152CCB94B0BBE442071A2AA0B
5E959DF4A0B9006E5BD9E1337B78603C268C81C7EE3132C969F2FFEB628D8CA334DD34EF624D7D
B536D79ABDADF3A4C335054CEF00DABE90DBF2B77D699908A2A5BB8A32F12A4483C90A43D653
A6ECA694743B9E5E1D4FD7AB42A4A3B00AEE1123DB9879BF5AD883AE50A341451637C622B7B97
E7880E72BEB46691535F090A9C47EA78EA696F1B7F3E423D15CCF91EC0B44E8EF4AE1D4065282A
CE10F898ECF0FC86EC3CBB8CE3FDF50A82FF01BA8B09EC5C3904D23A83F06B99C2C7EEEFEB8F
E1B3C3FCC4B4AB2715E32C22E3A015CBFD15F99902F030696C6D98EFEF66FBECD77B75D26B44F
A1C86D672C78A85ACF37CE5897124A56C1F34E74563850452C04BA941B41B70CDDDB9668D5E8BC4
BDD10E0CC19D28F93D3FADFEEA95A77AC3668E675D0139637FAF0A23E6B73C0DEE2B499E7892
16FDCF3E61AEDACA0EC00AF51B6F3E966A4460A0DF380F44AFD5FE707BDB19CCD2E748482C5E
3A0F67D0B3D6A64D95E6D94DAB4D90270C3F2EF5D2332F0A17767B402EAEDDEACB8D575485C3
CBF3B56E0E0FF732D68A595888D200377A2FC589D47F0E18308FA2CEDD436BE0CB8B2A38D93C8
A0351205A560B8E6517D396730F1A6398A4A301E886A7DD898688ACEC3636BDDDB680DBB9950758
2FCA4E60D18F70FDB6966E1B9DF30AEC316E78DD81FA017ACFE0432D2F556151BC4EA9EE5D433
1CAF8976D18CAE683EE9279AE222B2049278E61FC4222463272AECF9B8D53BA1EC01FA28951FA0
740E5CA1C2D0BEB78F0A1BFE12C8ECF18AA35F39EC36DEDB664EEB0D0171624D15AF9814973C
4C0273BB557BC05C527C3C6FAF6FA1F5EA0D30E776FDB89436AEBD1A7550F216B7CC78E6FD81
19A3524D0BFB445FA09DE7D6F5FAB1202EF1345C0A012E6D7B01150BC3F767748E84F9F3B0BAE0
DD6FF25339989EBBE6B27E4A2CF743170979D244807C6262110D8BF23605FA9EC74E7F69E206CC7
3001697A79F4AF5C0EC00A2E30167C8AC392663B522EFF48B2FE521433E9EFBAF61BA96D13FC
C69AFB2F0B5B64BC8AD09E8E797CC540FA02D510AB00029A86E701AE2457B32CED213B2FDEB1
26DA522CB6C258FCDD2988011C42CE704F967B73A10A6E9DA0A684F28A262ADA1EB1BC02C5D9
5B7235C00AB11C3AC4239065311E7A8976F0872ABC21DBD06085D219212DE291F7EE5DCC6A5C6
E5028D1A04442A83F53B3BFBBA39A7C5C4A2777F31C874518D72A39BF4DD4021971C585E587275
E65589A86F0647A2F089D396046B6DBFF292E9E26D6D07BF354DE03F923D62802E3112FBFD910190
226CBA1746C10CAC2FB2A5127F9597FB8AAEDA4B2F959B39236EEFC4553D47127D63726FF04A5
B2F50926BDC2FD9761F80CD79AC526C0C5D1FC948C463E1C91EF9975289D1FF768D0DF6E629648
A550AEFD87992A70A9E93EFCA8B8423062AE389DF3F8F86DCAB5CA25BF7831AB1D0E8484D1189
A0A032EA7DF8B8A446F2A04CE63A4088077C46560B450CC14AEC244B5CC0FD8A9546E180C4F60
26491021FEBE80E376D444B8DA2211210E65414981D48417ADB21C06B4B32877E414B8016ECBBF6
F95545AF9B6E6F806AFE9A9D48E9AA378CC1E5AE9ED5F01F1892C0949D68F4838A77BA131704939
5B230D44BA2FA403C0C4C63C091C4683CC138CF499751F1520B8C81FF3D98E419317B99A43414136
32DF1D86B48BC0A8DEED7186C6E225EB3ADA8002C40F32A2991B49C039E5BE20B626705D3C5A5
A502B88D2B2E90B4117A0FBAEB6691802EFE52F4858EE544750EC4DF8122E9880D2A2CDFD735A

B0CE15B6BF968282EECCCC7636179E64852C3C48E03D95F164A243A6FD6A37B984018C4F9D7A6D
086E46A619539025217ADCB842D68DAC0B485021FEF4A62E817AA3DA68D5493843365AC838035E
627399E29BE49F6D082A45F96F092ECC5C6AC5A4ACE980BD7D82B9DC73531DA559003AA5C809B
43FF4C4AFD2E72D44649B160A1DA52D0B56B62121B84E25425A870959A9537D0FFA3C9E835F70A
B2496DCD9F55C9C6EEF91E7247093482E5168C94CF5997D2E4642BC4D029F58293D29A7DB64FD26
7612FAB7B2286939DAB844A7EEF1B8E814915ADBE7E6871A182FB6AA945395198B4F31F156206A
22CF67CDED5A93C75CE54F7F53EEFB5D418F142B8234BB814F09F079456CA0F9A7132FAA1E9C4C
17B7CD4D5F12046097AABE785F3C9A4BC1256E8896F57681E801286B09212EA9866C8DCB136AF28
031480610EBE4842BD220E6E5225E7E4D5272C448CEB72A482F8107DE6DE783B16732523C64A20E
D178077A275B782B8214EF52F2DD0BDDBE939473458EB80245838D04207B608AC338C8D6EB83845
2118D4DD8A2D769689DFC12A80CEDFB53338CCD71297B242A9085B2C61B02C176AA60CE5A05E7
33124DE0AEA4D92FCBF7BBD8EFE703C1C6E94C6A15A73362447CF56940A459E8A5AA6FE04D2E
DAAEC50938E31677C519DC095AA47A676D3E2885648E3C0C2ACA3FA252575209B005B28AE7F13
CE074281716E21013BF9370C80D1C96ED667C751DD394AFD08ABC1C85FD81AC9982D922635458B
A72BEB88EB4768C40B21B2078E63F9A2361B8473379201FEFE56C3A49E19088C515433D179D4E37
D4CB2C25A911EF8A500BBA444BC5FC9EF7BBC2DFAF5D263A601C672798B8E60326BCE78A1EEF
4ED8877CB9B61A31D80E8355A96F8B267AA6F796B6F9FB7F44C9419F5BBBE9953CC1A7DFF0AF0
4FB3D6B54246C1301D23578D225C73952E0AB573768C15D82C25BE70CCCB163696596415B3A4E3E
96CB4CAD58334945A7753D2793036A3A3ED45560E18D59F504F3FC49B47F5939FD9A259EB6D0EB
B767563E75025C51DAAE3BB353AA6427202166B7801AA40D86E3CE11B607946805A626189915AE2
C112CA31FABE0D4372940AD23006EBF46E433E5154571D74BA1A0C02D195A4250227305B948ED1
4798BD6A4369384E210247010A83150EE891469A16CF3278DFF67AEE9A26C3C7EAEDB17AABF00A
021B368168FA6775316C128F8262577DB7586437D2C531FD03A23F5BABBE8D03690997FD43A28816
2C7B89111F2ED58BAC3CFA8773E6D0B6148D901CFCC9F79708F3CA96F079B101311EA6F88D205D
E884DF6C0527B825F4D742294AF3747A6904E47097D48F9F79A7661010E066F66C82CDC9B31A3F66
239022253E2C805E27037B3FB2C7595FD101ABACF53E07E349BF73AD541EFAAC79E543AB5127B9
6F6BF3B74E91940E810CE460B6DC20A14813AA0B1A17F810AB1DB16ECC4D81A93BCD1C532C359
A26829F58BCFCE27C306AA3CD8706E9FFE41C25BDAF527FDB422A7BDF5268F40B263D16358AB4
F824CE8F279D051C39EB6D4FBBBF6C89A9C1AFD7FE7DFD29612A911B6DEC04F67624D0BD91F7
3A4A726F37A19F40D751622076C09BBABE51B0B87CD595314DC0A52BAE887500BF24AD9B79C2F
DC8310E8A305B22E36CAF35D85E9C3E675051921C0A96FA08FF6F759C6834241C873609C05128929
EB59B330C11B807D26EF4A

ct =

FB5F976EFC220EDD7DFC77CA2FBD07233D03895D0ECF1EE0ACD596FF75924BDFC059F5A7FE37
9E67660062F6EE5505665C05E0C5A4304546948B5612347AF6963BA03F5674AEFEDDEC07083A5F66
3E31BB7E3BC44B6037EC91E4F833D5ED4F289455397264D9EA3DD32641957F3562665FF3061557AE
5D498862404B8D1B93815753F834AA0D2A273F865923B5EF71580C679F1511740A3F9BB41C758EF5
94FD2089665CA9C83A42D385E70C4A525C1957A2FF306C3ECD021DE5EAE2022C4E8A851C748344
DA0B5823CA87374B36317809924C8850A59EC71F36FDCD42C2D24E400F3A5737D041850F103618B
E833A91C721C4E6086A5FB6F50F9E001136478518323086DCEE70723267E2354CFA8A0614D265C28
81471B61776120ED73FA6861548D8919A2003D7CEB60188CD5A280D05A4BD5097786BB00420749F
1157FC3072F62FEC6F0970F33B61C3A6995F59EEA1D5D8E90E5ED21683503127ACA0282087D15D8
B925AD410443C02B29C47A738C5D25FC1FE95E0056AFF3004A31D3E951BB7FC0467543CEBE0A3C
7B4F787A5C39885DBB3EAC24A189AEDF01C3C849F3017E3FCE5ACDFBDAB2D39557144391066B8
1EDD6B66231532F32279C88E0BD80074D5D806A69934694854AE9C22004A8EE0997DE04A1123291F
F06DAA5DD00DED3456B1ED504C0735E59986E7AC2FC6F67F7301E463D8FD23A348736DBE152DA

5D44054874602D57B8CE1A78234B2A3F0F6C720F97B5FFB59552596F2811B15CE74377F91D46336A
DFC3C25A5FF91D8695BD57A0330EBC7C24A745812206277342B82F73F1763411B61B291F6419D337
64013AE2CD29801833419152B4165BB493483C0CB421AF7079EB8E9771A4271C733A4C12BAD25C7
74DB0F1F65D596FB2335FA11C312041A118E04572245450A6806EE80920E7DD3F7F5192A3027A361
53D538E86D34CA7E91E2E209E21D38DC45A066F972A7F803E639E4F194C90D0A602D55546079693
E311C3253D16235409A4141D61FC3920A6FCC9801D75F4E025D30308262C701FF6007B96F05197330
A1F41511B0CB6AC9562BC1350C2EAC8EA2D47B87B2AFA9B06E48739C06C63A77F9179234B07214
E7D28D4B90136F4365405AF90E0D99BEF1F323909D123F8F3D486235348E91711F0BE27195E30D89
30519E71845A0284031C2AFD531CDC6B684ABA4B7A4197D509D1562B5BF1A273DAD9FF8812FDA
7C50595EE246EC9B2402073FDA31D5B76BC1F0189CA860522977BFD12E43A6D0621C60A9B5409D
A869E6C79E27C31EDE953C732158811D23C919BD72F85A3E460038FF6BD6F8198ACA65123F5993A
212C0554841E17A07342C4AD85085C7939566B8AFEC4B229C9E8718BB63843BDE308ED31992D3FB
3AC18D1A87DBC527F5854641DA371CD14A565B025AB900815A43347D9D0D3570211952F87451BD
58B7C32A76E00A4F0F1DC6BD018230732A34BDD8FF5650163038F0792325D0574F1E56B90F564072
4E9183B66D3E1397005D858426B624369A9472C63BB1B37962A1ACE4165521955722620EC6174B2D
98FC4767E1EC0F85250648D233D5D0560D26BDE35790F446786CD28E9E64E6E5682E06DC26CB48D
86BA40A5E1366402481295A4F455DEB510362327E9775E348A0C5B3687B7D78989AE1C6DAC6A19F
4DEA7C9AC21FF060EA6710769E928F24481A0CB9654063FCA1F960602A182C487123ED792806A07
B75F4BD8266194AF2C7029349EFD73E9BCB1124809F8F11266EAD0A333BEE605384C2BF7F65CF30
7C9E399935A7F31CCB996D5B36DFF55636B182E092E7934F6298555A0D5FA3E154478DBDB2AC0A
6B1D70B2EE255139901547A93252B9E75074D40F9A96FEA69E294CAEF90704EAA82EC70E5381F41
77A3BDD6266C2CB01658DA55C532478E76277418DE7E26F424A2613559D6F9B95D75E30D2F60E94
E53E754B2DE003CEB2C2205E6319E31761D78B24C23FD9E6844987D767448D21811B0D33087601C
AA1E44136527E5F2A789C237BE17E9851000F9AE75EA013F879369EF596FDA78E8138EBB63F95F75
B790891C2DE31D2FBBBC34265689DDE506CFB220C9192B4D467F134A2DA3039593CE11EFA98942
918CC613465AAC77F08C4AF0820B2C40CA582636338D3A1B942EB1CEB1D33C5F084D8D45C39C4
5A802798586A001EDB3312B211B6553293385E96FCC1047392E56D3940A461DD355DCBC44061760B
1F94776DD37988DC25943B3F4252E6D12378D3FE7311FD6D22CDDDB9AF447BDFEAD43A71662E4
E5CC14478146D777519BB75D37411F227808894F3863645DEA1C686D60A5BF22D40A58E3A6055D2
D425E43793781B21C1B5DE7528DA7AE8BC60F59198C18B95B4080F15644F8667274CDC82F5E7BDF
F1D227944A698CF1E943082232CA4809FC16E197C2E6CE0F0778DE5D36EDF7FB6762F3E02D51C55
A4D6D088310E7C95161D48B1B0002F9F50D25BBF50ACAFE5088BE321D3A79758F00043FA530B53
28C8DB354D0BB01C06FC611881424AD7C797755D0148008CC4DAD9176344AC3772E8C8E331A1F3
9D67705E0AF33750CDC02BEA678C377CA0E9D62675639D24D4E0DA2108AF448A7513860E6A0C8
BB06B1104B044087B69C79F524FEB7D42BF633B102DCDF625A32A068F3D5DABFCD5E3B3A9A5
B15DAB90AB36EF7F55F62405D62761E10D1F4E09B06BCAB57372B11848493497614F0D12C908063
F97638528F72964F6053051928013071D644E77A76A7EE0216644B0CF88574DC05C7C80195F062F381
CA540B83F89B72AC2326071C3FEB9A22376C39D42B472CBB6186D58485463C1707357F818877A308
16E91446464F93C6E128CF7A8CAAB73065FBB3901AC273C4246384B586346FBF226682E91D15388F
2B084150D0A7454A0F5578A004E7E4CF1940A9D60326E1E1AA6B189440E7CC587A3D0F89E4F0B41
AE61A26214C7113117377771F0A31522660B065FC8D02A8FB0300AD5F5BBA01EEFD7E1034ADD39
95848D234570A6906C420D0DFA7A645FC04AB7E6B5C9422EAD3A73B7360A4A6D029AA13002BEE
6AC440B226C22906FDC2A1F17779EEAD4AB5A2B2E52DB48FC595B58A71F8E97BD8C809CFC0440
7949F08B76CADACB380404A24D259D04DFCF51C44018378680D7E67C6B22A8D3F451039E8183E7
DC87DF042D9B54FE811794ACF8AA5F17CEF89A2361498387253DB6FA921449F182864457F4D5F81

64DAE7244895F621C9D224B4C00D86192C9D0C6916A07BC0A9B3E45A50775FBD6D9987CF8934D4
5880E551DD234820916B51E9D736E9B28306FE3ACFBA63CE5AC0B051119

ss_enc =
4297531ED1788B9BF89FF08A9CA74642701EFC20BE658DAFE997A4590D4F192A9F705341F8D1A8A
25D6893E9B149B6C0

ss_dec =
4297531ED1788B9BF89FF08A9CA74642701EFC20BE658DAFE997A4590D4F192A9F705341F8D1A8A
25D6893E9B149B6C0

result = PASS

case = 2

pk_len = 3009

sk_len = 15617

ct_len = 2353

ss_len = 48

pk =
BB6A10549913FD73A7B20915218D1F8DA1781B942D004DDA95E89AB666B27439A1682878E6B4D9
97D96AC164DA8222B59D6DC4113EB3BB0FA6EBA AFC154369B417173C4615539F81BE4CAA27BF
AC0A28641331779DBE67DEC432FC08A163E93AA2BBE279B9B74775758557515C94B35856FFC93F1
99EC6410A5327CAB13B77D68B3F162386791F05EFC A94CD03AABA2005FC39C48087905EF7FF1134
1C84F78608878191B6E3C5AC7E4D676D88137A50B5DA641EEC9C882E147BD768CC371AFF11F716
C9EF07BBE5E4C872C9C724C8B0619E559CDE4FEDF62AF101786E2716E4D8348FE96CFE697A4A81
3538AF189AFB5BB13E838FADA9E676A3077FB4B58C39B7FA749725A2BFFD82F544351D607795B8
3B45654EE9F597E1CDC06F9C38E16BFB26C1256B1282B40431564DF229576C4BB7978E0AF58E8A5
7315E7340B7D9768EBD0F8086A412819E40B510F9C2DE19A13B6F34458211B54D4F05AB9CA79606
1C6CDEB8ED790FC0CACD3EC84425E636D9660041A8DF0313CC3A4E81870353C82931ED95D28AA
8D2BC09BB9142DDA0C94953662E3C81F05D96814DBA893AE2BBCAC54D81EF6844F225F3517542
AA1ECD80362098AAFF4409A3F2BE8E13BD15052C0CA4A824842D7A7C6F72BA93373266A324EE8
C309F323DD96EE92602D4549D325192002993C1AE31F19057B045DC07C2EF0BF285EF5214C70AE3
A18F26A700747160AB01142B5B036BAE0391F0DAAD7D296A7F9CA0B8472ACBD3F52C17751FC91
0A8EE67FA5E26CAA56ACB560482BA97E0B892FA690C53B39D7DBA913952060B866165314B06FF8
6EBE70DCACA86107950AC7B6FE65674305FF009942116A5069C12A8BDBE189E6BC7CB798523862
8E5BED4669E3AADEB1FC56184B4D52C3FE94439EA993F44DFAAE761BB1929F6C02A8BDB67FD9
AD20750D90B64A213252C1ED0EBA9030C1423461BDA4C8BB90E7B0D7FE272745EEB1FFDB83347
3134ECC3B4A71D949413D4DB0F35BFF937EE0520499C6E6F113FA491933684F5EDACE189F43FB61
5F881926768EC822B142E192D203AE303F0E9932BEACE33170C12D845D153948B6EC454A5E094797
1B8FD31F32CFAB8E3B39B431C9C15B7A9A48519CC5A0CDFA9AD08D135E99D41E721FC563A104
BB657B8D7A8F978B3ED4B80F0F2DAA4BD46632CCBF8667E8FE79B59790D31827AF2FD51CFA4E7
889D14E03A96165400374856BF23BA7426175164AA80B4882555DE748E044F31AD9512EE933DB302

3F883D9ABBBEFD98EDC10520D3BD020C8F2EECEFEA37729C7F8C64CB0E625AACD9D0A3A9165
7E12D251C4A8A0E5D012E2BFB4DBCCAC21888125DEEE369980333B37CEC90945638ED1632C73FF
3D70B6B48C3E99A2E27F4DF8A58A81DC0C52DC056A7420E7291C005B5CC77540FDCC995F042178
42DC89BDAD1757FBC61D036F46AD7F665F527CD240D14353833FB62F16DD308278E4AAF59A2F7
FC1561D5D3C3E864370F28FBDC66DC4B1972D4FD9EBEC37F65795910D3EDCB567D4FCE51C71C
C8E104FB0ACAD4FE0A3B6A846E2DE156FB360B94D2D55739EB7897D14C2ADCA8739F16E1C32C
EAF59691835018985CA9AFC8534338E1C4FEED593C14D9716312EB1C2E44FB2D23BFE9448C85585
FD026C1C71667AC55383D5D8D128203615591001F420F55680ADB69CEBE136DA94EF254B65767F6
343CC2B4A5750F9A6CA95642D7CC326946CD3E9EC8053C4554E5E030C00200446C3C15F0D2AD37
B61763F6A7FBB2F4130487B35657A37AE17BCED68942C52E437A8B3DB79C42BA544593F14247185
2954B847CF353259C26E15FD25132D114D5BD02DD130EAE930B1DE524CFE56D6EE8839FCD0779A
5C9FE60C51FB09FAE36D33ECAD877F08CB873F2637B06A6D53E9F92A61FF96F66A74E6640A6112
3C21ABCDC55C9FD1135AFA7F084ED5AEE3A0CF82121B76F64311B5F58F06E63D3B3DBDCE0AC2
2E033028A97370E7FCF1CF937EBF0B08F80DDF9143E40293DF40EF4AAC0B655501F5616C80F35362
40E85685E12921EDE481383820A2594E1BF53A236305A83C6FA49F4921035922D4AA8E4AAE28A39
BF7D7E6213523426C77064DAE519C81C854A699AB01C44C2F9A06A6DF21A1D5B73E5B1E1412015
88A1C5D8909B99F2BB0CE9D914B66704DC06D2E68A5AC9BEEFFE7916904EE79703A143A20F130F
56BA65C64FA73A7CC9A174E499D3A8BE0061A88500E969B175FB14315FA5E2E7D6C8D9C8A8EC9
E666000829E96EFFC8A98CF5A3D4725859FE670E0A448F14465B100AFF8C6313EDB304EF904E89A
359526B2DF9E4F4381EEA6AE9966F795B0283713BE797EF9CDEA6FB700A6B2BCD9173C7077B4C9
3C76A4116B226CC2D80428C39E55138B8353003B2BD591504E0938C49EB3D4794DDC23BFD3E5F26
A6B2C5E70AD46ED62F9FB67C1452F810497083A8497AE6B56BB384DBBB1EBCC1F73D9B0E3B990
678F52E75341DB9C87FAAC300A591855D96DC28178DF2AF793940D0460673E0AE32BB2CFBEA085
C5C4BFFE931DCACA4D830F0B36E7355D2407BADDC646813956ED60C4E33E894CC71279BA6D176
3CCDA47A6A173AF914B0E99F1007A22B5161112F573896103A6FC64E0E363FAAF167B4718915364
AD3BE65CD2B8E2F949FAB736936F07B92F1D387F55EB543E10D8CEE54869C13CDEF48A30945542
D0F933AD752F1D54A8171D04D40C88772B6EF3BA43AA0D67EA9F3BB585387EDD885A441F54C26
CE6C439AE0A52EDD2003096AC1E2D5F37A48F3E33AE45D099B3D1DA57D6815FD2425369B8054D
6CD91D2B25E60D9720D66B55D6A0E6D98B2B58F1AF64E1952494C5DCA6B06721F1DBB72D27D92
2DAB23E6C16DCF5976BB0F754705E0508DEA72488E7AD374D1D202BE819F362A9A8808F0B5ADF
8E94954EACD30906A9155BE6F422AB92C6CEFCFB43FF10C69DE99C2C4C9E824B3EBEE97420170
DCCDF32A877BF844A23D45595E8AA9DA52F14EF50CA5CC7AB8A4D1E9C0DC1AB014971E589589
C4996A89BBA2F0CAEFFFEFF87E7F12EC6B55B3C376EC52B6A9063B13140AD2BF7491489AB74A3
66C73EFE17FF894B0FF7F5E177AEB55BDE2493AA590598CA35CCC9147DAF377AB3BD74B3EC1B
3B85A4B8C5A3F0ADB772D3C0BDF183DE89D35DB5B24B1E80148BA76C055A7A2867D5D46A97B4
49B7FDE751699DB148AB67B0FE44543162158FF2C759E4B1EB3C3201AB0CB6A85828266CDFB739
E33A74EDAD64439B0D0A8260D45484923BA23A45BED4E3055F2D83B522E2073D86A7438B41F0D8
0FB26EE414418B198A5E0EB16B38A5C9F7553A3B7B994CD3D80E7A84F7A7BA2AAD739051DD109
9B12960EF43B68EA580B16AF7B38958BEA0A02B23D8B7ED851868EE7DEB6870A6B47854C88AF5
D45DE0E230DF6FB888EDD5E84262CDB4DCD26867A48446B9CFA3EA2A1FDF02EB0BCF7F82F496
BBC97190C0B692D5CBE5E7AFD304696C8D951CB4CDB4F53BAACACE709775AD0EE6371653E98B
4368988E29415136BC5E2BE7A4CA996B414F31197FD3B9D60DE986DE42052E9360347D30F18A450F
0B72FAF7D9C2BC7D8A9CA55E66DB5EC160B4FE592D01FAD4D06E032045E5A12F1D5CD546547C
E679F6A4D1092469F3A29C024E91B7765680599215BD4D7A20B7E521AA80CF5CC0BEB19512386C
A967A9A6F9E183A71A363FAA4F64D1428842512BD65D9080EA0ED959715AA5047820F62EEDD3D
4CF82A7183CE5C7538D6FF5E2FB8C7E2A419610E2B28821BECA5EA5DE032F318A0C17E24BAE8A

DBDF516E5CC288C115461E7BB8985E0D8E228529082CB96D1D54BD1E9B401E36EE4DF01DCAB38
93084B4ACA018DAD2E27B69F0C2FFDDE4552D02A97DD8BA25A2D5F919A1F2B42539974E2C5578
8CED56CB9BBFECC65AA2044DC13C33D6B6E91ECC19AC3C3068169D9F56EB3E06445BA724E08C
58CBAC8343F4EF78452F76F3ABD6B73A814DC29CE4BEA487848D64F07D306573847511C44D6782F
8A06139564C026D62B73630882D90A964E36431F69CB84ADC2E48E0B0635180D2CF1F0E733CF5D0
85BA2223B6A444E24AA2DD1CF67CF08DF524E283D3CFD1DCE8F1330F0C5FE8B3D0CEA8B27EB
BA18CFF8242C3F095F7A1051353190C0CA02E3321C673089CE6E767D699FD20E348DE559BE3D409
A876D29BDE7FDBD2A8A47DF7A45F78C383104CD018CCA0782700ABF6878698E1C46C773D1B62
DD684870C9CE2C6DF6892E9AEB55F3936195F46692FC

sk =

3BCC075D2E7E70976AC9F804CB71AB99A0ACC4AE2974C44E76BE660AF9AB5C3D6FA832C43117
DCAB8971F07896169D2B6E5A22A242D6724CB9CFCEFD03BE9780A5EF0E1F1FF000FF00FF011F01
1001E21001E0E1001F0121F00001FF1202001F11020F0F12FF101E1101FF1112F0FF01F001F0F01F100F
12101FFF0201FF101E121FF1E0000F0FFF00101F01F0F00F0100111F1E0FF0FFF10F0FFF0F101EF000F
01FF0F0EF0001F000000F200FE0000100EF0201F131FF1F1FF10FFF1000FFF10110F00112F1F00010010
1F0F00F000202000001010F0101F0202F011FF011000F1E1F00F11F010000001F010F0DEF0FF01E0F010
1010001F010012F001102F10F0000F102100F00001100EF02F1E00EFF1F0FF100F011F112FFE1EF0F100
1001FF012000100FF10103010000F2FFF10F11F0011010F2F001F1F00100101F00F1001101F0201F00EF2
FFE111000F0F1F210010020FFF0011120110F0F111F010001FF121FF21121FF00F10102F1F1010F11FF
10F0F1E101000F102E0110F010010FF1111E12101110F0F1101F0001020FFF0101001F2F0F0FF1E01001
001FF0000111FF1F001FFE1F01000F00F1F1FF10111F120F1011100010F10F1F0000000F0001FEF3F001
F00000F2EF101FF0101FFD01F0FF00100021F0100FF20F2F0FF1000FF011101000E12F10F0E0FFF1F0F
F11000010F000FFF000F00130001EF00F0023F020EF2F002F0000210E00F0FFFF0FFF1000FF120020000
0001101F1FFFF00F02EF20E101001110002110FF2F00001112F002201FEF2101FFF1101F0000F101100F
11001F1F10100FFF011021F1F10011E000FFF0F001F0E0FFE00F1000FFF02000201FEFF110FF11000FF
011021E10F000F10201012E111100FFF0001F110010001011F01F1FF000F0EFFF0FF0E0F0000201F01FE
01F22F0F0F10100101F1111110010E10011000F00F000F0F00F10F100000FE020FF001112FF1201010F11
FE110110EFE1FF02F00E210F002010011F0FF0FF100F11F0110121200010F0F0FEF1F0E011E01E01000
110000021212FFF0F1F0F000000001021FF020F00FE10101110200000F2FF11E100F0F001001D0011000
0F0111110FE101F0110F21E0101FE0001FEFF210F100111FF2010F101111F001FF01FFF1000000010FFF
2F0F0F12010F1F1EF0F110FFF011101E11F10F000D001E200000002E0EF00F00000F0F21F010E00F000
FF10F0F301FFE0F0FF0FFF20FF00F1F000FF100F2F02E00F01E0FFFFF0000F10F20F0E11100FE0100F
E00F1D102000F10012F0101000011E1DF01F0F1E100EF1200EFFF20010000011F01F01EE00020F111E1
0001F11F200100F0000F0F1F100E10F01F00F2FFFF001F1000F0FFFF1F1001000F01FF1EF120001F0F20
1F010000FFFFFFF00110210FF12FF1100000000FF000F00000F0002000F1F00F1E00201F111FD10FF0F00
00000FF000F0EF01001011FF2F1FF0FF10FE0F101001F0E000FF00210F20F2FE000101F0E00001E101F
111E12000FEFF00F000001200E010F110FF0FFF1001F1F0F0001101000101001E01F00FF0FF01EF01000
0000E21FF0201010011FF002FFFF0F020000F0F0F110F001010100F0101100F0E010F012F11F1001FF1E
0010F000F11FF001E11F120FEFE00FE01F001F00F0F1FF00110112F001F000FF2F111F21FF0FDF10F12
110F00101F110F0021101F111FF0110200FF0F001100000F12FE000110210020F0E1FEF00F1F2FF101FE
EFF00F1101D0111020210F3110F101101F00F0FF1110F0000F1FF110100F1000002110110100201FFF00
F011F1F1F000010011EF20010F00FFF0000100E10101000FE1F1F00FFF1E10110FE100F010F00000F001
00F1F0101FEE2F10EF00110011000EF000EE01F0120001F2100021F00010000120110E20001010FF0EE0
02F110100000FF0EF010FF10101EF21F1F000112FFF0110001001F1F110FE0110FFF0001F010001E31FF
1101F1F00E0F0F0FFF01110FEF020211F0FFFF0F0F00F1010F1F2E0010022E0F0200F000E1EF101011F
0FF2F0E000F0101F01200E001F00001010E1F00110201EFF01FF1100FFF1000FF0000F00101F10000001

F0FFF101000100001F00F00F01F0010F00211FFF0001F1EEFE2FF01010010E0F0F1F00000101FF1EF00
0110F1F0E0F021F00001F1F0100EFF001120001F12000F22120000EF1010E0F200F1001011F10E0FF1D
F0E0200000FF1FEFFF001F1FF01100F111FFF1100011E101000F0FF1111FE0001101F100FF21F0102F0
00F20F201F1F1F0FEF1100E1F0EE1FFE0E0F20EF0010F102FF2FE2F100100F10F0ED00EF2111FDFF1
00F20FF1100FF00011F0FF11F020010F1F0110F0FFF010FF2F11011F0FFF00F0E0F010FF1F01F10FEF0
002FFFF1100F20F0102F20F1FF1FF00F0FF0101111FF1FFF000EFF20E0FF0FF0FF1FE11F00F0F10EF0
1F000000F01F0011011F0F010110112F00F0120100301FEE0FF000101001F000001FF10100100F1010100
1F00F0100FF0101FF110F110F2001000000F0F101FF0FF011F100EF0F11F2F1110E21F011011F1F0F01
F00011FF100100FF101FFFEFFFF10FF100F000EF010000F010110000101FFF0EF1FF1011E101F1F100F
10100011F0F10FF0010F0010010FF110F01101100FE11111010F21F0011FF0FF0F000D10F010100F1000
F1FF02000013200F00100100F10F00100111F0000FF0E0101000FF01F01F101000020FF1000010011F020
00F0F000F021101F0001020F0FFFF11FF0001F0210F0F1FEF001FF210E1C0F11E011F0FF00F1020F11F
1F0F101010F0FF0010E1000FF0F101FF0F0FE2012EFE10110000101FF2F000F0E0010F1F0000E0F00F1
F12F0002F102221101F00FE010F000F11F0110FF0E1101F10F111F00FF11EFF0030012F1FFF0E0001103
1101FF10F0002F11F0F0F0100F01F00E1F100100F1101010F00201F1111003012FEEFF4008FF7F5FFB
FFBFF2FFFFFE003FF6FF9001002FFB018FFF00BFF301200000F002FFC00AFF301200F01300A015006
FF0FF2005013FFBFFCFFF00E014FF6FFD00C00200BFE900AFF9FFC001FF8FE4012FF600700FFFDFF
7008FF8004FEC015FFF002FFFFFEFF0005FFFFFEFF6FF2FFB00A017FE8005FFE00800700A014011FF
6FE600CFFAFF500001400DFEEFFB00B00E00A004002010FF500EFFFFF8FF8FEAFFFFFAFF700EFFD
FE7FF2FFA002006007017FF5FFB006FFD008014FFFFEFFFFD01EFF7016FED00CFFEFFD011FF700000
3FF4015FEEFF7003FE9003FFCFFFFFAFF400500A00CFFEFE6019FFF001FF500B004FFBFFA000000F
FBFF9003FFDFF700A00A003FF9FF100BFF7FFB010000FEDFF1FF8FF8007001FF401500100500AFE4
FFA003FFF00BFF5FFE01300701100AFF4002FF8014003016005FF6FFD00D011FFA01AFFB00CFF7FF
AFEE00EFF7FFDFF500E004FFFFE6013004FE8FEF009005FED00BFFF00900B01800C00400200500B0
0FFF8FEA00D014FE6FF700300B00AFF4010FF6FF5005001FFD008009017FFF009011017FF201600D00
8FFCFEFFFF4FFBFFCFFEFFD001002FF300EFAFF1FFFFFC00DFF000B005016003FEF006FFCFF8FE
B007FF7FFA013001FF9FF9009006003010015FF301300F01000B007FF500A004014FFA002FF3FF8011F
F4009001003FEDFF5FE3002001FFF001FF5005FEEFF8FFF00100300900F005013012024003008FFE010
FF2007FF300A004FF000FFFDFFF009FFEFF401AFEB00C01600D000FF4FFB002000FF7FFDFF9FF6FF
200900EFFFFF5016002014FF4010011002FE7001FEEFFFEAFE8001FEC016FE400C00100800300000C
FF0FFFFEEFF008020FFE006FFF00300201400001100A02500FFFFFEFFFEFFD0FEBFFB00AFF1007FF
3FEFFFB00AFF1000FF7003FFE00EFF8FF100F001003001010FF3011FED01401800EF024000001FFB
006007FF700AFFD009FE4002007FED006000FFFFEDFE600E00B011FF600CFF9FFC005025FFBFF601
400601FFFEFF4006007FFBFF900B007001FF2004002FE8007000007FF700C005FEFF5FFA003012FF9
011FF7008FF200F00701A009FF6FF3FF2000002017FFDFF3006FF900CFFEFEEFFFE01200A00DFE9FF3
FFF00AFE9005FF5FFCFE901AFFD00DFEE00700D00501A009FE8FFE00AFFCFF2FF3007FFAFFD017
FDBFF200BFF7007FF6003003FF2005FF2FF5FF0FF5000FFCFFEF9002FF7FFEFE01600200400A005
008004002008FFE005FF9012FFFFF5FFC005FF3008FF4FF3003001FF3FF9FF7007000FF5FFCFF600400
0FF0FF0006008008FF8007FFC00EFFC00F00000BFF101501601001300F008000011FF7FF9000008FF8F
FDFF9010FF3FFF004FF400701B004FFDFF4FFD01600BFFC010FE600A00B00BFFC001FF9004006010
00D006FFCFDF002002FF8FEF00E01BFF6FE7FFBFF9005017FE5000017FFDFFF001001FFA00000200
B008FED00700600C007004FF701CFFD005FFA008009FF8FED013002004FD500D009FF7FF4FE6FFAF
FCFE6FE2009020005FF9007FF800501BFF200900501DFEC001013FECFF9007000003FEF00300600DFF
500B011FFDFF7005003002009003FEB002FFB00801800901100C00601F014FF4014FF0006FFC015008F
EBFF100BFE8FF601300FFF6002018FF3FFAFFD001002004000001FFB01C008FEF00BF001004FE80
08005FEE00D004FF5006001FF7FF7FFB00500FFEE00AFFD016FE4003FF5FE1FF500500500CFF5FFE0

06FEFFFE007011FF1FFF00CFF000C009FF7014003003FF200B003008FE5001FFF01A007FF4017FEEFF
900AFFA00AFFBFF8FF700FFE900C009004001001FFAFFAFFAFF0FF9004007FF8FF3000007000FFBF
F7FFC015009002FEB00AFF1FFFFFFFF9005FF3015FFDFE0006FF4FEFFF7FF200C01A00000400AFF9
00A014018FFAFF2FF701400600000FFF2014FF800E018000FEBFF300A00E002FFB003FFBFF2011006F
FB00AFFCFEFFFCCFFCF8000FFF00B011FED003FFCFF0FF5FFD00E005FF9FF700AFF201800C00901
0FF7FF8019FF0FF4FFCFF9FF3013FFB01500CFF6FF6010FFD000005FEE016014FF700200EFFE005FF
F008FFA001FF9FF0FF8009FDD008007FF5FF6FEA00AFF9FF100800DFFBFF3005FE301B00A016006F
E200C00CFF301CFF8008FF8FE8010008FE7FFCFF7FF7003FEBFF6FF6FE7FE8FF8012FF3FF0FE60010
19FFA01AFED005004002FFA012FEAFEB015006008FF1004007FDDFF000800E00500BFF70070170080
0D006FF301AFF000F00A006FF200700300AFF6FF2010FFFFF3004000FF3003002FFD009005002FE900
6009FF5006014005FFAFFBFF900FFF6009FECFFF006FFBFF8FEA014FF3FEEFEE00E000FF301BFFCF
F1FFC009FFE00D00C009011FFBFF900F014FF7FF3015FE7FFE007FEDFED00100400C00000301600E0
0CFF8FFE01200BFF9006015008FEB000013001019FEC02500200CFF1007008FFAFFB005FFDFF00090
12FF7005FFCFFA005006006FF200B001002003FF400F010FEFFF100500200B002008009009FFEFFA00
8FFB000006008FFCFF1FF2021003FFDFEF002011006FF8FF601EFFD00E007000004FF8000006000FFD
01900E00BFF700CFFE00300FFF9FF200DFED000FFEFEEFEB005FF6018FFE00D001FE700C00BFFDF
FFFFD00FFFA005FF9000003001007FEC00B00300B00600E000FEB010008FFC012FFF00BFF9FE70220
1700200CFF9009FFBFF0011004003FEE00F01F00D019000FF1FF0000FF700C005002011005007007002
01F008011003FF800500A007FF3FF8FE8FFE001FEDFFFFF7FE700000700000DFF3FEFFF40050010030
1100BFF3000003FF200AFEEEEFFFB00000500900800200FFF1015FF0FFCFF6011FF5FFE00602600300
3FF4002016007016FECFFCFF301A00400CFF200500EFEE00A00AFFC01600EFF7FEC01200C009FF50
04010FF5FFFFFF8FF1FF7FAFFBFFC005007FE600CFF400900300A000002FF800DFFC004007FF900BF
FA005FEDFFFFE000A011FED016006FF8FF7006007FF200D005FF5FFBFFCFF3FFE00F012FFF000006
008FFC01CFFB002FF8FFB005000FFBFED00A00BFF7004000013021FF5FF7004FFD008FFCFF6FEB00
1FF901700300FFE007FEA015002009011FF6FE7003FFD006003008002FE800600C007FE6009008FFEF
EB000000005FF0009006FFFFFF00301800400500201600CFF300600E006004FFE006FF4FFCFFAFED00
2FF701FFF005FFC00401D007FFA009009FF501201001DFF2FEDFF5FFD007FFF00401BFE7001FF9F
FE003010FEC005FF5FFB00B00CFF7FF0FE700E011004011003FF400101200EFFC00B003FD5FFAFF6F
FAFF8FFFFE7006FF5FFB00501600CFFF00201200DFF5FFF00201101B002FF7007FFCFF3001FFAFF30
0CFF9009010009FFD008FDCFFCFFA003FFCFF5FF5FF5FFAFF701B01200CFFCFEAFFB00DFF8011F
F1FF3FF8005008FF0FEF002FFCFECFF8FFF002001FFBFFB016FFC01800E00600CFF100101500DFEC
011FEE005004016FF8FF6FF600AFF3FF4FF5FFCFF400FFE8014005FF5013018001011003006FF700800
F00BFF6FFB001FE4FF0FFC009000010FFD00A006008FEB00900501301700BFF2004012005002FF6FF2
FFB00F00B002FF000E000FFBFF0005FF3FEEFF500CFFF004001FFE016014FFF002008001FF20080040
03002009FF100E009FFCFEFA005FFEFF200601EFFCFECFFC006001FEB007FFDFFDFEFFFD00FFF
CFFA004FEFFEB00700200D004FFAFAEA003001FE9005007FFA006002002FF701BFF6012FEAFF6018F
F700A008FFCFFFFF801300300AFFEFFFDF700BFE7FFAFAEA01101100AFF0015019FF301A018FF0FF
E00E010FFB00200600EFFCFF4FFA00E010017003FEF00700DFFFFFF001008FFBFFCFF5FF3016011FF
1FFCFF7006015015FFF00FFCFF8FFEFF6FEC005FF900DFE8FE800DFF600BFFDFF4FF5012FF80040
0600F001FDF00B001001FE600900DFE9FFB00B011FF8004002004FF3FF000E00DFAFF9FF2018000F
FCFFCFF6FFC014FF8009FF0003007000001FFF00400EFEEFF200A007FF9FF1FEFFF8FFB00500AFDA
FF500CFF5FF6FF800201000AFFC00D00F018FE6FF5004005FFA00300BFF3002FE9002005FF800A000
FF4FEDFF1FE8014013FFFFFAFF900EFF4008010FFFFFFE01100BFF400600F008FFDFF5007FF3FF5002
FFCFE800C002FF8001FF8FEE002FFEFF7010FF400BFFEFAFFB000006FFD004002FF4FFCFF8013FF
2FFCFF4FF1FEE010FFA003008FF601DFE800FFF6FF2FEC01400D002FFFFFF3012FEC01201101000E00
2005FF7FFE00500900900800E005001006000FEB00700DFF4002FEFFE6FFDFEA000FF5FF7FFFFFFBFF

3FEE007002000009009002FF5FF9FFF00E00CFF4009FFCFF4FF9FF6009006FFD004FFF00901C002015
FFE006007013FFDFED00EFFF000FEB00C006FFAFEFFE7FE3007002025FFEFFFF006FFA006FF400300
2FFCFFA005009FE3FF1FF400F009007FF4FFC005FF7FFDFEC00E001007FF9FF4FFCFEC005FFBFFE
FFAFF600AFFA00CFFDFF4004011012FFC00C003FEB01CFEEFF9FFD00600B002FFEFFAFF0FEAFE
F00FFF9FEAFFE006FF0004006000FFA002FE5FEEFFE01100AFF0002005FF7FFD010002FFE00001000
6008FF0002003FEAFFCFFC00000901DFFAFF7000FE7000000000FF2FFEFEFEFEFF7004FE801800400
6FF9008000FF9002002006FF6FFD008FFBFF401600A00A009000FF1001000FF0FF3FE6FF000FFFD002
FE9FF1FFC01D001FEFFF6FFDFF200D009008000FEA006FF9FF4FEC00400000B000012FF3FFCFEAF
FFFF7003FF4FFFFE7FFD011FFDFF1FFAFF7FF5FF3009FFA004FF5FF300DFF6FEE003002FEEFF3FF
2FF600D00BFFE003002FF8012000FE600DFFC001FFEFF4FF9FF5004FFE005006005FE2000FFE00BFF
200A007006FFC005012FFB010FFC013FF200100A016FF5003003FF8012005FFE004011FFFFFFF1014
FFD00FFF600000500CFF6004009009FF6FF6FF1FFB002FFD004008003020FFC005001008003FEB0040
15FFDFF2FFB002FF9FFFFFFBFF3003FFFFFF6FF0FFBFEE008001FF701300500C010FFC003FF6FE9003
FF6FF3013FFF00BFF5FFF001FE9009FE700EFAFFFFFBFF7FF8FFCFFEFF0FF000AFF5012002FF9FF
6FF600401C00B008FF800FFFEFFDFF201C00A01601300D004FFFFFFAFE8010FFC01A00100700800900
9FFDFFEFF2009FEEFF1000015004FF5004010000FF6001FF4FE900CFF1FFD008FF1FF200500D000FF
300E019000FFCFFCFF8005FF9000FFEFF9FECFFE00000AFFAFFEFFF00800A005015FF1FF10020
03FFBFFB010FF4FF3005FED008FF400FFF9004FF6FFB00E01400E003007FDAFFBFE3007007001FF60
09003001FF600400200000300A008FF700800900EFDCCFFDFF6014FF5FF9FEFFDC01D01000700EFE7F
F9000FFF006008005006FEFFFC00B009FEEFF9008000003007006FF1FEC00A01B005FEEFF3FECFEEF
FBFF8FF200CFF701D00200600400100AFF7004007FFF00400BFFD006FE5FF0FFCFFB000013FE2006F
F6005FFC00300500AFE900DFFDFFC00FFFF00FFB00100E000007FF6008005003002003004014015FF
9FEAFF8FFCFF4007000006FEC00E012FF8008008FF8FFDFFA001007FF4FFB00EFFF008007FFA007F
FD000FF4005005013FF8FF5007005006014005FFCFF0FF9008006FEE00AFF3FF900300CFFEFF9FF501
5FF2008FF600A009006004010FF9FF8FF0FFA007009FF7FFBFEB00200FFF9FFC00A008FF4004007014
FF0FEFFF800600901200CFF6009008FECFF5018FEF00C01200BFE500EFFF007007FFD01A025FF8011
01700FFF3FFB017010FFFFFFDFF700DFF6FFA008FF700600FFFDFFA002011FFAFFA00AFEB00CFE30
05FFBFE5002FF7FF9FF600DFF3009FF3FFBFED00AFFEFECFF7FFD00F00100F00000DFFEFF1FEFFF
F00DFF7FEE00700CFEE002FE4009006FFCFE8FF6FFFFFF301B00400501900600DFFDFFC000FF100D0
00019008007FFC006004FFA00A00801D013FF7FEAFFFFF1FFA007FF4FF6FEF01700CFF200CFF80060
09002FF2FF5011018FFAFFB00800400DFFDFFC018004FF2007FFD002FFC006FEEFF3003FFCFF1FFC
00300BFFC00EFF500A016FF200B001FF9014FFCFEE005FF2FF100DFF5FEEFFA007FEBFF7FF500400
F01300101AFF4FEDFE6001FF700301000B006FEF00D00301300000DFF800F007FF0FFC003FF3001008
FF900DFFC01001401500500FFFFFFEFC014005000FF7FFCFEFFFBBFFB019FFF00A004005018008000
00300C009001FFF00600DFF6FF200201EFFCFEE002FEDFF500700100700E004FFE008000FFD008FF50
0F016FFFFFF5FE9006000002FFA027FFA004006FFE01500EFE8017015FFBFF8006FF9000FF0010019FF
F002FFE00A01500401FFFFFFC002FFBFFCFFAFED00400D009FF000900CFEE002FF901CFF9FF200D
011FFEFE002003010016FF0FF1FFCFF100C000FF9008013FFC01D00400100F013FFE006FF50020020
03FF9FF600000300CFF5FFFFFF009009FECFEDFF0007FFDFF5FF0009FFAFFC019FEFFF4008FF300B
FEAFFAFF8012FF3FD7FFC00BFFA00700600A002FFC013002FFBFFDFFFFEA00100E000CFFA009
007016FFCFF9FFB01C001016FF1007FF9013FF500A003007FEE002FF200000CFE9022008005FF101D0
0201000001701800CFF300601400200900E003001FF4FFB00DFF1004FF801200300500AFFB002FF8FFA
FF6FE8005FF9FE5FF400F007005003007FFEFE002FD1FF5FFC00DFF5002FF8FFCFF4009FFDFEDFF
801AFFDDEF013000FF7009FFFFFF8010FDEFF8FFB008FF8FECFF1006009FFF00B00C015FFEFF800BF
F500501300A007FFB005FFBFF501900E00BFF4011FF900FFFA012FFBFF2FF9FFEFF7009004FF5009F
F000F011013FF2009001003004FE6FE3FFA005002008FF7FFC00300CFEEFEC008FFBFF9FFB015FFFF

EFFEE002FF0009000FFCFE300CFFF010FF6FEFFF4FFEFFAFFFFFC001FFCFF6006FF4005FF8FFE000
010FF5FF9FFFFF7FF0000008FFC007FF100B005FF000EFEB014FFB001011FF8FFBFF400FFEDFEBFE
6FFD013012FF9FE7FF3FEF00A010FFA005FF7FFD010FFAFFAFF5019FEC007007FF7007FF0006001F
FCFF6FFF00B001008008FFC007002FFA003FF0FFD005002003FFFFFBFE900C005012FF900E003FE20
0E00EFF9FF1FEB007014FF6FEEFFDFECFFEFFDFEFFFFAFFEFFB00A000002FE0FF700A010FF1FFEF
F5001FF4FFBFEEFFFC008FFBFFC00DFF3002FFE001FF400400CFF401800AFF400400D00B005007FEC
00D010FFD010FF5FFC004004FFDFFF014005014FFBFFEFF700D002012010FFAFF7FEB00FFF601400
4FF300A013003008FF7FF0FFE00FFFF00EFFFFF7FF7FEC000001012FF8FF9FEEFFE000FEEFF8FF600
400FFFBFFFFD6009008002FFAFF7FE3007FF5FE3003FDEFFF001FFBFEF00CFE8001008FFC0100120
02FFB00600700AFED00B00BFF7FEDFF500FFF2FEC007002FF9003009FFEFF3003FFA00200C019024
FF6FFAFFDFFE001FFFFDF8FFFA00300400C00CFFE005FFF00C001FFCFFFFFEFF4FF7FF9003F
FC014FF8FF1FEBFE4FFAFFC00FFF7FF000DFFB008FE6FFAFE9001004FFB01500A01EFF101E012FF
4006FF3FE0005007007FFBFFAFECFF5005FF9FF100900200B01BFE6FE7003FF8FE5010FFFFF900200
FFDF00600EFF5021FF6FF2000FFA003002FED00EFF6FF7FED005FFCFF2FEEFFE00100C0130140150
03FEE005FF2FFFFF2007008FEF002001002007003012FF9008FFF004007FF401B004000FE7FF000FFF0
FEE00100601500AFE6FFEFFE01CFF7005FE8FEF003002001FE700100A004FF600BFF7FF3FF2009FF9
FFCFEFFF6001FFAFE3002000FFCFF4FEEFF2FFD000FFDFF4FF6FF800500101900BFF1FFAFFD016F
F9003FF2FF5003FFF010FF600BFFFFFA005FFD017FF300D003FFFFE2000002016FFF012009014FF8FF
4FF4006FEBFF6008FFC00CFFD004FFF00CFEF012FFCFF9004FF8FF9FF300701CFFCFF7FFEFF900FF
FFFF7FFCFFE007000010009005005FF2010FE9005FFEFA000FF4008000FF6FF800300E00EFF9FFCF
AFFCFF3FF3FF300500E017FF3001001FF8FFDFFCFF1000FEFFEEFF9FEB000FF700FFFBFF3000FFA
00DFF4015FEF004FFD001013005007005FDEFF400900DFF9FFE00EFF7FFA005003FF0001005FFBFF
D003FFF008FEEFDDFE9007002009FF900200D00A011001FF8FF7002016FFAFF200AFF7FF5FF7017F
F1000002000FFFFF8002FFE008001018FF7FF7FFDFF9FFC00400D011FF9000FFD003FE9FF3FF6FEF0
11FF8FEA00A003FF1FF2FE700EFF5FFBFD7FF2005FF7FFB019009FF6FFEFF300300500BFFEFF0FFD
FFC003FF200300900A004013000FF8006005012012000001FFDFF4FF000E002FEC008FEB00DFE3FF5F
EE00EFF700001000400B00600CFFA013FFFFE2FFBFF9FFD003FE0FFFFFFF002019FF9FF6FFE003FFF
006005019FEEFE300C000003014002FFB00D00400E00100401C00B006FFE003009FF5FFC00B0060110
0900800D018014010FEB006FFEFE8004FF9010FE500C000003FFD00F000FEC00B000012FF3FFBFFD1
3D8E8A6F118993DB0BB67F67C149DDAB5E6C6739473CFC39400D03FAD748BA1FDF9DDB3E30F7
02C47B50189FF62E7810DD022342F002F77C52734ECA1163C78EB21E8CFF3770110F633E66B37308
9F97AA0E18093802E7BABAA6FBCF20A711247098FE826CEFD768C6F0F986B85F06C1CC4C06BDE
A94FCCD0C41442B67036D891635A0BD2C1FFCD46C0BA940B61A93225D36F09BF0ADCB06F64771
4B83229B1DC00F2779B3C3F66522E4A21D588A1C239E636F78B828BE7BED87202B01ADC7970405C
C7E816A28D007D0E2CF80F5AFC6A14375CF6AC0570FFE80378FD42478C2F7739BE22F53D0A19828
2E6278F4C413F6F3BB5F50E1512EF447ADEA77FA1C3690D383063A768DD968E4EB8ED8032CF78E
1829BC6AFED86029AF640F72284C1110D4E30279A8271F6D065F7CCF733BC91CE13B61B83EF4D9
CEF139460C5F7989D58409A003DB066AE7D0DF48316A90085BECA91B8DEF67CE1267B73FFB1EA
A1CA7027B7BE5E0115382EA23B2AE63EEA8D074DE52D3FEBB7D7489E2BAC5214EFF0BF88C3BB
2C8023E8ECEBF5EB9D2627E29B3F89664135E07D6124F85DB6661A33F59A68C63361E4BC9031D22
45ACFC765FC9F0A81CE575BB17D1ADE150B4E4DF116284297167DA4B9C33C5388E2E4D58A491B
F1D5E384411BD9DB7CD567E6FC47844110182C7BEA80DB02DBE2D3DE38FC1D2FC602C05E75C6
C827DF1ACDC9905B1FC3724E12FF997AF5FF2FC3D3DD5FF7F1E08F323242CCECFD0786202074B
D338E7163E8ABD0430FC291400B2D7FD340C5E19488C69A41ECE5A812ED8755224BC29D9134483
C7010353981641BFE7A026E4137087E550009D518F790667A8C087D29AFF97E7167A0CE107FFD44F7
53510E839EDDBE6F8048DA326A81FFE3EA9EB53B9CDF9F54A97EB84A07DDB90E1516C672EE659

B2DBB6965945517E425C67F75C403D93D62BE404013B5062FB061C625001CF3D358508E74B1E8043
4B245EB30D0FC6F4F3E3D2FECF9F177FCC11E135426FA0C4C4270A22BBDA9D1134B9ECE323226
ECF9628E5FE0F31E6807D08374D18606D2C984CA0B2C24035E17F0247C20C3DE820EF3FE96C6AA7
89B423381BED762F5CA1CD201FF310CE41C8C4380E63ADCA2C62CB8ED5A6F4403E0BF52CD65A
9D7082C2446F9CD4698C3853AE2697252E68FD8D017E0E512E91CE101D5A63F10AA5043E832FA77
A132C1EA050C1404305DD017ED8479CE2AC1DC7B9922C4DE237E02F029A9709AD811AB0C9481A
7B23B396F2C1221DD17B11F8CFD3D4F86FF01927E9D2077A840D734DD9B4CC53E8F778A3D03309
2161FF30BBCC1BE73BEF27C10F4EB183E0E7420A6388ACF846ACB098762CF7C85814CF42983E4C
B081CD001D97BFF320CDCFC13112E3FA3130DE9080D9816378424802FFC5A894F277C1063B027768
238045385DB0D010895FCAEBC081BF3F18F7473A7E034F160B84D5753EB9CEE2C558C10F8E20019
E14531D5CD1429AE2916C9644E33B9B9E6DE58E83757C8B7A02A1895612D88FDE6796D26D169F0
B9019E7632311328C26B9EB3703CB99997F30F88FAD5367CBE7C647C06F2B13F99B334BDF176D29
6959D0DC6C0E2EF5331ECFC15DBFC6BCF690EF3BC67FCD47104493F7C28220B1EC5A90416BA2D
99AB9FF59748371667916103788F26C8BB843762EFB91E144294375E1F7982B1F93F3FD712BDF7A76
B56582156E515FFD1B244423E9D70E9C0C7EBD225601E38F1B5BB26E6D2E3D9F9C9E08AD3C0E92
D596D7F857165463DADE0C7B4F79662F639A88849BF7BBE2AD351D79A743049D6442F7BEE06822
E828B1C1D17626C4EDE51C039C9F62DB86CC088634480398C907081CB1DFB02068E1701978E41BE
7F9B09624B063C179BB131C4664250BA99BC63909203B120356BD0B100FFB9DF2B16BA106857DC9
B2A576CAF0C0F1B6632DA8F1C30C6305EEEEFC8769A39059D43A19D28AD9C32863946EE931B05F
749432AE1170FBEBF73BC6889259C9065C11142D8ED09D8B26BB4CFBCD1E545E28E07C46288BCB
44976BF20284417EA5F769788E839A2622960B63C1C6A35751E6FEA5BCE85392D907729952CA58A4
3821E6B996F8836FA07D40F11BC989751E0B9020A06C2BA71FAF94EBE5FE613945C54D41DD979B1
027AA37B7F10D8ED5831C07C6416B2EB56CEE1023503219F5813B14DE52538EF13B80B7ED96B4E
BDF9D49C90098DC120B468EAC22F66F342C8DEC82B3201EC5338E482B5BF4D79A7EC858FE5605C
DB9980159E630E7CCAFA1C1D55EEDBB79D5B85F4195B14644E41277430DE118EA7FD7654340430
059B0A5BE0A9BD26C7CE832E91EF06F96282A23CE33D5DA318615388810BC240D2CE9450B16019
6071A2F63253C3478231EA578135431FA439FD81F181489862410CEE8C630A0707856EB0F9BEF8EFE
0A3FCA0F1F07CC7CD83A294DF8607E1C338641011C9CA98850E9BA2A3C041312978FC5588083D1
B6FDDC633861741E729ECAB9685A8A5214C01373C0444C54F4B58BFEC2FD47120E2C18672CA69D
854B07F2F9034ECA467ECC60B6E6488CE23F6AD8C9F2EF81F726AEA1E9285B9B83C83F83C8EE81
B938B90111AD67904F141A4A3C83F0159169BA9DDC0B33EA6F76CC3354232630A19E7AF14C01D8
22DAD9082FB04A4C2B2EC11F85F073BE01220051072FC40E455901C56658366F24983CBF6850CFCC
8C27DC0E6767FE8E34A08110EC40977B945BE82DDC50698F8F52FF76010B4E7710EA5E05B22E45E
A133B079E5643AD26D74307B80EDCE5FC91BD73DEC71BCE7386ADD77DFA67C53BF1E225058727
99FC90EE1CC4661447EFEB6BE91F4123BA40BE8D2800947B11ED69DEE17BF2899806FC2D0F0167
EC133DB0AB13DD82557DFE3B0162B870720FA21E0DCBC549749133B3FB86014010FCB04455E792
C6B07AC1CF6D5B283F27E925B311C98F0BE9F2817F36E69B1FCCE31E8850B4240F2EFCDD97AF87F
696192915D670DA441184D65EE6862B4730FBE22414FF6A7B4E80BC668E7D303340205013260C949F
76F1A7659F51CF20A3CFCA37D9B7E61D00AF453C0492BEB0A7FC8CE601954802C093207B571B53
9F0477D7ABF51D9B57CE6A72CE31F1613F2F31A8F39B9639A0CCD47A08A97BB9E5BC8CAC47704
B3EB61B53DB9EF1BEBB7BA4D9E38C37D116FA3A4AC7FFD8056534CBE5CBF1D56EE17BF79FC8
0029630FC2C7345FA670E0A7C44CEB924218E386F73972733F6722FB70061387C3D72C1EE76A194B
8F3BECAF0E882437441DBEA2DE1015A70679448941BD8E97A0A0A38551F31C012E758DDDF27FE
A991FB7B4AC6D8AE5D9E83BFC0543628F5756D66B390E6A9D441F79CD78885E06B27FC0CE2477
A9842FE9C11BC7737D1F7D208A583D985644239AC9CF6AD0FCB37CDFCE879831474335ED17F03A
92EF788479EB02C7769DB49E87E2C2D957169AFE695C5FAE8D4F7BDF951810F7FBAA7E5B0B49B

CE9C0FBFFA68BA4BC1A76E0401583CA7B44E17E2A30B44CBF87F614E9518BE63F51C44F10B0395
89784185C06F1B8F62A3DE8E2417F8949F80C12261110E04C8E49821AA5D2CE1C27DB97E7D03FD6
EF04F975FDB7BB58907F21E5574705EB06D5040CA8713EDF22300FC7A7B0C43847AAD3D6167B8C
C2E1774FA5F1A3F7E0B454C2418C6B183FA2418FD04A909EA03BE7FC05E56383EFB753E323EC145
195A79C739356CC019B778E83C2083DC1C6C6BC39F2202333480E6E4C807946EE63B952991097819
C36C396B94E709BA96E67000690E581C0F639D20CD1B1DB7772BDFB98481B330E71D8BF3F07434E
B978008D615965F86DC86D5EEDD8FC147F2E7C35DF2C57A9672AC19A8D2DA37261CD212114ECB
890D61F584C3585F26D0455475CDDD8CD0B0A00E3125E46311D5806F9C3F61AB14FFE886D746241
D32CC0048FB645247CE16F346FEAF8412B8B0ED66BEEC2C4477F66DF2810C7D50D138B90EE3E80
4EC3FA79DD203A7EB23850B0742F410AF1C3B4A35EC45EF0A0E80CE0C428C4CD2C13028A083D7
B11AA584B6F6C025E0316BAE68EF32BBBE06D15498A1F5079D26016FBE2CE89C7DF446CD8F709
ACD01C12FA4395651CBE29481670F0CDDD236C20128B0975EBB7519F39A3A64DFA680B65A18
2DADEF629CA77A3C84FEC8185DB2DB3D8AF10EA7C6ECC655B5313164E204C77AC7F272DC9022
6C6B51CA84191D65D04E4A793AAE2F585047215E03F3310282E23FD762C0BD7B77056821AE1A722
42C7085B257805DFED887A988F358295DCB606FA734DA39E20FC2D691BC5FAC089FC349EDCA7A
E0479000B9AE4253C20C514BBF741F12FE7B061C013007355F89C030B706A905A043B39D51C0FF17
C087CA9580F52609629AB743848ABE8A91923B562E65372EBC9966C8D3E29E1DFDD86E2CA94258
5538956FCF7D12459CEEAC4F651BC1C3ACEF414F6369034FB897E2EBA9562C70A7ABE619229861
F33E17BAAD63405B744BC42ECB09077706B5B182C0224D97A53CD547C08F1823773A4288622AD23
7B7552412D9B79CEEA8312CCBD3820ED4730A4C35419FA04F1B7E901F65F473DDDE0FF218F73E
61BF63D6C69D2C2DE1EBBEECB41619B46864E1AAEE727953BFABBA2050FEB022B7C3D03C63255
A700C05FF44EAE07E1FF924F43256D7E1C7609F78387BAE42C2070F735839FA4D03B7F4058E37A1F
9CE9700C6DB6D100BB83E9EEA2D70F883162E0D1F293B4794325DC4EBF6283BBF2633B14EE7176
3D6246FA22E3469DC46AF21F3BC69692743EC97CB7EEB4883930381A1BABDD5A741F03D5E5138F
6278CFEAF0EA66B6F4F516B741C2FBB930AC16BCC0A8086A7A38D16FAE4018DB618F3728666440
F032792E3AE5088DE07DFA8187C6B133E1799B6135C916AEB78EFECD01B4D9E50337C03B9918AE
84C30CF326219D8ED924E33718116DF6C906EC00CFC4E76C55A15D24BA89A91C05CBF9CB505FA0
031FB2CE98C0710E639E033AF09FC9F3FBF56CA930CD83354BBBE7321B6455038C81D2F001E02E6
77DF65ED938C60C71BF8646B9AF9BB7E1602363B2A7163C12710F645F9E7F34A6EE0F1BDBA7DA3
B7BA0CB3A881CBABD57708120F4A66549C36F6E7CFBA07EDA5DFE38043C8C1DF5F110B46824F
C03685392610EC07B724B8AB7C23E662CF544BA61CE2788A4AB832C72F50E72976AB73C638E63F9
D9E07C4428670D31DA8BCBC851C0B94E583F8CA3A257926A05669D713097A4312EA7F116E359A6
44F1E913113084E5A4E40F9E16E22416660FE3242B0462FDEF44AA6E1B6729D0B73A4244EDE383F1
8B0C01D8D1563F88D3EBA7D162E6DF907077A60078BD705EF790B922441BD9BF947A0E4FE185BF
D86FC2747C6C7C523DC6153C426F84953F4B08A5075DDFA70FE89C61DD0C6C5EC47076716C93F8
1DE5B12DF9E9324F3CFE099DA40F076B387A3BD603B1959370CE666EDE7898070FF57FA038EA8F
7F16426CE09F4F62F822C82985D0A3CAC88086879B5F7BFC66708D175F7BE524F6E01919CA8D3840
7261008BA872E81F59E42FEC8C4482C68E4F923E468FAA4BDCEBBD1D79F5371140FA92F36ECC08
5FC5E159DACFE4E33C3761DE406379BBDBAF8A45A30C7E920ECF0BDFBBD020941ED5FE3D5CD46
E19B9B4D89085D90B2371FE6EB93245F2E0F0C9225751DF93BBF1E2F537381924070365B6F020D886
30693DF26BB2443525D6C1FA29D968D1F663BBD46C98D1B4D31F386FD2C905647D5C567FC1F963
58334B1B6DDFA5006A8F7EF3AAD5FBCB6AEB097104364CDAC052E0FD6235E53ACFEF1942C299
8191C8003938076440CF3F7AECE6CD56808FED9DF8DC20DA4DC74B38C5833D33219E358C1381011
83CA8220DF2CE8441054322CB916551A59F8001BF6CF207934F7DBD6596C2048141E729402F731F6
C83E645BD82183F84DE403C24166F33EBAB6301458E9D85FCDFE1448FC9DC4BC9FBC88543724B8
0422229B70E787C63F5F10D53137E8515056511523E7BA0612FD1CC1D8B0CABB6A8B962EDC37043

6C0C4C5BB3E16AEE977A9B842F7C80C9D4032B3180F622516CA982B03BCC2D5C6F4B74AE018AF
EF4F184D3FB291B0A4F7A1953BF189A7B28215FEDE53AC8DFDEEDFA47ECCFC4518E2C23739579
012BE68E1EC6B7375C23B1D11A6A10549913FD73A7B20915218D1F8DA1781B942D004DDA95E89A
B666B27439A1682878E6B4D997D96AC164DA8222B59D6DC4113EB3BB0FA6EBAAFC154369B4171
73C4615539F81BE4CAA27BFAC0A28641331779DBE67DEC432FC08A163E93AA2BBE279B9B74775
758557515C94B35856FFC93F199EC6410A5327CAB13B77D68B3F162386791F05EFC9A94CD03AABA2
005FC39C48087905EF7FF11341C84F78608878191B6E3C5AC7E4D676D88137A50B5DA641EEC9C882
E147BD768CC371AFF11F716C9EF07BBE5E4C872C9C724C8B0619E559CDE4FEDF62AF101786E271
6E4D8348FE96CFE697A4A813538AF189AFB5BB13E838FADA9E676A3077FB4B58C39B7FA749725
A2BFFD82F544351D607795B83B45654EE9F597E1CDC06F9C38E16BFB26C1256B1282B40431564DF
229576C4BB7978E0AF58E8A57315E7340B7D9768EBD0F8086A412819E40B510F9C2DE19A13B6F34
458211B54D4F05AB9CA796061C6CDEB8ED790FC0CACD3EC84425E636D9660041A8DF0313CC3A4
E81870353C82931ED95D28AA8D2BC09BB9142DDA0C94953662E3C81F05D96814DBA893AE2BBC
AC54D81EF6844F225F3517542AA1ECD80362098AAFF4409A3F2BE8E13BD15052C0CA4A824842D7
A7C6F72BA93373266A324EE8C309F323DD96EE92602D4549D325192002993C1AE31F19057B045DC
07C2EF0BF285EF5214C70AE3A18F26A700747160AB01142B5B036BAE0391F0DAAD7D296A7F9CA
0B8472ACBD3F52C17751FC910A8EE67FA5E26CAA56ACB560482BA97E0B892FA690C53B39D7DB
A913952060B866165314B06FF86EBE70DCACA86107950AC7B6FE65674305FF009942116A5069C12A
8BDBE189E6BC7CB7985238628E5BED4669E3AADEB1FC56184B4D52C3FE94439EA993F44DFAAE
761BB1929F6C02A8BDB67FD9AD20750D90B64A213252C1ED0EBA9030C1423461BDA4C8BB90E7
B0D7FE272745EEB1FFDB833473134ECC3B4A71D949413D4DB0F35BFF937EE0520499C6E6F113FA
491933684F5EDACE189F43FB615F881926768EC822B142E192D203AE303F0E9932BEACE33170C12D
845D153948B6EC454A5E0947971B8FD31F32CFAB8E3B39B431C9C15B7A9A48519CC5A0C DFA9AD
08D135E99D41E721FC563A104BB657B8D7A8F978B3ED4B80F0F2DAA4BD46632CCBF8667E8FE79
B59790D31827AF2FD51CFA4E7889D14E03A96165400374856BF23BA7426175164AA80B4882555DE7
48E044F31AD9512EE933DB3023F883D9ABBBEFD98EDC10520D3BD020C8F2EECEFEA37729C7F8
C64CB0E625AACD9D0A3A91657E12D251C4A8A0E5D012E2BFB4DBCCAC21888125DEEE36998033
3B37CEC90945638ED1632C73FF3D70B6B48C3E99A2E27F4DF8A58A81DC0C52DC056A7420E7291C
005B5CC77540FDCC995F04217842DC89BDAD1757FBC61D036F46AD7F665F527CD240D14353833F
B62F16DD308278E4AAF59A2F7FC1561D5D3C3E864370F28FBDC66DC4B1972D4FD9EBEC37F6579
5910D3EDCB567D4FCE51C71CC8E104FB0ACAD4FE0A3B6A846E2DE156FB360B94D2D55739EB78
97D14C2ADCA8739F16E1C32CEAF59691835018985CA9AFC8534338E1C4FEED593C14D9716312EB
1C2E44FB2D23BFE9448C85585FD026C1C71667AC55383D5D8D128203615591001F420F55680ADB69
CEBE136DA94EF254B65767F6343CC2B4A5750F9A6CA95642D7CC326946CD3E9EC8053C4554E5E0
30C00200446C3C15F0D2AD37B61763F6A7FBB2F4130487B35657A37AE17BCED68942C52E437A8B3
DB79C42BA544593F142471852954B847CF353259C26E15FD25132D114D5BD02DD130EAE930B1DE5
24CFE56D6EE8839FCD0779A5C9FE60C51FB09FAE36D33ECAD877F08CB873F2637B06A6D53E9F9
2A61FF96F66A74E6640A61123C21ABCD55C9FD1135AFA7F084ED5AEE3A0CF82121B76F64311B5
F58F06E63D3B3DBDCE0AC22E033028A97370E7FCF1CF937EBF0B08F80DDF9143E40293DF40EF4A
AC0B655501F5616C80F3536240E85685E12921EDE481383820A2594E1BF53A236305A83C6FA49F492
1035922D4AA8E4AAE28A39BF7D7E6213523426C77064DAE519C81C854A699AB01C44C2F9A06A6
DF21A1D5B73E5B1E141201588A1C5D8909B99F2BB0CE9D914B66704DC06D2E68A5AC9BEEFFE79
16904EE79703A143A20F130F56BA65C64FA73A7CC9A174E499D3A8BE0061A88500E969B175FB143
15FA5E2E7D6C8D9C8A8EC9E666000829E96EFFF8A98CF5A3D4725859FE670E0A448F14465B100A
FF8C6313EDB304EF904E89A359526B2DF9E4F4381EEA6AE9966F795B0283713BE797EF9CDEA6FB
700A6B2BCD9173C7077B4C93C76A4116B226CC2D80428C39E55138B8353003B2BD591504E0938C4

9EB3D4794DDC23BFD3E5F26A6B2C5E70AD46ED62F9FB67C1452F810497083A8497AE6B56BB384
DBBB1EBCC1F73D9B0E3B990678F52E75341DB9C87FAAC300A591855D96DC28178DF2AF793940D
0460673E0AE32BB2CFBEA085C5C4BFFE931DCACA4D830F0B36E7355D2407BADDC646813956ED
60C4E33E894CC71279BA6D1763CCDA47A6A173AF914B0E99F1007A22B5161112F573896103A6FC6
4E0E363FAAF167B4718915364AD3BE65CD2B8E2F949FAB736936F07B92F1D387F55EB543E10D8C
EE54869C13CDEF48A30945542D0F933AD752F1D54A8171D04D40C88772B6EF3BA43AA0D67EA9F
3BB585387EDD885A441F54C26CE6C439AE0A52EDD2003096AC1E2D5F37A48F3E33AE45D099B3D
1DA57D6815FD2425369B8054D6CD91D2B25E60D9720D66B55D6A0E6D98B2B58F1AF64E1952494C
5DCA6B06721F1DBB72D27D922DAB23E6C16DCF5976BB0F754705E0508DEA72488E7AD374D1D2
02BE819F362A9A8808F0B5ADF8E94954EACD30906A9155BE6F422AB92C6CEFCFB43FF10C69DE9
9C2C4C9E824B3EBEE97420170DCCDF32A877BF844A23D45595E8AA9DA52F14EF50CA5CC7AB8A
4D1E9C0DC1AB014971E589589C4996A89BBA2F0CAEFFFEFF87E7F12EC6B55B3C376EC52B6A906
3B13140AD2BF7491489AB74A366C73EFE17FF894B0FF7F5E177AEB55BDE2493AA590598CA35CC
C9147DAF377AB3BD74B3EC1B3B85A4B8C5A3F0ADB772D3C0BDF183DE89D35DB5B24B1E80148
BA76C055A7A2867D5D46A97B449B7FDE751699DB148AB67B0FE44543162158FF2C759E4B1EB3C3
201AB0CB6A85828266CDFB739E33A74EDAD64439B0D0A8260D45484923BA23A45BED4E3055F2D
83B522E2073D86A7438B41F0D80FB26EE414418B198A5E0EB16B38A5C9F7553A3B7B994CD3D80E
7A84F7A7BA2AAD739051DD1099B12960EF43B68EA580B16AF7B38958BEA0A02B23D8B7ED85186
8EE7DEB6870A6B47854C88AF5D45DE0E230DF6FB888EDD5E84262CDB4DCD26867A48446B9CFA
3EA2A1FDF02EB0BCF7F82F496BBC97190C0B692D5CBE5E7AFD304696C8D951CB4CDB4F53BAA
CACE709775AD0EE6371653E98B4368988E29415136BC5E2BE7A4CA996B414F31197FD3B9D60DE9
86DE42052E9360347D30F18A450F0B72FAF7D9C2BC7D8A9CA55E66DB5EC160B4FE592D01FAD4D
06E032045E5A12F1D5CD546547CE679F6A4D1092469F3A29C024E91B7765680599215BD4D7A20B7
E521AA80CF5CC0BEB19512386CA967A9A6F9E183A71A363FAA4F64D1428842512BD65D9080EA0
ED959715AA5047820F62EEDD3D4CF82A7183CE5C7538D6FF5E2FB8C7E2A419610E2B28821BECA
5EA5DE032F318A0C17E24BAE8ADBDF516E5CC288C115461E7BB8985E0D8E228529082CB96D1D5
4BD1E9B401E36EE4DF01DCAB3893084B4ACA018DAD2E27B69F0C2FFDDE4552D02A97DD8BA25
A2D5F919A1F2B42539974E2C55788CED56CB9BBFECC65AA2044DC13C33D6B6E91ECC19AC3C30
68169D9F56EB3E06445BA724E08C58CBAC8343F4EF78452F76F3ABD6B73A814DC29CE4BEA48784
8D64F07D306573847511C44D6782F8A06139564C026D62B73630882D90A964E36431F69CB84ADC2E
48E0B0635180D2CF1F0E733CF5D085BA2223B6A444E24AA2DD1CF67CF08DF524E283D3CFD1DCE
8F1330F0C5FE8B3D0CEA8B27EBBA18CFF8242C3F095F7A1051353190C0CA02E3321C673089CE6E7
67D699FD20E348DE559BE3D409A876D29BDE7FDBD2A8A47DF7A45F78C383104CD018CCA078270
0ABF6878698E1C46C773D1B62DD684870C9CE2C6DF6892E9AEB55F3936195F46692FC

ct =

FB0FC8FFF09358442CEF1C712F84F70A7360E52420942D63774C2EAF19AAF885F0B2A6B60D690E9
7B9D1CC300F0651428956617F8FDD294E998A83177BA54D5B2ED7065283B49700BF0B4C1B5A706D
B3A9BE2D13070207D5FAC1461F3214A1F6EBB54DA137FC53B0472D5A13E53978EF0D27FF05BBD
594074FEDA1B92628FD74745D96C075091165623E8EC647CD4217E90B6CBAF2F7AF8F0E7940A1CE
60931FE37C0A406B351A06EFBB11C43F0ECD490F217D43F0774716072C57D17F94B57E36316EB8
0B0D36BAEA0F187BDCE4B8B8EE41BC63444F1DFAA433E4610147C442CD80B6D0D8318275F1F13
2F738221C7542BB4DA8C61A8E8B76280BEA959A359B1167F1D47422039BDE2400A4C600348FCA3
F1B76E4EC19A293E05F7E0EB968BC20793A8AA26BCFDB629717494777781F664BF0C445FC4A0767
CD81E22959B6DC0CF33F81301B7F21D75F671B7452B5317665500F7BADDC36549FE3C67A75C44C67
2AA821060066956F25BC017865E1DDBAAF25E84183F4462A189665A480CB33C159B96485013756E6
5D76E75769F286E98005E5F68A85E71B13A8150D439A63300D0E4070829F94306DC85322FEA769CF

3B7C7400C8AA000DCB3A8415B514F21227927B9A1CD854D26E61E32F0F451A88F655ACBC1EBEC
DBB242128A2DD25F8187A9594BE15045D8A49DDA4CBA61ADC5F9606F8419337ABE36FBA963920
B9824B8C351DFBC966229732156BDE32B5D73EF8E6082336F41F7429456F8F026C1BB4121F7497391
681BD1A13673BF96558F1555C225B2BE0E351164E75C25A71A94D61383A07D64B03B35567DD3B59
81EA74CBC2604E4B0440A895DECE3923DBC071A058BF1D4FF3639CB1F27F86E425174A5FE79A81
64B219E26ABC66706CA67E35DCC7E1148DEB7121146093C1A2E167D1D1143300C6879ABEAB396
D093108979CE6D4130C2900004260DDA7906F680708A2AF46117A313391BA0142D0518459C1A2F7D
3254BB4AA77CE18E7D4BA598D52517913292580714C2B178AF14FF1D399CB53609FACDD936A6C6
3DE6BB9B203B3E655629126837011B085C47209274EFC74E63A9F25CD6029963CC7A6863252731542
D8E31B190821326602E027BC8890DF2F8B3A2B751303413562AD2A7575756AEF177327052A0521541
B66254DBB59E5BEC954DA5093B81D44A17CE1697E904D6925AB8FABB20F8FC973D16FB03BA230
8E4EB0D0613A381E4C40995F918AAB486F2883DDEA475BCB91224442D9F6C38ACE197017E29BD1
233D1080A458882554A6091AD5BB5B4205A606C414CF5F79F5560E24A1DCCCE5B2CBC4A43E38
801175B149FC8B4FC9A43484522256603CAB687A224C111DD059CEF99254E2874E4E238CE4B2C8A
83AC1662B7CC0EA74EA1C59A20CB8BB4A30AC70FE433FCB971F554B3C67DD6C5073C9869C158F
7B48D559C0C70BC17837220499B770B684AC1F90B55DEE1510CC649D6FC94552864280DA6024C51
419780A31D6EC4B5802DF63DBC4155340F6FA60F0C7653CC22232C79566108048C01943E5C7148BB
2491A7D5F4356D94618CC256ED607C6272A0E56F35D36EDCDD751D123E709A0B8D2A3779FD6AF
6B5E3294E17A7A698D1A3AAB4D05F4ED19AD7DC452AF14685FE4756D472330A0006C9968331D04
9E10445BCCCA762C7D65C28A01A68D93E5417FB3B3C5A19F7917585046E8DD2AC66F28BD90E337
569B990A8F9544D270DB72D15AB761E7D0887CEC450757E310B4A88EFF14394BDF7C33AD1FFAD
378887CECC051327BD15A231C03570C3136A357B92177D4931EA5571D026A12B1CB8FD9B41EC4B
75E4376704C6C70CA674A9050239CE944850D1A81596C3EE471DC6FC5F8F24E23DD0FC13273A01A
5061B707853D1E358241255E3A99EE754593748D5D25B53C896727065E384C770D7F3243339DE0442
88922A665C02ED28D95175FA0BED654E4569D8DA090FFBA613031A473B1A10FE5E7BA66BD1BEE
546F8E730510B4731B38885F4EE1378AB531826C64A9D64522651706645D52077C77DE62473FB7E57
F56B62CC13685B1C6C07BCA9FA40796BB245374809411645286BC4C669DE170284D42E2E5461ABA
6041FF2DAEE7055C3B69F1C05264692BDDA6F6E16301E3D539DCD17F147C9CD9671234BB68F086
154CDA4A29353E045153991B2A3A4DA4C44B9D194E6E64BDCC093E25C26A4AC0198D0169E74AD
1487B3F63B08C5F94CF66E6379D304274369478EF94E8947CC808440A6497C9615B6A670A3390F4B
D40914B96D95B18965DE1C07C4276502F369D341A1D07F781F673CE9203A9EE6B482460C416C539
DA2E41F0D5F08691A8808F1077BB8355B009AF9256406CC2269D88519E0E598D04F0BA69C039414
C6BAD06C8648B2890213F1B177B1115975D53BF6345E47F80D570A5ACDD268201C4EA43F92EA89
067FBA2000A9D52FFD1D05F45707C8DEEB1B0F05E101C0119D59B0498E3E0FB4B06197144AFE3C
C661EE3F02AB48755BB9A61E3CA084642BAE3987BF0FF7AD703A4085689736C1E875DBF96C92CD
5FB540354657CDF71F56A8FF1C10C0B19503A2D77439DCB40F50FB803D4C0BF20E30A40F636E026
C25961C12ABB1F0960070FBC1E01F23AB5C469CD12746BDFAEA5643D1C202F02E58B0DC664A83
7DA3317EA9D83A3E358EB08E3D94815044D40B3423666D54716B5366576B9E1BD148DE430060D1E
DB79B531F014D33AA8C42A14A0ED1C7E70C965B6F7BA786F9A70DFCBB146B1F097844F1517D20
6D9EDF53E5EDAB52D22E7C28CCB62185EEB31F29EA4304698723E10590CDAC7578132BFD1F85D
657F4DB9E709C492EF938838416AF604CB2EB432313BF7AC14B865F05677477C5007BA60525746EE
62C072C47E0B984F3AE975E6E92A81DD6E072420E09074B6366D42E6924241A3315E000DAABD384
4742DC502D58703D4D0933C3B5D2B1A1C544F35102E277CD982C1FC74875250C7B4DE13F229CB24
3875216118D810AFF7040C585607AF5CD71581F0ED40F49D3808093B6F5239C19709BE8F1C1B7FCA
445AE2D3950D0906B1F47C14D56B4DFEA487B5A7B7245040AB6ED2D1968CAE7B399535D9641791
4B5D53C483AFE1CCAA9F148F4025A6E11271138A26D8A85F75DFD365417FAEDDBA93535A7ACE

94C5C559488507429D71995884710F0E189D0360A05C4625A4C72F73C5B7A4C8AE1C736B60107B86
A79BD887D03882034B29260E61AD7521C7BE276D5981E641A2361376846A280D9B0A16C2D852BE9
4CFE906C9750CE2D103EA5179E596B24A2E3C59F94C190C7762BB6399557B1F7AE43D08D

ss_enc =
192971980D869F1E0A18481FE7ED968C56588C6CC5C8663D7086D0F1F70A089C9876859368DEFD6C
EBA0BD4D5095F08C

ss_dec =
192971980D869F1E0A18481FE7ED968C56588C6CC5C8663D7086D0F1F70A089C9876859368DEFD6C
EBA0BD4D5095F08C

result = PASS

case = 3

pk_len = 3009

sk_len = 15617

ct_len = 2353

ss_len = 48

pk =
BBC8398339915E97E9DE7576C9EFBA90EE4F4865AB03AC2BFD0BAE34D23B55CBEA3CF6CE54FD
15C9CCABCB42B5D713B2D1ED06CA86EB8DD88088FE1DE3F9B93B67D0697512919B5379E59E4F9
06F3B254AFF6729C070407FEE8D63AC2E1633ABFD13669CD103C5CBC08B34351B14D71C62943C
DC43CBBC87D0FD11D826AE533136645B5E9ABC26C84348EEBACEA702B32B53298BA7F1A76896
F2E9E47FF4B8A6400E7980686EB72E8AAC5D44FF4C44C1002AE290BE7DE80CBE3ABDF204B24F0
04C63328BB4C7B59FD05BC1741B489DFF19CCDA26CE6125C1A1343164A970A077063D8500F37AC
FE5A166E6F6113A4BAA99070E07798C600D5AEC7D474D839627B0ABC24027F07A0BD3B478BB4F
B19A91D0363F303D89E6C6701F54226C114D6F361AB7DFC2211510B8779A7ED849FBAD97047C2F5
AC258F3A8D113868431D58B1AB696703AB9AB8E4F60CD57D01F4AD511EF5AC46C5726ED7069BF
6D86CEC7ECB86A75B99AC276F5D97B7904C93BC7B35711261DAB2BF909B28A8A408C45E3C2C06
76F2241BFC96497AEA795212A45E690DE6948FE358F6C2580877D64B6904231B17EA511A09B676D7
807465EF79C476D5EFD50DDD55C68B4C78FA6B05FE252AC18D5CB7EA55F471C6C9F17A34E3883
F12BD7941B5560A83CE5B1BA8731AB1DC165863DD0DFBBC361C1535978DFDCB3B6AC65466DB7
8BA0FA1CDC19D500F14EC12EA566EB5C4780C290B3D89CDCCD63456D11E8AAC0CEA0F5B929F
D46B60A1E9B42FA1852459DAD0F4FF0B5A4CA81F7DB19FDD3D3ADB2000D24B1AAB3FF0578A1
B165FD7E5B0338A4FF1C95A22486D37701296C3A49767A91FF94B9430E374C6094AE53E3BCFCF45
74FB3952FF30C4C7F3591079B54EEC1FDE819BF23D0FCFB9057BC95A6F079A7D52B7F704245FCC
7578D0623CC4DDF7FBB6C2987090C463839E37BBB833D39EDD879426694F8D63645B1566C1384D3
8AC7FB3EF36A8369B8A2A2A9F6ABE949DD08FDC8524E067ABDE7AFB0F02E5E02BD5B26093AC
AF54F9FC0E0E60C4402A7358E3F48721708382BA031E5DAF07893F92521F0C9E88922C8B3AC46434
F738A28A44079D8B8FADF961C1D95D818D01DDC6DE8E6B623FFA168367CDB70C5AB802360D715
360B8B80F9F1EC66D7B26A9D6B7A034EF4F3B1C5F8282C6555C0DA8D58ED6685A94C29D44D68F

E267CAFFEA6F9A9AC94FEC9F9062B62EDC3678A149A95339C7CF02DE95B0CBADF86727AAF78B
86DE79E41AA2DE20F94BB25840F8E4BD16E2CE36699F4CE82E697A83997AAEB68136F7701684377
FB5B31F599E1CDB1E9CAB6D79234E6487A3BE0D5059A5F214157CC2F567D255FB86881541CE0D7
CC3AEE51946FBBA4DC865C715D1F76E44E5E5055B826B2329AB703E5A948A7A25F9341A2DEB77
F9126A10EAC176B679A359546C5CB105DB2C211EFC3CA0482DCBD20C6DA9AA54C64267255C1A
F1E8BE30C20E072A43545D5375E59450C5E72227BCE2BEFB8A97429BD9D0E687AEAEC447A862D
CC28A4B96991A205E2C1D9BD92B08A69A1B6C5C4BE4D7C5361E4358E628820E2479D9C446FBF8
BF35D3E33C1E4A9FCF725DEA4A32BB0419A7422BB2F3806980C0C4D9D30A27026AA28A70E1201
AC3DD59C7C09C01A58F185D9F5487C10C0A7141BDA68E2471C6C6F255483196A75CEF5212340B0
345356616DB94976A48B2CA6951F7D82AC8F3B3842192038CE043FF3AA75A90F6B0B2A113550F7
FA35751C9871F2085A4509CF1BEF2271F3D9CB8D02A99B5300B2DB75BBA8624101C2FDDE9CA94
817FB01E36254C39F6FE427864F51723A9A387F290494802E375356BC3C5B38E187B35F0C0BB516F4
65D017C060BFE554C5E03900D561F8FAA8592805E20F239B70EA5170061A98C02C30A07118E650B9
24A543D7A6CEC6126B0B652D2B53705C6671E5C66031500EF833BF16FC7FF2307443AD3742B9DAF
8C05E53D5A1BA085335B9980D921780E2F81A1087C674E56557B2516F81CAE5929AF3033A4CDAB
CF21BC344DACCCA5A3149F68BE9305F3B313234036F80A01318604BB0C6F7BE085FCC9B7EE6839
F625D701864C5EAC5DF72CE8816CC73B79CBFA11D18054817265F97C2A69BEE5FD3955FD5BC50
BD0EF48D6256DD0FDEFBB939BC374F5675D414C1A501A086C2379A5F2AC2D933FEE465D65CE93
3BFA8403214EE64C4C8B15404BA30577CD6FC51388F0CD41DEDD7A429D94FD51C12BF79484EE5
5256C0AFE5179AB9AEA916BF99F137A66059860F0EB4FF2C4807C73B5CDD7B4786D4674E8112A2
C91CDA62D63A5C05955970210EE52CE2EEE1E6FAC5C96C421A79A8EF6A70625815B8E2EE5895F
AC4D339FC4DBC0EA548653655E755D8EEF00A2AB7617B11DE4C4B066802E64B0E63C75FE1C8454
684A40ACDBE68B4BDD77CDD52017575AD6CED5AAEDF00B1B9156689AE18ED6A6B6F40166335
8F01EF639C9E2123ADACBB9A36C1BE8B871207F830EC3175503D6EBD1140C19184AE0BBEEF3281
CB3ED3061816383147524FB540CC35B542C9DDF16BECFBA346E4A5CB906562D6BECB0AD528247
9BE0DF34E6C95B35DB4475A4E369452C735EAC3332CD9443FB4ECC1B76DB1D2C5E76B6188EB66
360CAC84AC50A338F0F3F5BFED50214059C608EB65B92A5F6B6A324D72E95D148F0921A943F7C4
322C70E03831E717C6E34F598CD8F7DDD80CAFC683F4FA3C2E7E0BEA93299BE95C6A82C5B5DB6
AEF52865CA22750F4A96EA17CB34DF1DB042AE600827CD1A6B7769B5CE2CBB708D519F99EA969
2A314035F58A343A9BEF58A4412D2FA4D34EBFBE17DC1C74AFE62F980D2C03498E0FD0C8AC52F
E3C4D1A086B1AB1B193B5F7BAB65614F5AA7A1AFA9DF1420ADF2A14431C737C59BA2BDB09F7
84E207D0B4608AAE4D96E91B74F7F503FCD133ACF61F5E420EC6693BD6AD6AD9C8023269938D7
E49E1269FE8C97458D2740D97DDC868786F2A6A06D9BF3A640737E1FA55B3A9AD1E6CA78553CF
62908EBE856ADEB8BA174C677FAFB74AA388D72217222B4DA3537956DA5647564186CD8621B544
9C32937B0D2120B61C971A82E6A081C4054392B0920760435F40F3D60718CCC0676FA4CEB8F1E01
D064DB49C0405A2F274610E97120A26F05B6DBC6B55A2A411E7633D6C66E90A4B163450DC15184
FF80FF4FF4CAFD8368D3422A104CB7AFE1601CC23051ADC98BA9E8AB9F296F6DB3C1003C6DBD
0F84EF8A3FBC048AF1CCBF9F8A2016C1012EE38925BB8EFB8044B621587626AD3371FF90ACD5EF
21F91C99C93A0D9F105865C9D849630780C8A3AB45B8965DA9053478D033CC13B45C3C34E3B8346
05278DBE8CE1D0B4AF00B4B886D1BAA5F8A68D41502CAA2A01162EB8C09D2897403DA48F1A22
61E2E1363E0E6F21B72708A2F9E51C4909C42889DBD0ADF8F0B8D88439AAF573DE87993C75976A
D48CD9260347E504E989F004464E810291A6CF6A64C416D028EBF5BDE5CADD7F077F63AC99672D8
812E43896206BA6FCB8C91601956DB53E49F970F66B7BF9856FAA5CA43C976C51FBFE008AE4435F
96472CB7662675CB5C70319AA89EA7DAFEAB753F6C1BF453C3CFADFA97F36B816B508D5E9D276
D3645D2B0967E4367B5BC174AC4A57DAF36009D37A0FB09D6DBD497B809A73AA8F50080C71009
A1D2D15B16590165262434C7E86F176A404EB4A73F5FCBE1720017347772798381A2481B95B906932

A2B5D70820F0544C3DDB7F26517072148346B623E0353A10BFFDB6A4057D3AD78281643D085DB44
DE0EC60DFB093316230B8356CB0E456A946389E8F0E7EF268BB56A48F0C9235B8B38D6EEEA0B90
6E12192C4E004A4B3B069F610710E14E7A4014CC441ED13D38D7EB05BEEBEC5CC6E4BA9FDD24
DF4CE90C21864C09DCC01D23C83B35B555195A8F8DB7817256EB3C16B97AD52CF4F0A91BBA696
47B05AD4FC4A86E90B5D11C0D671556BFF6548A99F03FFDAED2A5FB32B00BF68EF3CEB0812E54
5C8DCB793A6DB754BE6801E263FE3C2B7D6C964C88B1A8B4DABB94667302C04726A46083B0D63
0351AD0B184C8599EC7EFA14F39BBAC2A575D264E3AB89C26E7124A31D9F7BF8EE7914DDE36A6
74CA22FE1DABD333905D770FB1C545A7525297963834729C0BA991811FE91DD44D3D49AF3D26FF
05284B44DF2553B6624D64081A3C28A6DEDC5788B4CD56E97D049E71FAD721389C41D78620210D
999310B9F31085F9499FA60D31F55890EAA981E002F246A3499

sk =

3B339B724A0425C86BEE58E15B05C4206CC0E63D9009AAAC8CB446C37870C69B37089D8820122B
E3CE84C893BA3CD178148E8699117DF023E283E3B5E3BA9AF11E1FF001F0110F0FFFF000000FE11
EF11D0FF0100F0002D00F01FF30F10F111200111001EF00010F10101F0121F000020F110F101F0F10110
FEFF11110F12F111F0FF01F1120FE00F10F1001000F10F100102101F0100F0E10001F1F1F0011FE10100
F101F1F11F00FF0000E1000FD001F00F1111FF1F1F0100001F0F00F00E10D10101040F120FFF10F01F0
0F00F010FD21000010E12100F0000F00F1F211012FEFF10FFF000F21101F000F100121FFF0F0F100FE1
F00F21E01100F010F0201000F10110E11000110FE002002FF01F01F011E00F11FF12F111F1000120F110
20F11011101020F0210F00111011110100FF0F010F0F110E1110010F10F00000000EE000F10001000E201
00F000F000F00F10E0F0F10011001110FE10E1001E0010FF1FFF00E12101FF01001F0000E10F0F00FFF
F100E1010110010F000101F001F0E1F100F2FF0F0000101F10110001F0110110F0F2110000E1F01FFF2F
FF0200FF10000120FFF1F0F01F0000F0FF21F1101F000F1F200F1E020010F01111FF0F1EF0FF0E02111
F00101F01FF1010E01FF0FF11F10E1FFFE00000F0000EF02011FF101F100F100100F0FF1100100F00011
0E01F010F0011F101F002F0010F0000E100011F0010F0101110100FF00F0D11F00000010FF0F002111FF
0FF1120F021FFF00F0E10110E00FFF2010F0101FF01E0F1EF01F10FF100E000F0102F0F0010F20EF00F
000E0110D1F1100010E01F10200001FF002E200010101010F10F1F01001F10010101FF110F01101101FF
FF0F10FF000E12FF10EFFF002F00FF00FF0002001100F00F0F0200100001FF0FF010F0FF100100F1F02
F000000F11F2FF0021F010002F010F011F0F00000FF0E10FFF11110F010FF10E0F1000F1000E200FF0F0
1F0F1100100F10F111E1E0FF1000E00001F0010111100F100FF00F0001E0000F11FE00110FFF0200F010
0FF0F020F0103F001F0002110F1000100FFF011210F10F11010001EFEF000F1FFFF1F0EF00F00000F100
10F01F00110001120F0FE0100D0F10000F0F10F1100F110210E0F00010F1F10F10FF0E0000F0EF010100
F110FF0FF1F00F1001120F001EF020F11F0000F101012EE0E011FF00FF0F11111EE30F10000FF00F122
021E02100F11F2000000111EF0F1F020F0FF00E02F00E110000002EFD011FF0F01F011011E000F0FFF1
F0DD0010F0E0F201F0EFEF1112E0F1000FF0001E000E000011010F10EFF1000110001002E100E200F12
001F0100F111021E0FF0E0FF0F2000FF0101110000FF1FF1001FFFF11E00FF010FFF01F0021000F01F0
FF0000F0E1001F1200010E010002000000000E00FF001201E01010FE001F0FF102100000F1000F100020
FF000F10E0F00DF02000FF01F01000101001120F100EF00112110F001FF00F10FE21E10F00000FE011F
0F00100020000EF01001FE0F1F2000F0010EFFFFFFF1F10000001000010F02001E0011F1F0F1FF10F0010
00011210FE0F21100F1001FF11F100F10F100000F0010E0020100EF2F00101F0F0F13000F11F1100FF11
001100F00F00FF000F10100F000110011F20100001200E01EF00100F000110F0012000F0F00002F110F00
F0F200E1FF10DE0000F11F000FE01F1000000F100000FF201F10F00FF2FFEE3021010F00110F10F0110
F00102E1F11EF001F1F000EF101011F100EF010FF00E00E000000010F00001F0DF1F01FF10E000F0F00
1010010FE1E11EF0000F0F01F2112F1FF00F0000001F0F01011101100E010E00011E011FFD1F0000EFF
FF0FEFF00110001000F0012F0F2D1F10FE10F1FF01100021FF00E1EFF10FF0FF000F110F102FF1F1012
111212F1F01E0021F2F00112122E00EFFE1F1100EFF1F010F021000010002F1121000FE21000F11100F1
F000F000210200010201010000000F00F0110F1F00F000F001011FFE200F02F001010FF11201000F011F

00F01FF01001000000FF100F100101EEE10101000F0FF11F1F0FF011F1F0F011F022011010FFFF0FF111
00000F010101F11E00000F0FF0100FFF110FF001000F1100EE00F110011FFF0012FF0E110011F10F1E00
FF000F0020F2FF00010000EF00020F1FFF111F01020F000F00F12F0FF20121F0FEFE21F10F0E0FF0F1F
000111F0100000F12010211101F110F1100FEEE00FF0F20FFF220001001F01101F0F1FF0F00F1FF00F01
00F20000F011010F010F1E0EF000F1FF110FE0E0E0E1F001F00F201FF1101000101F01020000002FF0F0
0E0F0110F1100F1FF00FFEF0211FF002F010000EF0E11F1F00FF00F0F0001E0F01F0211F101F0EEE011
201F1100FF0010F000001200FFF01FFE0FFF10101F00FF00EF1F0F0E001FFFF0000F12111F1F01E01F1
00E0F12F2F0F1101FE100E0EF0001FF1200E10021F0F0EFE2000F3FF0F0FEF02E102FF00100F1021D0
000101EFEF0F1F110001100100E001FFF00F12E111F1F000F000FE001E101100000F0000210F00F01000
0F00F0F010F01F01F1F000FF1FF00FF00F1100E1000210000F0F01F10F2F00201F0211E00020F00FF000
E10F0FF00F110EF01110F1EFF010F0F0E1010001101F001F10FFF0CFE00011010F00110E111FE21000F
0F1000F000E1FFF0100F1E0010E1003F0E21011101100F000F2210F101FEFF0110F00F20111E0201E111
FFF0F010011EF00101F10100F100FFF0EF100F01111001F0F1F000302100F11F11120111F0FF1000FF10
0E0EF000001EF0001110010100FFFF00F021001FF1F000F0F00010012001102212200000F00200F001020
0001010F00EE1F000110F010F011100F1000E01111F1F0FF11221210000000012D1F2E101FE01FF00320
010FF0E1010010100FFF1EEF010F0F00FF0FEF11000F10000001F0003011FF701300CFF1FFF012005FF
200000FFF600D000013006013FDC006FF5FF7FF500CFF6FE4002FFFFF9FF0001FEFFFC00200C00C01
C011FFD00000AFECFF5FF9006FFF009FEFFF9FFFFFEFE9FFDFFA022006FEDFFAFF7006FEE00C00
8FF8FF6FFF007002FEB00DFE8FF2003FFFEDFF0010010FEBFFF013FFBFFAFEAFFCFF600700A009
FE8009002012FEA00401AFF5004003003003FFBFF3FFEFF600300001DFF8FFA00D000FEFFE400900E
FFE00200CFDFF1012FFDFF3FF3002FE9FF400C000002FED000FFA001FFD011FE9FF4FFCFF500200
101000BFF2FF3010006FFF000019FF2FF5000FFD005018FF6FFBFFE008FDFFF7018FF6FF0024FF8FF8
FFAFF8FFFEFF600BFF800300C002FF7009007FFCFEF002010FFDFEC001FFDFFBFF7001FEBFFF001F
FE004FE3003005FF800B01AFF3000FF8010008003FF8003006019FF100200600F000FFBFF9000FF8FFA
006FF6FF2001FFAFF8FFBFFF01000BFF900EFFEFF300100FFE9FF7FFAFF700D004FF9FFFFF5000FF
B004FFDFF4FF7000000FF000100FFFF00EFEB00CFF3006FF8FF301CFF6FFF008002FFD004FEB00AF
E5FF100AFF4014FF5001017003FE8FF9013FF4FF2010FF1FF000500CFEC01A007007FEAFFCFF601A0
0A013009002FE9015FF0FFC000FFE019004008FFFFF901C00B005005024FEFFFB00900100400C013FF
7FF6FFD00100C017FE3002FE9FFF008FEAFF300D005FE7FF0FFDFF8FF1FF2007008FFF00100EFFBF
F3005FEFFD00100AFF3FF2FF4004006000009FFBFF6005003FF7FDA00F00B00FFEAFE800B013FFE
FF700900400D003009FF500700CFEB006FF6006FF5FF7009008006FF7001001FFBFF4003FF301401700
AFF800200D009FFA002FFD004FF6FED005014006004FF501CFEB005007FE5FFB008FF4FE7FF7FFE0
0CFF7005FF4007015FF2FF3FFD01800600101100FFFEFFFFED00B00600A006FFCFFD00C00BFEC005
006FF3FFEFF9016019FFFE500DFF1FFA007003000FF7014FF0FEBFF2FFAFF5000FFB005FF4FF9FF9
FF4FFF007FF0FFA00DFFEFF7FEBFEDFE8012007FE9FF1004FFDFFAFF5FFC007FFAFF2FEEFFCFF4
FEF006FFC003FFFFE7FF6003FF6FF8FFB002012004006FFDFF8FE5022FF2FEBFFC003009FFD002FF
A000005FF3FFDFFDFE1FFBFF400B008FFB01B009002FFDFFB00300701000900401400300E00B008F
EEFFB00D01AFE8013FE800600100E012FF3FFAFFDFFD018005008004FFA00C00AFEB006FF4FFA00
4FF701E013FFBFF800AFEC00BFFBFF3006001FFF00AFED005006004FF3001012FEF00200D01101401
4FFC017FF2FF700100D006FFC001FF2000FFEFF600700FFF8000FFFFFEFFA005006FF2000027FF6011
00100BFF7FE3FF9FFFFF6FE9FF1018004002FFFFF0012FEEFFDFF6002FF2008008002FEE00700A009
FF100E007004FF2FEEFF601BFF800AFF1FFEFFDFFD020FF9FF3FFCFF3FE9FEBFFAFFCFFDFF6010
FF7003009FFEFFFFF8FFCFFF00BFFFFF4005FF400AFFD008005FEBFFFFE9FF701600700300E010FE
FFFCFFBFFFEFE1FF800B012FFDFE700700000300F009004001006FF3009008FFF008FF800A004010003
FF8000002FEA00A01FFEFEF009016FFEFF9011011008FFFFF900E00DFFF00B00F012FFE003FFA01
CFFF01EFF0006FF8FEEFFCFF8FF301300400CFEFFFCCFFD00DFE9009FFC00700300A002FF100C002F

F5FEF008FF9013009009FFA00DFE600600AFFAFF9FFFFFF8010002FF0FF000401DFF7FF6FFFFFF4FFE
FFDFF8FFF007000FEE00400BFFDFFD010003000FF800CFFBFF501EFFBFEA008005014002FFDFF1F
F8FE900FFFCFF701000CFEE004FF8FF1FFEFF5003004025004004FFEF501CFFCFEFFFFC008FF200A
FF900DFE600C000FF8003001018FFBFFB000004FF1FEB004FFAFE6001FF4005FFE01CFF1011003FEF
FE2FF2019FE9FF800701200100600AFFB012FFCFFF00100B017FEC012FFE00601500AFFDFF9FFF00F
FEF00A00F00000B004FEFFFD003000FF0FF401B000FF2FED00800401000A00100400AFFE002FEEFF
BFF100CFFE013FFBFEFFFF302001200000700101C01100500800C005000FFAFEC00BFFB00A00FFF800
0FFA00F00700500501200BFF0FF900300BFF700EFED006FF2FF5001003FEDFF8006004000005FF2FFF
FF6001FF700100AFFB002005019FF5FF1001008016FFA005FFF006FF5006FFB008007FFF00C002FFBF
FB005004FFC001FF500E00EFBFFA00EFF9FFC00F00A006FFDFE500E008006FF0FEE014FFCFEAFF
B014FFD001FF100CFF200A024008FFBFFDFFC014010015FF7006009012FF4FE4FF8FEE006000010FE
B006FFA00BFFE00200E00400AFF7FF5003001FF2FFFFFF2FFB001FE7FFE009FF7FF7FEEFFA008FF7F
E4004FFD00200D01B00A01CFF2008FF6002FE9009003FFF003FFAFF8FE3014019FF4002FFA008FF70
1BFFD008013FFB007FF8FFE00400CFF0FE8003001FFB01EFFEFFFF4FF2FF3003FF9FFF000008FE1
FE800F00C008004008FFCFFC009FE3FE0005FEB002FF1FF3003FF6003000FF1020FEF00CFF9FEC00F
005004002FF9008FF000D010FFB001001FF901DFF4FFC000FF6FEE00A003011019003007006FFFFFFDF
F100201501500A01000A01B002FF3FF3FF0000FFBFF400E000006FF8016FFA002010002FFFFE9002FE
C00700A014FFBFEFFFA00DFEEFF7FE600700CFFE008FFCFE5FF500E00000A012007008001009FFD0
09FF300500B009004001FFCFF200FFFCFFD004FF8FF3002FF5FEF00EFF5012002013FDBFFC0010010
0201CFF9FF8FFBFF8004009014FF7FFC00300400600200FFF2FFC01500100B002FFA00200CFFC00A0
07000FFF001004FFE01200B010FECFEDFFBFF6002007FF9FFF00C006FF7FF900B007FFE003001FFA0
01003FF200B008FEBFEAFFE700100300100C005005FFF019FF4008006FFA01200C003013004FF6FF
4FF4FFBFEE026014FF2FF5010015FFF008007FF600CFFDFE800F00A00C007FF6009FF3FFDFF000FFF
AFFD00E00F007004005FF7FED00AFF1FFEFFA00600000300600100600800F006025FE0006FF4FFFFF
D005001FFBFF8022002FFC002FFB00100EFE2000008004005FFDFFAFFA021FF2008000FF9010000FE
4FF2001005FF9FE8FE000E008003019FFA003004FF7FF6015FEE00700AFF400100200BFE5FF6FFE001
FF000C014000FF7003004000FF0006FF9010006003FF9002008FFF00B00400700B009FF600300AFFB00
8FFDFE6008007002FE0003009000FF1FFBFF7000003002FFF020027010FEA018013014FFE006FF7001F
F000A000FFB007006000FED008019002FF6FFB00C00200BFF4FDE00CFF8FF900C006005FF7FF0004F
F901301100600B00800AFFBFFD00CFFF00CFFDFE4009FFDFFEEFFD00EFFDFF4FF1003016001001FF
D015FF3002004FFE004FFBFFD01300A013009FF700DFFBFE7000FFF00BFF2FFF00600100A00A0040
07FEEFFE00901600B01AFFFFFFB001016005FE9FFD01C008FE300F008FFF006FFCFF0005FFDFFDFF3
001FFCFEA00300FFEE00800200401900DFF7FFC00DFF8000FF4016FFCFFBFFCFFBFECFFDFFD0090
03004FF4013FFBFFDFFA007FEF01200100C007023FFF010FFAFEBFFA002005017008010FEEFEF011F
DCFFBFF8004FF9FF6006001020013FFB00000DFFBFFA00AFF1000FFF00BFF401F00EFF3FE1020FF5
FE5FF4007FEC00B01A000006009FF8FFF00600BFFE018009FF5FF900A00C018006005004FFAFFDFF
A00DFFE00EFEAFFDFF100AFF000100EFF7FFEFF0000009000006FFA00800A011FFC008FFF02200B0
03FFCFEC014008FFDFFE00CFF3FF200F024FFCFFF004FF500BFFAFEEFFF8FF000EFF6009002FFEFF3
018FF5FF400F00A00E00C009FFCFF9009000FF8FEFFFAFF8FF4FF2FF7007FF7013FFDFF6010000FF2
FFD003FF9005006009FFD00C00E01600AFFE00300700FFF800B00000500500CFF5FF401900EFA0000
0EFF6FFBFFBFF5FFBFF5FF100C002FFAFF5FF500400CFECFEAFF3009FF3004FFFFFF900BFF7FF700
401BFFBFF9FF9FECFFE00E003FEB008007017FEEFF600CFFFFE4008010003005FDFFFCFF9FFE006F
FBFF8FFA00E008011011001010008FE900DFF000EFF0000FF600DFFE00EFE9FFD00200200600101A0
11FEE00D00A002FF2FE700A004008FE201D00700DFE8006FFA00B00500AFFBFF300900C003FFCFF
B015FF0FFA02301901BFF5019FF1008000FFC00900800C005006008FFDFF2010004FFB00BFF8002016
FF200E001FFDFFEFF400E001002FF0006FFCFF8007005007FFDFFEFFFD006FFDFFEE000FFE00EFB00

5FFD000FFC00A00C00D00D00AFF5015FF1008FFCFF8003FF2FF1FE1FF601200700100000CFFDFFBF
FEFF8018005FF6015FE800000600B002004FF8FDEFF9007019FFE00A00AFFC00200101600C004FFA0
0500E00A00700DFEBFDB014FFB005FFE014FF8FF500500200600F002FF9FFA001FFFFFF701400EFF30
04FED007003FFE00EFF2FF0FF6FF4FFBFF3FFCFFDFF9FF5009010FDD00200000E00200200501E009F
F3000FF5009009009002027012004FE900F003FEC003FFFFFF100DFF5004003005016FFE00500B002009
FFCFE7FFDFF7FF4019FF9FF5007FF6FF7FF8FFEFF8004FF8FFF002003006FFE008005FF7FF2005FF0
004FE6FEE009021004FF501000AFFF014FF2FE2001000008012002FF8002FF3010001FE6FE100000B00
700E005001FFA000016FFB001002003FDDEF2011002006018FFFFFFCFF302000BFF4FF5FFB00701AFF
AFF0FFFFFF2FE4FEBFF100C00401300AFF6FF4000021FF0FF5FFDFFF00A009019FF6005000FE6FFC0
09FFDFFFFFFCFEFEEFF5006FE5FF6FFEFF400500A012000FF6FEF003FF7FF9003FFFFFFE000FE
DFFBFF3FF400500C008FEFFFF700D00BFF3FEA00800300A012016FF9FEDFE8005FFFFE500FFFE012F
F8FF801C004FF5008FFA010FF2FFCFEE00BFFCFF7007FFA00500A000FFFFFFB006009FF8013FF500F
FF4009FFDFF900A004FEE001012FE901200000FFEA00000AFFFFFF5FEFFF500100C005FF0FF6FDDFE
F012FFD005FF2FFFFFFE017FFB00F00C003004FECFF7FFDFFB002FEE00800A01B012FF2014FFC0070
0600E013FFDFFBFF6006FF8009FE400A005FF4FF3FFA006000FFAFF4FF7FF101500FFFA004FF8FFB
FFFFFF7001FFB01000F003FFB004FF9001FECFFAFF3FFB011FFB002FFC00B01D00AFFEFED00AFEE
00B00DFDFFE4FF7FFBFF3FFC003007FF500D004012003002FF8FE9FEF00BFF8FF0001014005004FF4
00AFF4FEEFFB00E02100D003015FFEFF3FFBFFC00100601800101D009FFEFFFFF010001001012FF3
00C00500600401301900A01AFEF01AFE800CFF6004FF9000000FFDFFE8FF7001FEFFF2FF8004009FED
009016FF300BFF0FFF00D01C012FFEFA00200FFF800D006005FDCFEDFF2FFC01101800B010003FF
EFEB00C007FEA009008012005FF7004FEDFFB003FFE00D00500D00D004FF800D010015FFB00E002F
FCFFBFFEFFFFFA01600BFF9FEAFF300D009002FEF006FE2014007016FFEFE01100900300EFF100E
012FF4FF9FEE00900F01401300300EFEEFFDFF3012FF7002FFDFFE00400AFFAFFFFFFB006001FFBFF
D006FF2003FF4004007FEFFECFFF020015FF8001FFAFFCFF5FFDFF2007001008FF8000FEFEFE00DF
E9FF9FF500C00000700F00CFF9FFDFFD002011014007016FE401200100BFFA01300CFEC017FF6FFDF
EDFFD005FFEFF8011FF8000FF7FF2015004012FF2002006004FED004FF4FFA00C00400FFFF01300901
0013016000003FF4004FF8002FEF016FFB012FF0003018FED004FF9007002013FF7008FF9000FF800A0
1AFF8006FEFF2FF6009FFB000002FF6FF500BFF4FFC00C00300700F00500100BFF1FE7FFA0270020
0800E00BFEEFEB00A007FFA006003016001FFBFE3FFEFF400200D00800CFF8FF5FFBFF2005FFE016
00000AFFF00D010FFB000017008FFB00B00E00E000FDFFF9FFFF400FFF3FF2FF8018FFE01CFFF00
0006FFE004FF5FF0011FF9FE8009FFFFFFFF5FF6FEFEFAFDB014003FF3FFFFFFE00CFFA00CFFFFF1
FF301400F00AFED00600A00C012FFE000FE9FFA005FF9FE1004005000FFF00EFEB01B002FF9FF0019
004007006009FFC002FEF004FF4FFFE8001FF900000B009004FFAFFAFF6FFB001002011009FFF0050
07002004000FFC00101100600B019001FF2FF5FFD00C009002005018008FF101400900101200DFF3005F
EEFFEFFCFFC008FE800600B007FF6FF7FFA000FE5000013FF1011014FFF002010FF2FEC015FFFFFF20
04FF8007FFA00BFEAFFF000FF8000006FF8FF4009FFD004FFB00A00D00D003009000001FFDFEBFF6
FED025FF0002001FFD009FFE01BFECFF7005013FF5003009011FF7FF3003FFD010006007FE7FF4FFE
012FF9FECFF2FFFFFC003FFA016FE2005FF900A00100D009FF6FFCFFE004FF9010007FF0006FF2011
FFC003FF7FED010FEFFE3FF1FFCFF9FFEFF1FF8010004FE8FF8FF1FF900A00BFFD003FF1FF5FE9F
FE009FFB00200AFF6024003007FE9FFF00AFF1FFA00A014011007FEB012003006010007FFFFFFD0080
17007FF6FF5FEFFFEFFFFFFF9FFF01500500100300C00500BFF400DFF5FE700001200F00AFEDFFBFF9F
E8004FF8FED00301500900B00901C00700900E01400A005010001FF9FF5005003006FFAFED01000401
C010008008007004008FF500B004002FF901CFF9FFEFFC00900A008FFC003004FF6FEDFE9001008FF2
01D00700200DFFDFF2FF3FFF00200C007002FF8004FFC00601100AFE4FEFFF6FEDFED01300000900
1FFEFF5004FE5003FECFEDFFB014008003FFB002FEE00700101200500DFF600A00BFFB00C00801E0
0100D00DFF900000BFFD01400BFFEFFCFF000F00C01200FFFA010001011002007FFA00400A00E0260

1300DFFE001FE5000003FF8FFFFED01001CFF000A017FF9009FF900300D00D01B004000FF6004FFAF
FB00AFEBFF0FEC005FF800AFF8014011003006FFC01DFFF004FFDFFCFFB00D00CFFDFF9FF300CF
FC00EFAFF7FEE005001FFDFFEFFB00B000000004FF200D000FFAFF9FF800500CFF3FFB010FFE00
200700B00500700C017FF4FFA002FF600C00A013003FFBFF6FFD00701FFF7005FFE001FFD00C01601
200800500900601000300800AFDC00BFF7001FFF00000B007FF5FFAFF7FE8FF700E00F008FF6FFA00
7FFAFF1FEE01000300AFFCFE7001012009001006FEC01E003FF800AFF4015FFBFFAFFC008FEBFE0
FFCFF3FF0008FFB007000FE500500400AFF500C006001FFD001FE3007FF8FFA006FFC001FFEFE9008
00B003018FF9FFFFFF600400EFF6012FFEFF3008016FF5006FFAFFFFF1007FF402300B000FF7000FE9F
DDFF6FEEFE4FFB013000010003FFE00D00DFF0FFEFF500CFFBFF001B00700900500DFFCFDFF0FF
3FF6001006000010FF6012FFAFFFFF0C012003FF9004006FF300B01D009003FE0FEF008FFFFEB010FF7
FEDFF9015FFE01DFF9FE7FFC005001FFDFF9FE9FFCFFDFEFFF1000004004FE6FECFE2FFC00E0050
1CFF4FF6FF2006001FF100BFEDFF9FFA005007FF0FECFFC00DFF202000A010010FEA00700500A001
001FF5FF500200BFEEFF2012FF600D017FF5FF7FEBFFFFFF6009010FFB003016008FFA008007FEF00D
009017006FFF005004FFF007FFB00B005FFFFFF7FEB000016FEAFFB002FF2FE7FEDFF1FF9FEF009006
00400BFF7FF2006FF4FF6009FF2002003FFD018FFDFFA005FE900BFF8FFE01900A017FF700EFFCF
9FFB010FF1FFAFF8012019FF3005004FFA008004FF8FFEFFFF003006006FF0002FFCFE004FFD01000
2005004FF2007FFEFFCFFC004FE5FF4FFDFF2FF5FFF008007009FFFFFF8FFBFEB00CFFF00F00201200
B001FF8FF3FF3FF1FF800C003FEDFFC00C00600100800800E004FF1FFAFEEFF5FF2FF4FFC00C0000
0BFFE007FFEFF5002FF3FF300C015FFEFFBFF702100AFEB01DFFF003FFA005015FFEFFA009015FF
BFFD01000CFFCFF1FFFFFFE008006FFE006002FEF00500FFF5003FEB004009FF4FF6FF100FFFAFF7F
EAFE9FF4FEE001FE7FFB00900CFF301600B001FFCFFC014FF7FFC019005004FF7FEEFF801600CFF5
FFF001FFC020FF400801300500801B00BFF700F00100A00FFFEFFAFFFFFAFED000FFC005003005FF
BFF8FF5FFDFFC00301500801C003018001018FFC006FF6FFC00B007FF6FF2FEE002010007FF0FEE00
1FF800B003012FF9004FFAFFC000FED010FF2009FFFFFFAFF800CFF0FFAFF7FF8FFB002FF800EFFB
FFC009FEFFFB00DFF7FF8FF9001007FF6009006006FE300300C004FED00C00901CFE8007FFBFF5000
FE4FFA027FF9020006006FF4006FFDFF0003FF901100201301AFF3FEAFF801300700AFFCFFE001000
FF2005FFB00800600200000000100F01500C01601AFF007FFBFECFF1FF7FF9FF6005000FFEFF9FF40
02FFC00F00E009003FF7001FFA001FFC006FF9FEE015FED008FED00B001FFC00E00D00800AFEEFF
A00FFE8FFF00AFFD00AFFE00100D01200EFF9FF9FF8000FE8FF3FF500BFFAFF7FFFFFFAFF2FFAFF
5011FF3FFFFFF700DFF4FF7FFE00900500FFF6FF2FFBFF7FF2007FFF008FE400FFF8FFB00AFFCFF3F
F601F00200500600D2CCE1C8AF1DD23F35DB0480B705FE3E1BFFEBD604348D53877014DE9884C11
639707FC489D25F1027E5A9FBBA13E735D18521EF5AA8409C945C1029F23E7AA5B73945C80CFA59
D3070AF683FD09E38C3C05CCC02BCCFFD00E737873985FE22AF68777B261C76E30681FAB2C93EF
BAF8AA6B178115859F91BCEF1F5BFF4248DCBF7C739F302BE49A3C75E0B55B0989B9DBE6461D2
8098504501EE1BE37E43DEDF2C22952E9878C8399C22C55C0533C091D7A7A0FEC8C6D3C0F3BAE1
FB0081008320C3F373AF5018332B22430CDDDBDEC320AD7DCC8103A39A3BF46F77F50DBC4F9E28
EDA7D5B132EE05F73B4A63938DBCD6DF0F366597CC3B9007BD74B5E5A25032B8B45B6C03D2FE
F6D091DC421925869D25C7B29011F875CFF4BB6282F67EC61231E86E325BCA125551050BE0EE4A2
4C62A3C80C89C0F22E5C158D98C1845C9E55E31430A804F0C3F223007DEBD937DF7720348622416
D46456583EA2C85FF5C503E4B9ACDE8E5F6F0785B38CF9F9F897E9310CFBC40CE2B1F18FC8D3B6
DFE6BF13054B64BC041E3B1161C4979273D51113E2D4ACA62386D8DBFFA380BB074E5D91383B
B44A45C98729017EC204DDFA5693BC9FF933F54BE214088EF2A589CDDDB1EDCB5D7F85798E5D4F
F0DAC610B8981A7DCF16D97240C1B93C595E255F2C0C580FEE7CC45715E2FA621A741E8E0FC688
1BE413B5403606A3BF89DA33DC4D399F3C37AE2DDAE800171400F01E3E234D1E485448E0558702E
C74D1874C3888B072A8DB17AAD26A4833110B2DB1F89A3482DEA76AF479AFCA4CE7BDBFFF248B
3C7079174C62FF377D18B200A15978EF4DA3BE4B207076753E8DBC013FAC8F744BDBE647F7E8DA

6DDD815EDA9C5F58B9FA5B60EBF073DF676ED7FDA6B7DB2EE5CF400BA200037A3BAC66EEC1
C2A790F33FD21F7335E23EA92C2C8114F9D6E20D7F796E396782B41AA2C71D72FE08B49C15EFAC
F9D1FBDDBCC4CB71FBF960C4D72FEC1ACFB5A0A03FC7862291EA287E539D58628400E58B12474E
1A03D346E7F3234E4E18C0BC5E90128C7138FCF36B86920FC2C7BFEC09EBFD643F8DC01530355F4
AC7BD806F9304EA08ED78C17A24EE6EBE63774FEBD372E8B79875CE11B8309C7BC7A708A35FCD
9DC607971684E0C627D160D794A3413DAA6619009A27C719E462212F30742FA14E51CBFDC5F6ECE
FD4C489BC6C475F16FCAC14EE35386E36351C640DB13C7073E8E04434101EA60EB576CB1C1DC0B
0E94AE3470E2399EC40EBD77CC419154DEF51BC64190FC396C0B0205945D7C3D521C67942671B6F
A1EDD3A20CF1FB91C359EEB26F4F5F9FA98B5F08410CF97EA09D2B4EFEC631AB98B6166A20040
52E86CAE5D350790658BFBF35CA79C362824CFE2A1921C1BD49C2BB54F591BF3F1B0713B46F9952
937A3F83735FBC75F3F3906E73381C41F9C345F16710BA3C70A01BC9986C33DF7F535F392952A33B
4C67F21BBA66D34F8B27969EFD918F1B1677AFDF1B4381A185F8877772F6F1C1241127CDFE36A57
B9D740744F890F6CE9140207E4D9E15041B2509437877FD0196DA9C0FCFA5B438A10360821498D10
32F2EBA1B7CA3CA7AC443EF1A3DA681FD5D423833A7E9284106FBFF03E781A9FCE01E6B59FE3C
D9B21DC82F59869633FE805D7F07693090137825D1110A9A86BD61056B81FB96BC4FE91025CF3B5
EE53DC0E20084CD439EBA0B7DA22B1F0CC100F63B2FFA7A03980728F828D3B777C282E689C4C10
7CBC25F4130ABA317BDE41EDC0DFAC99261B7D06B886E1F30085439DBCE1E7EB5541C1F1EDDD
0326419338B8330F9F97AC38F58B81EAA99432082ADE22242910A46B372FBB4B244096EB8C00E5B5
932268834873E7E08BE610196A784DA30E4E2DAED4072785DCCDD09A00607CC1409E62EF7FD018
29F990E8E557FC9DC59803608D8F4F49C770BC70640CFEF629360EBD21E26BA8D59B6D903463EC9
F0E53C8F45B86803658D0948609C8632BD286164D1D617763C1EF3458E333F33032C7FB6C3B977D4
BBFBF1E3A23B8E62D5D32176319BA71DDA3D30A8F631794A24D6814650BD605C0453D9E365FCB
7BE1CBA058F36671B63161DEAAC06978F24465EE6C8DE62FEDDDC77FE4C85E909875B77D0455A
ECED7000437F1B8EA1060518D1BD03C6BDDE8279CD9B8563A0C8DCCE627B7D2A19A4EDAEB8D
2A725DFC8CC2B177EDAB0251D1E6D9646322C5791FF95D6506850B8C4C18C507BC83E6723704243
37076F9675C4582F80C2A1A63C31EFB666CF6F0CD53DEBBC39808C307EB72B9C30B1D4AC0825B5
236179B260C63A18E0A62D302D6AF87538007F98A08480023FB262432B3CC508E9B9E900D792956F
DC3C2141AE2F1F475DE2CED65B9CA1C2215D43E594522A21F605DD0903C6806C188FC9B77CC7E
6C529D71F83065444E0E0765F1E672351A3CAD99E6F0D09E3DB50BF4EBFE3036C29AB03D7490834
4B3D1F6900BA9189781CF1A2460DA4054F339F3AAA811A7E5EE387CC44EFED4A653AB69CE0A23
DCDB95A93C7DBA242F332115BB271408C9D36698584AC340AFCDE2EF08477D4BB0F7BDA488C3
0CC738E61E4F7FAB38C68C212FC7F1E4937C87C06C9DD1D60DCB373C2E86032CEED301725B9972
2E876F8706F860B604BFE49C05121A1D412D0ED9AB464C5A4EC66A77A83707820C626211E433DA0
FF8851246FB9F34BCC34202A947F7908E3E074CD17F91943B98E3D603B3B9641F09E5FFB7077F7503
8393471D3394C7FB238A3A498332306B8DBE621DC6EE13D88E7C67B07B7E00A375CE4BF206FBD2
8BAEABB90B5E270A039D961FC8B55A6D705BB7B57FF405B9CBCF809C88C58A1EB960ABBEB297
EE2FB183DE8EE11BC5FCBCCE41E91160A7AC441E28DFCEFO3AC6667BC10234875E481A63AFEB
F208F6208A8C5CF08316BE895B6D5C3B23F61005D7B816456713A9402D3011B65C44150F9410B0693
9DB0FDBE110E34F742519FCCA7EBBB9D4BE6AEF6966C183C22E2BA0DC82E81ECBCD45AF918B
6FD978CEC43790CD258C3504739F355DD9EE2A141B92692CE211FC9FB651E0F92383AF7B4C3D825
6F2D87E773749643C870EBEC6A810C66614C3FC44061E3714BAA3F23B114E6F2DB597AE298D85BF
7811E55C03795EB6973D017869E19A79D6B2FD796BB00BAFC1B6F073ECAB0877F02E8A54241C923
BD0F5B7B3998954E793E888CD9B1C42013249C174C1258833D8E467FD02D91DE023B8D0BEC337D
A463316EC0F29399F7A108FC1BF3B18A007E6AD23CE16AD4A422734CC2A2F286323BFD74CD6D9
952932E5258BCF22E93B3C35C9EE77F423019A32997AF102679066610E17BF69DE1173ABB85B62A2
93025723C7AA50463F1765F5F9083B3900BFB6C1DFB87C0DCC701FED0ADD8F86853B70508F3738E

6446C011F0715F9C6F2A689EC05C10EC37577D1CFE8F875CB888EC3815F83BFE4ABA1CC2FE8311
B74A871D71F6742538F3318E3C3E97409BF77618494868371F8D93C706A684B77D9B281A2402E41E
D10ACC9702C6DA63A10001969C8F302E624A0C30F34F7DDD5DF6B1CE9171CFB9FE32422E637526
3A9F85CE9FDE1973AC75E87AC22B676C54427EBA857D559695CB3C8E5AF6D5F16119FE150042F0
2620CB84A7BFBADCB1DFD657CF6B9E4634801D3E76C29FA7DC28B0AA0036EA17BE1015E2820
1A7E747AD07A73EBB8DD00AA32440E52302065D5BE1ACE9270F80702E612D9C7DCECF463FC65D
AD18CBE04857DA6E058A96068C66BECCCF64F0295466C813C026731701316D9807A1FF28C7F305A
E0851EA7611C10588A40D316A40E0177FD5BFD9EBE982C68753FA6FC23057B990AF343D79D8BE8
BF0F0E141624C9310DBAF764E70B5F033157307F77A8E30E6817A880F65947AEC07876DF0B0DE3F4
CDC39E08BA9A047DD081B97D2F8A5DAC5CD70A06F58217B95863A870BDE89D87DCE5230D9826
03CB85DADCD99EAA54A7BBBA49C2A1817F74C224B0CDAA0F5588D85863F80B8D038A0CC545B
9223CD7FC54DB381DB402AD7DD2C3EBBEBFE0ABE15EDFC0F5B7E905176D9287C65998731B101
523EC59B82F33FF1A4E092260F3FC1029CDA05221BC7B2F53FEEAF3585C53B143A6D6D4E2C9D91
0F78983043E7EE489AC1920BF0FEC3A1897ECA21FD7C8805A24C258865B47FA5F88A9EA960C1CC
6EE5AC7B3CB4F35FC1D8A35470DA38EC2F7CB1B6D9DBDDE37E8F4DF91070EEC1F9C42EA5A6C
9C12D95464D8F1A08B02C5E2E0F919A572DE8376C193FB45A02FC71AF38C37A300BEEF4EB8F893
8AB27DE8E596892A41B4BB6B9122304FA7DAEA1B5FE7C516817B271ABD4DAABF8EEC5932634A
18D96B740385C4F37E8AA5EE106EFBB89EFF639AEFB431E49E9E62C030CF5DF43521E683FFA3BF
CDD846E10FB1C87F6E9597E1D58789C2092803A420708048886BC7B76F360F8B8AB3F7668FFEC286
EDC01F95845F1C41EB534FA9BF2ACD2C308250C649AE9EF2057B18B58243F9E1B131237C73D2643
2F45AEA237782A4BAF006D7B34B63F462942BAA9372B7490609F3098EB783B5CC73DAC5589B8B3
13E675EFE14184E0BA5723F21C9D363B2BA4B9968ACFEEEE7D60ACC421E0E60578BBFEBD0152B3
43D0EFE0F02E62FBFE889AA0CB0A1FAC3DDDC92BA80B478C41DD35EAC94D1580C8A2DB98AC
30B06268C7FDCC77160E93BE12DCB98E411007B867F908AE89794DDD068821A32E9A7B34B9D2BF
198BEACE35CCD9EDCE0221FDFF906C29642399E8737D302368DD52B7E36BF6ABB26CDC78157E3
3655E409EC4293E4C0549B4E6CDFD7AD2BCDEA4E880EC261BFBD16854CB03231CDEFE607D62B
EF4771170B36FEB41EBDD89510E36C70ABE0CC18BB0CDBC003C23499E2B748467A0ADFFB6B40F
028E27D9FC490F3ECEDF5A479225763AC8C5A9C6190871FB644816DF61514AAF9C24BA22181CFB
BF86B0D0857E37BD0920D0239F0A3FE8CFC49865C7372037D21B5203536684501ED65D16888150A6
C5026F8E579801D1A8520C3A873E4021BCBF4365BE4E67A8EBFDCE23B79CE082D6E084F30949227
BD73C3030364E0F6C90FC058F03E898C92C53D7B526F001907DB9545CC972B3DE92101E543E3FDB
53ECAB7439459A0ED2F83CB40538770329CC81F2B2CE80BA73568E622313A7D2561AA1FA21C522
E63EFE5BA9690372D206E33761724788BCFBF92D3B3FBFA45E35BC69E0750437CD89DA349F559E8
6648F6A3FA73E4EF1DE4C621F62F3875E8F50D2A9C113F99B7531F6D881EB9137AEAB77887278C1
120AF4BA48215A341E56B9FA0F5E4BEE2C07CC15434DD1076BB334A77EE842723CBE33D26BC58
AA9DACDF304062808D623E346066B950AF8F8BAAFF40461463C2226C9DC57EE44D46CC40E9ED
62E058CC6F4C6ACAF897803F773724A4EC407674C03BC8334F2D459E798C8D435F6D173632D41911
E6CCED619C4F43AAE1EAAEEF788133844BD4D81019135F003A1AB33024CEA60F12C05A76157703
02C889C22F22C9A7896B4AA84283D8F7ECE98E01A6FCEDF869D937A91C2B4AC7AB64287CB5A32
A2173913C5C97F7A6BACB20F54B89DD3D7A303C51F0E4845247B5F64B9426899E445F52C379FEE8
D2498F5A2A24736E0DD2849601E6F696480218358B1F43940AF3A6FAE8BF92601EEF42921DD833F
C6F5E4BB15E616DFAC81584D74E2164297DECA841781BDD2230F88D535BEBCA19DE687902460D
47B94ED420088FF12D9587E95283B40F5ADC9FC0A4E1E617AC1DB8090AB604FC8310C2DE5C740F
BA4BB1385C84E8F1F613C158BD12521E60E5C8C68A1830C6ACE3F6F3E201E823791727BF670185E
9E855744BBC363D718FE119D6F72187F7C7F537FC3E7C12C2BE8EEE584D67E863F0520214A043892
5A04B3C99E84858B60C3C551EEA79D7BAEE378263C466CBABB44C1D8221331AFE3D9359040D28

F02B4B6DEDAF3E3370941889C3A4D5B26341574791B63B9EC24F7164F2D3BA881C36C05484B79B
EE76F58E1458F8D437EF0E5DF1B7286ADD91F222A69DC8C12721AC91065FBCD848D800D415491D
4B4B8E5E586FB4402933C0DDB6D02D1AF7BC4FC35EFD8F9221B99C696079B3D5B375137812A810
3F7B189ABFA1C985FDFBB207F28BCE27AD21813631FC9F786811F44FF4CD3B11E48FBC839833991
5E97E9DE7576C9EFBA90EE4F4865AB03AC2BFD0BAE34D23B55CBEA3CF6CE54FD15C9CCABCB
42B5D713B2D1ED06CA86EB8DD88088FE1DE3F9B93B67D0697512919B5379E59E4F906F3B254AFF
F6729C070407FEE8D63AC2E1633ABFD13669CD103C5CBC08B34351B14D71C62943CDC43CBBC87
D0FD11D826AE533136645B5E9ABC26C84348EEBACEA702B32B53298BA7F1A76896F2E9E47FF4B
8A6400E7980686EB72E8AAC5D44FF4C44C1002AE290BE7DE80CBE3ABDF204B24F004C63328BB4
C7B59FD05BC1741B489DFF19CCDA26CE6125C1A1343164A970A077063D8500F37ACFE5A166E6F6
113A4BAA99070E07798C600D5AEC7D474D839627B0ABC24027F07A0BD3B478BB4FB19A91D0363
F303D89E6C6701F54226C114D6F361AB7DFC2211510B8779A7ED849FBAD97047C2F5AC258F3A8D
113868431D58B1AB696703AB9AB8E4F60CD57D01F4AD511EF5AC46C5726ED7069BF6D86CEC7EC
B86A75B99AC276F5D97B7904C93BC7B35711261DAB2BF909B28A8A408C45E3C2C0676F2241BFC9
6497AEA795212A45E690DE6948FE358F6C2580877D64B6904231B17EA511A09B676D7807465EF79C
476D5EFD50DDD55C68B4C78FA6B05FE252AC18D5CB7EA55F471C6C9F17A34E3883F12BD7941B5
560A83CE5B1BA8731AB1DC165863DD0DFBBC361C1535978DFDCB3B6AC65466DB78BA0FA1CD
C19D500F14EC12EA566EB5C4780C290B3D89CDCCD63456D11E8AAC0CEA0F5B929FD46B60A1E9
B42FA1852459DAD0F4FF0B5A4CA81F7DB19FDD3D3ADB2000D24B1AAB3FF0578A1B165FD7E5B
0338A4FF1C95A22486D37701296C3A49767A91FF94B9430E374C6094AE53E3BCFCF4574FB3952FF3
0C4C7F3591079B54EEC1FDE819BF23D0FCFB9057BC95A6F079A7D52B7F704245FCC7578D0623CC
4DDF7FBB6C2987090C463839E37BBB833D39EDD879426694F8D63645B1566C1384D38AC7FB3EF3
6A8369B8A2A2A9F6ABE949DD08FDC8524E067ABDE7AFB0F02E5E02BD5B26093ACAF54F9FC0E0
E60C4402A7358E3F48721708382BA031E5DAF07893F92521F0C9E88922C8B3AC46434F738A28A440
79D8B8FADF961C1D95D818D01DDC6DE8E6B623FFA168367CDB70C5AB802360D715360B8B80F9F
1EC66D7B26A9D6B7A034EF4F3B1C5F8282C6555C0DA8D58ED6685A94C29D44D68FE267CAFFEA
6F9A9AC94FEC9F9062B62EDC3678A149A95339C7CF02DE95B0CBADF86727AAF78B86DE79E41A
A2DE20F94BB25840F8E4BD16E2CE36699F4CE82E697A83997AAEB68136F7701684377FB5B31F599
E1CDB1E9CAB6D79234E6487A3BE0D5059A5F214157CC2F567D255FB86881541CE0D7CC3AEE519
46FBBA4DC865C715D1F76E44E5E5055B826B2329AB703E5A948A7A25F9341A2DEB77F9126A10EA
C176B679A359546C5CB105DB2C211EFC3CA0482DCBD20C6DA9AA54C64267255C1AF1E8BE30C2
0E072A43545D5375E59450C5E72227BCE2BEFB8A97429BD9D0E687AEAEC447A862DCC28A4B969
91A205E2C1D9BD92B08A69A1B6C5C4BE4D7C5361E4358E628820E2479D9C446FBF8BF35D3E33C1
E4A9FCF725DEA4A32BB0419A7422BB2F3806980C0C4D9D30A27026AA28A70E1201AC3DD59C7C
09C01A58F185D9F5487C10C0A7141BDA68E2471C6C6F255483196A75CEF5212340B0345356616DB9
4976A48B2CA6951F7D82AC8F3B3842192038CE0438FF3AA75A90F6B0B2A113550F7FA35751C9871
F2085A4509CF1BEF2271F3D9CB8D02A99B5300B2DB75BBA8624101C2FDDE9CA94817FB01E36254
C39F6FE427864F51723A9A387F290494802E375356BC3C5B38E187B35F0C0BB516F465D017C060BF
E554C5E03900D561F8FAA8592805E20F239B70EA5170061A98C02C30A07118E650B924A543D7A6C
EC6126B0B652D2B53705C6671E5C66031500EF833BF16FC7FF2307443AD3742B9DAF8C05E53D5A1
BA085335B9980D921780E2F81A1087C674E56557B2516F81CAE5929AF3033A4CDABCF21BC344DA
CCCA5A3149F68BE9305F3B313234036F80A01318604BB0C6F7BE085FCC9B7EE6839F625D701864C
5EAC5DF72CE8816CC73B79CBFA11D18054817265F97C2A69BEE5FD3955FD5BC50BD0EF48D6256
DD0FDEFBB939BC374F5675D414C1A501A086C2379A5F2AC2D933FEE465D65CE933BFA8403214E
E64C4C8B15404BA30577CD6FC51388F0CD41DEDD7A429D94FD51C12BF79484EE55256C0AFE517
9AB9AEA916BF99F137A66059860F0EB4FF2C4807C73B5CDD7B4786D4674E8112A2C91CDA62D63

A5C05955970210EE52CE2EEE1E6FAC5C96C421A79A8EF6A70625815B8E2EE5895FAC4D339FC4D
BC0EA548653655E755D8EEF00A2AB7617B11DE4C4B066802E64B0E63C75FE1C8454684A40ACDB
E68B4BDD77CDD52017575AD6CED5AAEDF00B1B9156689AE18ED6A6B6F401663358F01EF639C9
E2123ADACBB9A36C1BE8B871207F830EC3175503D6EBD1140C19184AE0BBEEF3281CB3ED30618
16383147524FB540CC35B542C9DDF16BECFBA346E4A5CB906562D6BECB0AD5282479BE0DF34E6
C95B35DB4475A4E369452C735EAC3332CD9443FB4ECC1B76DB1D2C5E76B6188EB66360CAC84A
C50A338F0F3F5BFED50214059C608EB65B92A5F6B6A324D72E95D148F0921A943F7C4322C70E038
31E717C6E34F598CD8F7DDD80CAFC683F4FA3C2E7E0BEA93299BE95C6A82C5B5DB6AEF52865C
A22750F4A96EA17CB34DF1DB042AE600827CD1A6B7769B5CE2CBB708D519F99EA9692A314035F
58A343A9BEF58A4412D2FA4D34EBFBE17DC1C74AFE62F980D2C03498E0FD0C8AC52FE3C4D1A0
86B1AB1B193B5F7BAB65614F5AA7A1AFA9DF1420ADF2A14431C737C59BA2BDB09F784E207D0B
4608AAE4D96E91B74F7F503FCD133ACF61F5E420EC6693BD6AD6AD9C8023269938D7E49E1269FE
8C97458D2740D97DDC868786F2A6A06D9BF3A640737E1FA55B3A9AD1E6CA78553CF62908EBE85
6ADEB8BA174C677FAFB74AA388D72217222B4DA3537956DA5647564186CD8621B5449C32937B0D
2120B61C971A82E6A081C4054392B0920760435F40F3D60718CCC0676FA4CEB8F1E01D064DB49C0
405A2F274610E97120A26F05B6DBC6B55A2A411E7633D6C66E90A4B163450DC15184FF80FF4FF4C
AFD8368D3422A104CB7AFE1601CC23051ADC98BA9E8AB9F296F6DB3C1003C6DBD0F84EF8A3FB
C048AF1CCBF9F8A2016C1012EE38925BB8EFB8044B621587626AD3371FF90ACD5EF21F91C99C93
A0D9F105865C9D849630780C8A3AB45B8965DA9053478D033CC13B45C3C34E3B834605278DBE8C
E1D0B4AF00B4B886D1BAA5F8A68D41502CAA2A01162EB8C09D2897403DA48F1A2261E2E1363E0
E6F21B72708A2F9E51C4909C42889DBD0ADF8F0B8D88439AAF573DE87993C75976AD48CD926034
7E504E989F004464E810291A6CF6A64C416D028EBF5BDE5CADD077F63AC99672D8812E43896206
BA6FCB8C91601956DB53E49F970F66B7BF9856FAA5CA43C976C51FBFE008AE4435F96472CB7662
675CB5C70319AA89EA7DAFEAB753F6C1BF453C3CFADFA97F36B816B508D5E9D276D3645D2B09
67E4367B5BC174AC4A57DAF36009D37A0FB09D6DBD497B809A73AA8F50080C71009A1D2D15B16
590165262434C7E86F176A404EB4A73F5FCBE1720017347772798381A2481B95B906932A2B5D70820
F0544C3DDB7F26517072148346B623E0353A10BFFDB6A4057D3AD78281643D085DB44DE0EC60DF
B093316230B8356CB0E456A946389E8F0E7EF268BB56A48F0C9235B8B38D6EEEA0B906E12192C4E
004A4B3B069F610710E14E7A4014CC441ED13D38D7EB05BEEBEC5CC6E4BA9FDD24DF4CE90C21
864C09DCC01D23C83B35B555195A8F8DB7817256EB3C16B97AD52CF4F0A91BBA69647B05AD4FC
4A86E90B5D11C0D671556BFF6548A99F03FFDAED2A5FB32B00BF68EF3CEB0812E545C8DCB793A
6DB754BE6801E263FE3C2B7D6C964C88B1A8B4DABB94667302C04726A46083B0D630351AD0B184
C8599EC7EFA14F39BBAC2A575D264E3AB89C26E7124A31D9F7BF8EE7914DDE36A674CA22FE1D
ABD333905D770FB1C545A7525297963834729C0BA991811FE91DD44D3D49AF3D26FF05284B44DF2
553B6624D64081A3C28A6DEDC5788B4CD56E97D049E71FAD721389C41D78620210D999310B9F310
85F9499FA60D31F55890EAA981E002F246A3499

ct =

FB2DC6B73740AAAD817328BFC5CD00851F217C3A570BD6E21FFB8F7B0863065A055D3DFA791BE
CF01F144980BB6062C0716130788697FA5091A09B54B294227F2F5BC0CF47866EB52E67E86F26C0A
33C74F579F4E5A288556B4476381DBA258386028404218ECF7C615251353B4FEDF21570A666F6E388
44E2DB41C73D87B70B54C9BD416ED5ECC851FA8EFEF490BCC60C3B6C8FA1543FA2746A4882D60
2A6944DCAFB4101E332A3197838037CB0F21430C6AAE47B0A8DAF3C44F7E5C6AF262F014E62E49
6EB9270347AE788E909FAC54D54670ED1919FCDBA3D93C926134334A29E35F3111C160FCC280A00
4A21262370EBB5BF04A46DB413DE0906D41E430C16A4227E269C0C4B8EBA43C25ACAD7428A518
957DBD9BE0E24CB394D8020BDDDB281AB1B1EA81C394437523B28F7AB2CB31735B79EF58B7C067
7F08333C3BDB4F882B30791F336758C02210E95E967C574192C1547574CC46BE8335368234C4DB578

175818224906D982258E943E72CAFEDC583A75D7405C5CEDCD562AE21DAE90110BF5F78A5B7E5E
44F70BED046EFA19CF5493CF5097F40DC5948C2278CC144C7E7F7A0264452D85DFB6AE5E6750555
3915935D52608F8AC08523F3459F40C2FE2055764632D780B35EC35E04D52E61B8B463AFF32E11ED
2A932455EDD326DF67A173A7CFCF956C9A8D3A82FCF8EFC008ACA125685E71D0066D527CA3
1992EE47434EC622B2D6FDB36038F10F26C7D38AF4472BE02F96071D8F84EF00EEE12E905956169F
62CAC918433187E55726B8256B82088E087E8844D953442BB85F7F04BA5B47D7181FE4B638AA8BF
EC100ADDAD056B07C0EA2BD29C6DE15B0EFEEB70C5CE9637903F61500B6C19DC7004DF21C018
B3AA8D023CE32A7764FB860B32FCC492A122B50E0121073F2E5F45796DDB374795BE24907041497
4CF8DD2E09022B44E94D151089D591A578B57020F89454C97966081DAA02DA12F665B5E916CE03B
B227857F23D7873EB8AE0F3C334D51B5BAB36518D86B05D50043D44009B566B9506A25490A3EC3
C09CA7931940D763F45BDCB00394330439593DBF672E5901C251C7D70043C86E93DA25DAA79702
B21C59387152F92AE162CBA358832331EED5764088372F2F84FE83C6DCC6361C077BB7A853251FD
7672619C6C5045F3DCB81212FA5D033391A2632CEA4858113FCD9BC03897547D672BB707D503D08
E6C169303FA62BC9DE30F345522BD2774FD97B6062A0CA25164AB70D94227D029A5C19E5A39828
42FB5468A4D025F0773E5CC54E2401A8E0D12DC70801D54F37961733805A5B760D5B331C19FB174
C9AEA84A2FFD30F771969ADEE47029F71DD3926075CA89B25135277B55C9B8217D696EB8E6407F
305A4A5D26E90E00E09611A1EAF564F0EF538F6523362873295F987866B7F2F496627FC10A6E9380D
12096DB1E6A0538E5D51436ADAA3123124CA6C3C060AE2B308F5F70F20FD0A5423828A769130926
04F239077C87C2C0081B892EFF4BC7D450AF20AC8A6044CEA55A5805352D31A9F51008195F9C423
D446ED8AC39F3181AD4C9DEC632712AC879B53EEAE48B8286E31B89215034DF1E2D341CC131D1
76DC52324367C71F6195A0979FA95B5340D767E8A002F88320776A174D13E77A5215D850903C9C12
652F8FAD1EC42CDB72457DCE5D9390D34E7ADC3FB81F5B64E8ED154D3BCF630348EA26C9A12C
C583F6F82B7EABAC299353DCE5E289FFF98730E61CC68C0DC913365E204B7624B8585A2BD33FC9
67C844DABD17374D5AC0A9D159D11D35B3A8137360B8A94765F63FD5F45972FA0541ACA939382
D674E5B198AD3E0CCF1FB353642E426456A4D15CF42462E8E42810CD25C4EAEA955A06434A6207
DB64265C85E5DECD06797E5C013E261F0F382A6B852A6C4EF163D59757303368375ABA953E554F7
A6AE241166208948910522D47DE47C6BF17B332FDDA2734EB10B6DF702952D5C8360E645D3A83A
2D575BA30D0852A45DE3DD0D2A1DEB0590FD3782047AEAFB67B7813BFC2097693F0182DF70095
0DD311E86B440F81340EBA285F46A5AF7414925663FD2B08F20190244637A388663BE9062417A5584
3F4B251E4CBFB7536660B007463B623F3221A4F877DB16100EF8872CE512486EFA97987566E1CE0A
46953D33A24CC4941F03E166C0062DC2101C0862E171B32A65F15541D50211A40591F6921AD6E7A5
6221320E547124B57562F1209A49561A46F9A3FF5CB5336DA7B97D61EB17752D6D9D805B95E1C71
81C82B78E96D77F8571F5757BC86FC1CAC234BF0E5048F2E078D5C694226D6425A4B87B6FB389E0
E6AE0361D8BB282F1BECF9686244D0365435CCC2A89E788D942558432E06381A061F4B27A3FFC03
2BF64E03C3FBCF47091FBB25145B61BE9F88D4F557819105967E2FA66FE10575C43A7037991DDB97
489B52F495C00F4EB445CF4E2217F84991722D46A1D50F65A9CEB112901B1401E15CD9D238EEC81
A6C6D2C4B87F76D0A157CFD2D996418FA79227583522EE365BB42ED115524FAB34851DF611DE6B
0F42036BAE7593F44C392C424D506934F3D055630328BA2F425103892915A49413A1ACD74AA91FC8
4826D1EA668D7522433FEEF2ED12F2A74A4EF3F3825430B936249FAD45E8387A77788E34D5B0231
F43AD2A727EA53F1429C7AE542CA912F015C0B6B1DD1A626394611BBFCF0D2CD022A24475B75D
5B50DE22C1203CA35ED2657738E1D4D5EEBEA2317A943CC467D371297D36001D5797E599D63423
00F1C44131E92543AC39426664480C9C50DFAB39C20AC97D598E42918614362112F23CE82F8006BF
27A3DD52FC1B2813899D315F29A4D5B1C769A7FC3F2B1EBBD834074AFC23251E5AEA43C75F58A
320B902399167045851704F1843B2DE12A199191B4FCF1083F87D1135AD9B8A3071613DF32FD4AAB
36772F7027C6C93A251A3B477D9D2716224BA45C72984A047E85C7F108A5995281928B162F05079E1
48694813EC765FCE92A52F52C972E9085AD7D432968AAACE23EC11DF545F139E0A1C2B4A335738C

815E3045B024130E2818B133CE3B60808BE0EDB987070E2C5C0625E454272FE1DC371CDDF931D41
5F0263817EB548FC55F9E4B0A235BE31395D986EE19529D1466E50C487A0709FA63C5C85707745276
204B5534F12165E560CC1813A10914995260FCB669758B3C57CA4B76CA3B77D710771C75F214C4F5
D7FDFB8C22B3C9DCC2233FA2120A381AEE02325B5FFF124EFAF09611B3859F0D9ED2D3FC9484D
A9095830D73B21EBDACF132D592D8CDA9406AE4FA7D3859EE5A654B00871E40

ss_enc =

0FA47F93CE8DAD32F2FE0D7E78D147DA29BEDDDD6F0BF2D4AE7AA192C3F014DEC7FA458B69
ADA01CDDC0338FDA62B9C6

ss_dec =

0FA47F93CE8DAD32F2FE0D7E78D147DA29BEDDDD6F0BF2D4AE7AA192C3F014DEC7FA458B69
ADA01CDDC0338FDA62B9C6

result = PASS

3.2. 性能与资源消耗测试

测试编号	REF-4
测试名称	CTL-257-512 参考实现版性能与资源消耗测试
测试目的	测试 CTL-257-512 参考实现版密钥生成、封装、解封装三个核心操作的平均 CPU 周期数、吞吐量、静态内存占用和峰值内存占用。
预置条件	1.CTL 参考实现版程序代码运行正常； 2.benchmark_selfeval_ctl.c 能够正常调用 CTL 算法接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-257-512 参考实现版程序代码目录； 2.输入 make all 生成 benchmark_selfeval_ctl.exe； 3.运行 benchmark_selfeval_ctl.exe 程序分别统计密钥生成、封装、解封装的总耗时、平均耗时、平均 CPU 周期数和吞吐量； 5. 程序在算法运行前记录静态/基线内存占用，在多次执行过程中记录峰值内存占用,并最终生成 ctl_kem_benchmark_raw.csv， ctl_kem_benchmark_summary.csv 结果文件。
预期结果	程序能够正常运行，并输出算法的性能测试结果。包括密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量、静态/基线内存占用与峰值内存占用。
实测结果	程序能够正常运行，并输出性能测试结果。程序分别统计密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量静态/基线内存占用与峰值内存占用，测试次数为 100 次。 密钥生成平均 CPU 周期数为 7694794 cycles/op，吞吐量为 314 ops/s； 封装平均 CPU 周期数为 29607 cycles/op 吞吐量为 81973 ops/s； 解封装平均 CPU 周期数为 159884 cycles/op，吞吐量为 15138 ops/s。 静态/基线工作集内存平均值为 5712865 Bytes； 峰值内存平均值为 5717524.48 Byte；

测试编号	REF-5
测试名称	CTL-769-1024 参考实现版性能与资源消耗测试
测试目的	测试 CTL-769-1024 参考实现版密钥生成、封装、解封装三个核心操作的平

	均 CPU 周期数、吞吐量、静态内存占用和峰值内存占用。
预置条件	1.CTL 参考实现版程序代码运行正常； 2.benchmark_selfeval_ctl.c 能够正常调用 CTL 算法接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-769-1024 参考实现版程序代码目录； 2.输入 make all 生成 benchmark_selfeval_ctl.exe； 3.运行 benchmark_selfeval_ctl.exe 程序分别统计密钥生成、封装、解封装的总耗时、平均耗时、平均 CPU 周期数和吞吐量； 5. 程序在算法运行前记录静态/基线内存占用，在多次执行过程中记录峰值内存占用,并最终生成 ctl_kem_benchmark_raw.csv， ctl_kem_benchmark_summary.csv 结果文件。
预期结果	程序能够正常运行，并输出算法的性能测试结果。包括密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量、静态/基线内存占用与峰值内存占用。
实测结果	程序能够正常运行，并输出性能测试结果。程序分别统计密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量静态/基线内存占用与峰值内存占用，测试次数为 100 次。 密钥生成平均 CPU 周期数为 30364861 cycles/op，吞吐量为 79 ops/s； 封装平均 CPU 周期数为 62111 cycles/op 吞吐量为 39012 ops/s； 解封装平均 CPU 周期数为 316103 cycles/op，吞吐量为 7655 ops/s。 静态/基线工作集内存平均值为 5911744 Bytes； 峰值内存平均值为 5922570.24 Byte；

测试编号	REF-6
测试名称	CTL-3329-2048 参考实现版性能与资源消耗测试
测试目的	测试 CTL-3329-2048 参考实现版密钥生成、封装、解封装三个核心操作的平均 CPU 周期数、吞吐量、静态内存占用和峰值内存占用。
预置条件	1.CTL 参考实现版程序代码运行正常； 2.benchmark_selfeval_ctl.c 能够正常调用 CTL 算法接口； 3.测试程序可正常编译并运行。
测试步骤	1.进入 CTL-3329-2048 参考实现版程序代码目录； 2.输入 make all 生成 benchmark_selfeval_ctl.exe； 3.运行 benchmark_selfeval_ctl.exe 程序分别统计密钥生成、封装、解封装的总耗时、平均耗时、平均 CPU 周期数和吞吐量； 5. 程序在算法运行前记录静态/基线内存占用，在多次执行过程中记录峰值内存占用,并最终生成 ctl_kem_benchmark_raw.csv， ctl_kem_benchmark_summary.csv 结果文件。
预期结果	程序能够正常运行，并输出算法的性能测试结果。包括密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量、静态/基线内存占用与峰值内存占用。
实测结果	程序能够正常运行，并输出性能测试结果。程序分别统计密钥生成、封装、解封装操作的平均 CPU 周期数、吞吐量静态/基线内存占用与峰值内存占用，测试次数为 100 次。 密钥生成平均 CPU 周期数为 1585861935 cycles/op，吞吐量为 1.5ops/s； 封装平均 CPU 周期数为 4412249 cycles/op 吞吐量为 548 ops/s； 解封装平均 CPU 周期数为 36017786 cycles/op，吞吐量为 67 ops/s。

	静态/基线工作集内存平均值为 7443486 Bytes; 峰值内存平均值为 7519150.08 Byte
--	---

3.3. 传输与存储开销测试

测试编号	REF-7
测试名称	CTL-257-512 参考实现版传输与存储开销测试
测试目的	统计 CTL-257-512 参考实现版在密钥封装过程中涉及的公钥、私钥、密文和共享密钥等数据尺寸。
预置条件	1.CTL 参考实现版程序代码运行正常; 2.算法接口能够返回或确认各类数据对象长度; 3.测试程序能够输出传输与存储开销统计结果。
测试步骤	1.进入 CTL-257-512 参考实现版程序代码目录; 2.输入 make all 生成 benchmark_selfeval_ctl.exe; 3.运行 benchmark_selfeval_ctl.exe; 4.记录公钥长度、私钥长度、密文长度和共享密钥长度;
预期结果	能够得到 CTL-257-512 参考实现版全部相关数据对象的长度, 单位为 Bytes。
实测结果	程序能够正常运行, 并输出传输与存储开销测试结果。程序分别统计公钥长, 私钥长, 密文长度, 共享密钥长度, 测试次数为 100 次。 公钥长度为 521 Bytes, 私钥长度为 2953 Bytes, 密文长度为 473 Bytes, 共享秘密值长度为 16 Bytes。

测试编号	REF-8
测试名称	CTL-769-1024 参考实现版传输与存储开销测试
测试目的	统计 CTL-769-1024 参考实现版在密钥封装过程中涉及的公钥、私钥、密文和共享密钥等数据尺寸。
预置条件	1.CTL 参考实现版程序代码运行正常; 2.算法接口能够返回或确认各类数据对象长度; 3.测试程序能够输出传输与存储开销统计结果。
测试步骤	1.进入 CTL-769-1024 参考实现版程序代码目录; 2.输入 make all 生成 benchmark_selfeval_ctl.exe; 3.运行 benchmark_selfeval_ctl.exe; 4.记录公钥长度、私钥长度、密文长度和共享密钥长度;
预期结果	能够得到 CTL-769-1024 参考实现版全部相关数据对象的长度, 单位为 Bytes。
实测结果	程序能够正常运行, 并输出传输与存储开销测试结果。程序分别统计公钥长, 私钥长, 密文长度, 共享密钥长度, 测试次数为 100 次。 公钥长度为 1230 Bytes, 私钥长度为 6030 Bytes, 密文长度为 1006 Bytes, 共享秘密值长度为 32 Bytes。

测试编号	REF-9
测试名称	CTL-3329-2048 参考实现版传输与存储开销测试

测试目的	统计 CTL-3329-2048 参考实现版在密钥封装过程中涉及的公钥、私钥、密文和共享密钥等数据尺寸。
预置条件	1.CTL 参考实现版程序代码运行正常； 2.算法接口能够返回或确认各类数据对象长度； 3.测试程序能够输出传输与存储开销统计结果。
测试步骤	1.进入 CTL-3329-2048 参考实现版程序代码目录； 2.输入 make all 生成 benchmark_selfeval_ctl.exe； 3.运行 benchmark_selfeval_ctl.exe； 4.记录公钥长度、私钥长度、密文长度和共享密钥长度；
预期结果	能够得到 CTL-3329-2048 参考实现版全部相关数据对象的长度，单位为 Bytes。
实测结果	程序能够正常运行，并输出传输与存储开销测试结果。程序分别统计公钥长，私钥长，密文长度，共享密钥长度，测试次数为 100 次。 公钥长度为 3009 Bytes，私钥长度为 15617 Bytes， 密文长度为 2353 Bytes，共享秘密值长度为 48 Bytes。